Derivatives – Study Notes

Conceptual edition from the 2026 course material

20 October 2026

This concept-organized study edition complements the source-faithful `Derivatives_Notes`. The theory integrates definitions, assumptions, derivations, figures, tables and examples into a continuous explanation. Repeated slide frames and repeated exercise statements are merged at their conceptual destination. The full question sheet is interleaved with clearly labelled generated answers. Source-page references and the coverage manifest record the reorganisation and explicit administrative and contextual visual exclusions. OCR screening and coverage checks do not certify every original formula. No recordings were supplied.

Contents

I Theory **6**

1 Market mechanics and compounding **6**
1.1 What a derivative is—and what it is for . . . 6
1.2 The price you see is not always the price you trade . . . 6
1.3 Short selling: a sale followed by a purchase . . . 7
1.4 One cash flow, several rate conventions . . . 7
1.5 Why log returns add across time . . . 8

2 Forwards, futures and the cost of carrying an asset **8**
2.1 Start with the contractual cash flows . . . 8
2.2 Derive the fair delivery price by replication . . . 9
2.3 What changes when a forward becomes a future? . . . 10
2.4 Margin magnifies cash-flow pressure, not the underlying payoff . . . 10
2.5 Why use index futures? . . . 11

3 Options: asymmetric payoffs, parity and insurance **12**
3.1 A right is different from an obligation . . . 12
3.2 Payoff versus profit . . . 12
3.3 Put–call parity comes from equal terminal payoffs . . . 15
3.4 Bounds are quick arbitrage checks . . . 16
3.5 Build the payoff you need . . . 17

4 Cross-hedging an imperfectly matched exposure **20**
4.1 Why a related future can still reduce risk . . . 20
4.2 Choose the hedge ratio by minimizing variance . . . 20
4.3 Turn the ratio into tradable contracts . . . 21

5 State prices and risk-neutral valuation **22**
5.1 A two-state example fixes the intuition . . . 22
5.2 The general finite-state framework . . . 22
5.3 From Arrow–Debreu prices to probability weights . . . 23

5.4 The same rule under objective probabilities . . . 23

6 Utility, the Euler equation and equilibrium 24
6.1 No-arbitrage prices versus an economic explanation . . . 24
6.2 Trace one extra unit of an asset through both budgets . . . 24
6.3 Connect the three pricing languages . . . 24
6.4 What equilibrium adds . . . 25

7 Forwards and futures: exercises with worked answers 25

8 Binomial pricing: replication before probabilities 31
8.1 Why a new pricing model is needed . . . 31
8.2 The one-period example, including the financing leg . . . 31
8.3 Dividend yield changes the replication, not the principle . . . 32
8.4 Delta is a holding and a local sensitivity . . . 33
8.5 Two periods: calculate terminal payoffs, then step back . . . 33
8.6 Three periods and self-financing rebalancing . . . 33

9 Puts, path dependence and the information a tree must retain 34
9.1 The put uses the same stock tree . . . 34
9.2 The three supplied interview questions . . . 35
9.3 An Asian payoff cannot forget the path . . . 35

10 Volatility and risk-neutral weights 36
10.1 Square-root-of-time scaling and the forward tree . . . 36
10.2 The pricing weights are risk-neutral probabilities . . . 36
10.3 The n-period European formula . . . 37

11 American exercise and real investment options 37
11.1 Exercise value versus continuation value . . . 37
11.2 A real option is a right to take a business decision . . . 38
11.3 The supplied expansion exercise . . . 40
11.4 Other real options and limitations of the model . . . 41

12 Options: supplied exercise series II 41

13 Binomial model: complete exercise set 47
13.1 Problem 3.1: replication versus immediate exercise . . . 48
13.2 Problem 3.2: American call in a two-period tree . . . 48
13.3 Problem 3.3: call and put hedge quantities . . . 48
13.4 Problem 3.4: dividends and early exercise . . . 48
13.5 Problem 3.5: three-year American put . . . 49
13.6 Problem 3.6: the first integer strike that should be exercised . . . 49
13.7 Problem 3.7: full node values and parity . . . 49
13.8 Problem 3.8: two-step stock tree . . . 50
13.9 Problem 3.9: three-step values and the exercise decision . . . 50

14 From binomial trees to Black–Scholes 50
14.1 Discrete steps and the continuous-time limit . . . 50
14.2 Refining the tree changes both the outcomes and their weights . . . 51
14.3 Normal log returns and lognormal stock prices . . . 52
14.4 Model assumptions and inputs . . . 54
14.5 The limiting derivation: two binomial tails . . . 54
14.6 European call, put, replication and beta . . . 55

14.7 Precursors and the replication breakthrough . . . 56
14.8 Historical oil hedges: basis risk and feedback . . . 57

15 A measure-change derivation of Black–Scholes 58
15.1 Historical dynamics and the discounted stock . . . 58
15.2 Eliminating the drift changes the measure . . . 58
15.3 Radon–Nikodym and Cameron–Martin–Girsanov . . . 59
15.4 Computing the call expectation explicitly . . . 59

16 Greeks, market-making and dynamic hedging 60
16.1 An unhedged dealer has an arbitrary position . . . 60
16.2 Delta neutrality and daily rebalancing . . . 60
16.3 Definitions, signs and reporting units . . . 61
16.4 Analytic formulas . . . 61
16.5 The complete numerical Greek example . . . 62
16.6 How the Greek curves change with moneyness . . . 62
16.7 Gamma neutrality requires another option . . . 63
16.8 Calendar spreads and unequal time decay . . . 64
16.9 Developments after Black–Scholes . . . 65

17 Discrete hedging, transaction costs and the charged premium 65
17.1 A quantile-based charge and its sign . . . 66
17.2 Proportional transaction costs . . . 66
17.3 Funding the additional gamma-hedging option . . . 66
17.4 The supplied experiment and its interpretation . . . 67

18 Measuring volatility, implied quotes and variance exposure 68
18.1 Unconditional dispersion and conditional volatility . . . 68
18.2 Implied volatility is an inversion of a price . . . 69
18.3 From a log payoff to variance . . . 70
18.4 The volatility index and its discrete option strip . . . 71
18.5 References for the variance-replication argument . . . 72

19 Beyond Black–Scholes: stochastic variance and jumps 72
19.1 What changes, and what remains from the benchmark . . . 72
19.2 Heston: two state variables and correlated shocks . . . 73
19.2.1 Pricing decomposition and its probability interpretation . . . 73
19.2.2 Characteristic functions and inverse Fourier integration . . . 74
19.2.3 Euler simulation and the positivity problem . . . 74
19.2.4 An alternative using the exact CIR variance transition . . . 75
19.3 Merton: diffusion combined with compound-Poisson jumps . . . 76
19.3.1 Why choosing the risk-neutral jump law matters . . . 76
19.3.2 First call-price representation: condition on the jump count . . . 76
19.3.3 Equivalent representation with tilted Poisson weights . . . 77
19.4 Calibration in option space and estimation from returns . . . 77
19.5 Simulating jumps without changing the stated model . . . 78

20 Credit risk: claims on firm value and default intensity 79
20.1 The option-pricing argument used by structural models . . . 79
20.2 Equity is a call and risky debt contains a short put . . . 80
20.3 Inferring asset volatility from traded equity . . . 81
20.4 Distance to default and the KMV interpretation . . . 81
20.5 Worked example: two observables, two unobservables . . . 81

20.6 Capital structure, incentives and the limits of the model . . . 82
20.7 Reduced-form models and hazard rates . . . 82
20.8 Altman's accounting and market-variable score . . . 83

21 Black–Scholes: worked problems **83**
21.1 Problem 4.1: call, put and parity . . . 83
21.2 Problem 4.2: the price of a block . . . 84
21.3 Problem 4.3: a forward-start call . . . 84
21.4 Problem 4.4: prepaying rolling insurance . . . 85
21.5 Problem 4.5: a physical-measure prediction interval . . . 85
21.6 Problem 4.6: European values and an American-put conclusion . . . 85
21.7 Problem 4.7: very long maturity with and without dividends . . . 86
21.8 Problem 4.8: a positive dividend yield dominates the limit . . . 86
21.9 Problem 4.9: the stock–strike and rate–yield duality . . . 87

22 Greeks and hedging: worked problems **87**
22.1 Problem 5.1: recovering a price from delta . . . 87
22.2 Problem 5.2: the size of a break-even daily move . . . 88
22.3 Problem 5.3: portfolio elasticity is value-weighted delta . . . 89
22.4 Problem 5.4: changing time while holding the stock fixed . . . 89
22.5 Problem 5.5: reading the capped-stock payoff . . . 89
22.6 Problem 5.6: a stale hedge and an inferred funding factor . . . 90
22.7 Problem 5.7: the Greeks of a bull spread . . . 90
22.8 Problem 5.8: a delta-hedged short call overnight . . . 91
22.9 Problem 5.9: an overnight delta-hedged bull spread . . . 91

23 Credit risk: worked problems **92**
23.1 The structural calculation used in Problems 6.1–6.4 . . . 92
23.2 Problem 6.1: default probability and spread . . . 93
23.3 Problem 6.2: conditional loss and a yield approximation . . . 93
23.4 Problem 6.3: recovery and the exact logarithmic identity . . . 93
23.5 Problem 6.4: maturity, default and recovery . . . 94
23.6 Problem 6.5: a jump to zero . . . 95
23.7 Problem 6.6: four-year rating migration (supplementary) . . . 95
23.8 Problem 6.7: other initial ratings (supplementary) . . . 96
23.9 Problem 6.8: independent defaults and a CDO waterfall (supplementary) . . . 96
23.10 Problem 6.9: perfectly correlated defaults (supplementary) . . . 97

II Code **98**

24 Reading the MATLAB payoff explorer **98**
24.1 Inputs, positions and the terminal-payoff curve . . . 98
24.2 Why the second curve is payoff minus financed cost . . . 98
24.3 Pricing and implementation limits . . . 99

25 Efficient binomial implementation: European and American puts **99**
25.1 From backward induction to a terminal weighted sum . . . 99
25.2 Recursive and logarithmic probabilities . . . 99
25.3 Black–Scholes comparison and early exercise . . . 100
25.4 Reported computational times . . . 100
25.5 Supplied comparison script . . . 100
25.6 All eleven source addendum listings . . . 104

26 Hedging and advanced-model implementations **108**
26.1 Discrete delta and delta–gamma hedging 109
26.2 Heston call pricing, Monte Carlo comparison and calibration 111
26.3 Comparing Euler and exact-variance-step Heston paths 114
26.4 The Merton call/put and implied-volatility surface 116
26.5 Physical-measure jump-diffusion return histograms 117

A Course references **119**

Part I
Theory

1 Market mechanics and compounding

1.1 What a derivative is—and what it is for

A *derivative* is a contract between two parties whose value depends on something else, called the *underlying*. The underlying may be a stock (Apple), bond (a T-Bond), exchange rate (EUR/CHF), index (S&P 500), metal (gold), or commodity (soybeans). An option gives a conditional right, a forward or future fixes a later exchange, and a swap exchanges specified cash-flow streams. The contract itself is not the underlying: it specifies who receives what, and when, as the underlying changes.

The same contract can serve different aims. A hedger offsets an existing exposure; a speculator chooses an exposure; and a trader may use a derivative to change exposure with fewer transactions in the underlying. The course contrasts exchange-traded and over-the-counter markets with historical charts of contract counts and notional amounts. These measures describe different things: the count is the number of open contracts, whereas notional is the reference quantity used to calculate payments. Neither is, by itself, the amount at risk or the current market value of those contracts.

The historical market-size illustrations cite the Bank for International Settlements' Quarterly Review: June 2013 for contracts outstanding and June 2012 for exchange-traded notionals and OTC derivatives.

1.2 The price you see is not always the price you trade

The *ask* (offer) is the market-maker's selling price, so a buyer pays it. The *bid* is the market-maker's buying price, so a seller receives it. Normally bid $\leq$ ask; their difference is the *spread*. Brokerage commissions add to a purchase and reduce sale proceeds. For N shares, proportional commission rate c, bid B, and ask A,

$$\begin{aligned} \text{purchase cash paid} &= NA(1+c), \\ \text{sale cash received} &= NB(1-c), \\ \text{immediate round-trip cost} &= N\big[A(1+c) - B(1-c)\big]. \end{aligned}$$

Thus an unchanged quoted market can still produce a loss on an immediate buy–sell round trip. The spread and both commissions matter; applying the bid to a purchase reverses the market-maker's perspective.

Flat commission example. For 100 XYZ shares quoted at bid \$49.75 and ask \$50, with a flat \$15 commission on each trade, buying costs \$5,015 and selling yields \$4,960. The immediate round-trip cost is \$55.

Worked example

Problem 1. ABC has bid \$40.95, ask \$41.05 and proportional brokerage 0.3%. What do you pay to buy 100 shares, receive from selling 100 shares, and lose on an immediate round trip?

Solution. The lecture's ABC quote is $B = \$40.95$, $A = \$41.05$, $N = 100$, and $c = 0.003$. Buying costs $100(41.05)(1.003) = \$4{,}117.315$, displayed to cents as \$4,117.32. Selling returns $100(40.95)(0.997) = \$4{,}082.715$, displayed as \$4,082.72. The round-trip cost is \$34.60, calculated from the unrounded values. Of this, \$10 is the quoted spread on 100 shares and \$24.60 is commission.

1.3 Short selling: a sale followed by a purchase

To take a short position in a share, a trader borrows it and sells it now, then later buys a share to return to the lender. Ignoring collateral interest, borrow fees, commissions and dividends, the cash profit per share is $S_0 - S_T$. A fall in price helps the short seller; a rise hurts. Unlike the maximum gain on a stock bought outright, the loss from a rising stock price is not capped by the initial price.

If a dividend D is paid while the share is on loan, the short seller must compensate the lender. In the supplied IBM illustration the share is returned after 90 days; the following ledger records the opening sale, dividend ex-date and day-90 close:

	Opening sale	Dividend date	Close and return
Share movement	Borrow, then sell	—	Buy, then return
Cash flow	$+S_0$	$-D$	$-S_T$

The broker locates the borrowed shares, may obtain them from its inventory or another account, and manages the lending and return. This operational step matters: the ability to short an asset is conditional on locating a borrow. If many short sellers need to buy scarce shares to close at once, their covering demand can drive a *short squeeze*. The lecture uses Volkswagen's 2008 price surge as an illustration, not as a general pricing rule.

The broker may also locate shares at another brokerage, receives any interest benefit from lending them in the source's account convention, and is responsible for their return; margin requirements protect the broker against default.

The historical Volkswagen example reports a 348% two-day gain in 2008. The accompanying quote is EUR945 (about \$1,200), up 82% on the Tuesday and 244% over October; its market capitalisation reached about \$346 billion, just below Exxon Mobil's. These dated Wall Street Journal/Thomson Reuters figures illustrate the squeeze mechanism.

Worked example

Problem 2. Short 300 XYZ shares at \$30.19 with 0.5% commission, and close at \$29.87 with the same proportional commission. What profit remains?

Solution. In the course example, short 300 XYZ shares at \$30.19, cover at \$29.87, and pay a 0.5% commission on both trades. Initial cash received is $300(30.19)(0.995) = \$9{,}011.715$; the purchase to cover costs $300(29.87)(1.005) = \$9{,}005.805$. The profit before any borrow fee or dividend is \$5.91. Rounded to cents, the two cash amounts are \$9,011.72 and \$9,005.81. The raw price move would have yielded $300(0.32) = \$96$; commissions consume \$90.09 of that amount.

1.4 One cash flow, several rate conventions

An interest rate is incomplete without its compounding convention. If \$1 grows for T years, an effective annual rate r_{eff}, a nominal annual rate r_n compounded n times a year, and a continuously compounded annual rate r_c produce, respectively,

$$(1 + r_{\text{eff}})^T, \qquad \left(1 + \frac{r_n}{n}\right)^{nT}, \qquad e^{r_c T}.$$

The continuous expression is the limit

$$e^{r_c T} = \lim_{n \to \infty} (1 + r_c/n)^{nT}.$$

These are alternative descriptions of the same accumulation only when their numerical rates are converted consistently. Equating one-year growth gives

$$r_c = \ln(1 + r_{\text{eff}}), \qquad r_{\text{eff}} = e^{r_c} - 1, \qquad r_c = n \ln\left(1 + \frac{r_n}{n}\right).$$

For a zero-coupon bond priced P today and paying face value F at time T, discounting and its inverse are

$$P = Fe^{-r_c T} = \frac{F}{(1+r_{\text{eff}})^T}, \qquad r_c = \frac{1}{T}\ln\left(\frac{F}{P}\right).$$

Worked example

The lecture's five-year zero-coupon bond costs \$62.092 and pays \$100. Its effective annual return is $(100/62.092)^{1/5} - 1 \approx 10\%$. The continuously compounded quote is $\ln(100/62.092)/5 \approx 0.47655/5 \approx 9.531\%$. The two figures do *not* disagree: $e^{0.09531} \approx 1.10$.

1.5 Why log returns add across time

The one-period simple return from S_0 to S_1 is $R = S_1/S_0 - 1$. The corresponding log return is $\ell = \ln(S_1/S_0)$. Over successive periods,

$$\ell_{0,2} = \ln\frac{S_2}{S_0} = \ln\frac{S_1}{S_0} + \ln\frac{S_2}{S_1} = \ell_{0,1} + \ell_{1,2}.$$

Simple percentage returns do not add this way: gross returns $1 + R$ multiply. This is the practical advantage of logs when aggregating across time; it does not mean log returns and simple returns are numerically interchangeable.

Worked example

Problem 3. Over one year a stock rises from \$100 to \$200, then over the next year returns to \$100. Find each year's effective and continuously compounded returns, and compare their symmetry.

Solution. A share goes from \$100 to \$200, then back to \$100. Simple returns are $+100\%$ and -50%, not opposites, although the final price equals the initial price. Log returns are $\ln 2 \approx +69.31\%$ and $\ln(1/2) \approx -69.31\%$; they sum to zero. A log return may be below -100% because it is the logarithm of a positive price ratio, whereas a simple return on a limited-liability asset cannot fall below -100%.

Why it matters

Keep the cash-flow viewpoint throughout this course. First identify *what* is exchanged and *when*; then choose the relevant transaction price, fees and dividend obligations; finally use a consistent interest-rate convention to compare cash flows at different dates.

2 Forwards, futures and the cost of carrying an asset

2.1 Start with the contractual cash flows

A *forward* is a binding agreement made today to exchange an underlying asset at a specified future date T for a delivery price K. It fixes the asset and quantity, delivery date and place, and payment. A long forward is obliged to buy; a short forward is obliged to sell. An ordinary newly agreed forward typically has zero value at inception, but it has a payoff at delivery:

$$\Pi_T^{\text{long}} = S_T - K, \qquad \Pi_T^{\text{short}} = K - S_T.$$

Here S_T is the spot price at delivery. The two payoffs sum to zero before counterparty, funding and other costs. *Payoff* at expiration is not the same as today's contract value; nor does it include the initial cash flows of a separate stock position.

Worked example

Problem 4. Enter a long six-month forward at \$50. Find the payoff if the terminal price is \$40, \$45, \$50, \$55 or \$60.

Solution. If the agreed six-month delivery price is $K = \$50$, the long's payoff is:

S_T	\$40	\$45	\$50	\$55	\$60
$S_T - K$	−\$10	−\$5	\$0	\$5	\$10

Its payoff is a straight line crossing zero at $S_T = K$; the short's line is its mirror image. In the lecture's separate index example, initial spot $S_0 = \$1{,}000$, $K = \$1{,}020$ and $S_T = \$1{,}050$ give +\$30 to the long and −\$30 to the short per index unit.

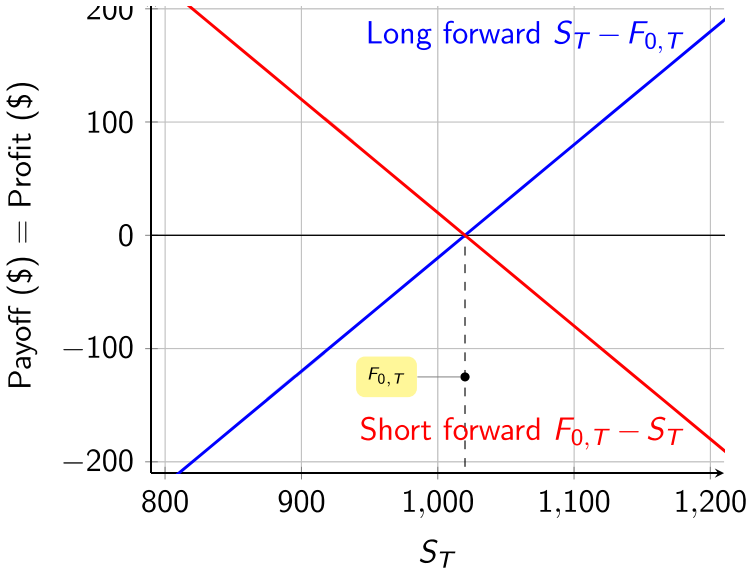


Figure 1: Forward payoffs: long $S_T - F_{0,T}$ and short $F_{0,T} - S_T$; both cross zero at the delivery price.

2.2 Derive the fair delivery price by replication

Consider a stock index with spot level S_0, continuously compounded funding rate r, and continuous dividend yield δ. Assume, for this classroom derivation, that borrowing and lending use the same rate, dividends are reinvestable at the stated yield, and trading, taxes, margin and credit frictions can be ignored. To deliver one unit at T, buy $e^{-\delta T}$ units today and reinvest dividends: the holding grows to one unit. Its present cost is $S_0 e^{-\delta T}$. Financing that cost to T requires $S_0 e^{(r-\delta)T}$. Hence the no-arbitrage forward delivery price is

$$F_{0,T} = S_0 e^{(r-\delta)T}.$$

The exponent is the *net carry*: financing raises the forward price and dividends lower it. In the special case of no income, $\delta = 0$ and $F_{0,T} = S_0 e^{rT}$.

The following ledger makes the arbitrage argument concrete. If the quoted $F_{0,T}$ is *above* fair value, buy the dividend-reinvested asset position, finance it, and short the forward:

Position	At 0	At T
Buy $e^{-\delta T}$ units; reinvest dividends	$-S_0e^{-\delta T}$	$+S_T$
Borrow the purchase cost	$+S_0e^{-\delta T}$	$-S_0e^{(r-\delta)T}$
Short forward	0	$F_{0,T} - S_T$
Net	0	$F_{0,T} - S_0e^{(r-\delta)T} > 0$

If the quoted forward is *below* fair value, reverse every trade. The terminal net becomes $S_0e^{(r-\delta)T} - F_{0,T} > 0$. The forward-pricing formula is therefore a statement about equivalent cash flows under those assumptions, not a prediction of S_T.

Reverse cash-and-carry when the forward is too cheap.

Transaction	Time 0	Time T (expiration)
Short tailed stock position ($e^{-\delta T}$ units)	$S_0e^{-\delta T}$	$-S_T$
Lend $S_0e^{-\delta T}$	$-S_0e^{-\delta T}$	$S_0e^{(r-\delta)T}$
Long forward	0	$S_T - F_{0,T}$
Total	0	$S_0e^{(r-\delta)T} - F_{0,T} > 0$

2.3 What changes when a forward becomes a future?

Futures are standardized, exchange-traded contracts with specified settlement procedures. Their distinguishing cash-flow feature here is *marking to market*: gains and losses are settled periodically, normally daily, rather than being left entirely until maturity. The clearing system requires margin as collateral. The *notional value* of a position is the current futures price times its contract multiplier and number of contracts; it is not the cash paid to purchase the underlying.

The lecture uses a historical S&P 500 futures contract with multiplier \$250 per index point, quarterly contract months, and cash settlement. The corresponding Bloomberg screenshots also show a natural-gas contract with a different unit, tick and physical-delivery schedule. Those dated specifications illustrate why one must read the actual contract terms; they are not current trading specifications. In the historical index example, eight contracts at 1,100 points represent $8(250)(1100) = \$2.2$ million of notional exposure, even though a 10% illustrative initial margin is only \$220,000.

Reading historical contract specifications. The index contract in the example traded on the Chicago Mercantile Exchange in March, June, September and December. Trading ended on the business day before the settlement-price determination; cash settlement used the opening S&P 500 price on the third Friday of the expiry month. The dated contract monitor distinguishes each expiry's price, bid, ask, volume and open interest. Its HELP and MENU functions explain fields and related contracts. The course points to CME contract documentation for specifications.

The historical natural-gas contract represents 10,000 MMBtu with physical delivery. Its tick of \$0.001 per MMBtu is \$10 per contract. The displayed margins are:

Margin requirement	Speculator	Hedger
Initial	2,310	2,100
Secondary	2,100	2,100

2.4 Margin magnifies cash-flow pressure, not the underlying payoff

For a long futures position with N contracts and multiplier m, a settlement-price change $F_t - F_{t-1}$ adds $Nm(F_t - F_{t-1})$ to the margin account. In the course's simplified weekly example, the balance

also earns interest at a continuously compounded annual rate of 6%:

$$M_t = M_{t-1}e^{0.06/52} + Nm(F_t - F_{t-1}), \qquad Nm = 8(250) = 2{,}000.$$

The lecture uses weekly settlement for exposition; real exchange settlement is daily. Starting from $M_0 = \$220{,}000$, a fall from 1,100 to 1,027.99 produces $2{,}000(-72.01) = -\$144{,}020$ of variation margin. The week-one balance is

$$\$220{,}000e^{0.06/52} - \$144{,}020 = \$76{,}233.99.$$

The futures price fell about 6.55%, but the margin balance fell about 65% because margin was one tenth of notional. The account is collateral, not a price paid for the contract; this is why percentage changes in the account should not be confused with the return on the underlying.

A broker can demand a top-up when the balance falls below *maintenance margin*. The displayed lecture path continues without modelling the cash contribution following such a call, so it is best read as an illustrative no-top-up balance path, not as a fully executable account history. At week ten, its balance is \$44,990.57; subtracting the future value of the initial margin gives an illustrative financing-adjusted profit of

$$\$44{,}990.57 - \$220{,}000e^{0.06(10/52)} = -\$177{,}562.60.$$

For a single forward with the same entry and terminal prices, but no intervening settlement, the terminal payoff is $2{,}000(1{,}011.65 - 1{,}100) = -\$176{,}700$. The difference here comes from the timing and financing of interim futures cash flows. The exact relationship between a futures price and a forward price depends on rate uncertainty and settlement assumptions; periodic settlement alone does not justify declaring them universally unequal.

The illustrative maintenance level is 70%–80% of initial margin. Falling below it triggers a request for additional collateral; the complete no-top-up path used for the calculations above is:

Week	Multiplier (\$)	Futures price	Price change	Margin balance (\$)
0	2000.00	1100.00	—	220,000.00
1	2000.00	1027.99	-72.01	76,233.99
2	2000.00	1037.88	9.89	96,102.01
3	2000.00	1073.23	35.35	166,912.96
4	2000.00	1048.78	-24.45	118,205.66
5	2000.00	1090.32	41.54	201,422.13
6	2000.00	1106.94	16.62	234,894.67
7	2000.00	1110.98	4.04	243,245.86
8	2000.00	1024.74	-86.24	71,046.69
9	2000.00	1007.30	-17.44	36,248.72
10	2000.00	1011.65	4.35	44,990.57

2.5 Why use index futures?

An index future can change broad market exposure without trading every constituent stock. In the lecture's temporary shift from equities to bonds, shorting an index future can offset some equity exposure while the stock portfolio remains owned. That may reduce transaction costs and simplify a temporary allocation decision. The hedge is not automatically perfect: a particular portfolio may behave differently from the index, which creates *basis risk* and motivates the cross-hedging chapter. Choosing the number and direction of contracts requires the portfolio's actual sensitivity, the contract multiplier and the target exposure.

Why it matters

Forwards and futures share the basic linear exposure $S_T - K$, but their cash-flow timing differs. Pricing begins with a replicating stock-and-financing strategy; risk management begins with the actual contract multiplier, settlement schedule and margin requirements.

For stock-pickers this separates broad market exposure from individual-stock selection. Further course reading is McDonald (2009), Chapter 5, Section 5.5.

3 Options: asymmetric payoffs, parity and insurance

3.1 A right is different from an obligation

A call gives its buyer the right, but not the obligation, to buy the underlying at strike K; a put gives the right to sell it. The option seller must perform if the buyer validly exercises. European exercise occurs only at expiration, American exercise can occur up to expiration, and Bermudan exercise is permitted on specified dates. The strike is the amount exchanged for the asset on exercise; expiration is the last permitted exercise date. For a European option at expiration,

$$C_T = (S_T - K)^+ = \max(S_T - K, 0), \qquad P_T = (K - S_T)^+ = \max(K - S_T, 0).$$

The writer's payoff is the negative of the buyer's payoff. A call is *in the money* when $S > K$, a put when $S < K$; at the money means $S = K$. Out-of-the-money options have *zero intrinsic value*, not a negative exercised payoff: their unprotected exercise expression would be negative, so the holder simply does not exercise. These definitions concern exercise value, not whether a trade is profitable after the premium.

The dated WFC option-chain example organises market quotes by strike and expiry; its call-valuation screen displays contract inputs, model value and sensitivities. These are historical illustrations.

3.2 Payoff versus profit

An option buyer pays a premium at inception. To compare all cash flows at expiration, carry that premium forward at the relevant funding rate. Thus a purchased call's terminal *profit* is $(S_T - K)^+ - FV(c_0)$, and a purchased put's is $(K - S_T)^+ - FV(p_0)$. A written option receives the premium first and has the opposite terminal option payoff; its terminal profit includes the future value of that received premium. The buyer's maximum loss on a purchased option is its funded premium, while a written call can have an unbounded loss as S_T rises.

For a six-month S&R call struck at \$1,000, a terminal index of \$1,100 pays the holder \$100 and the writer -\$100. At \$900 both option payoffs are zero. This comparison excludes the premium.

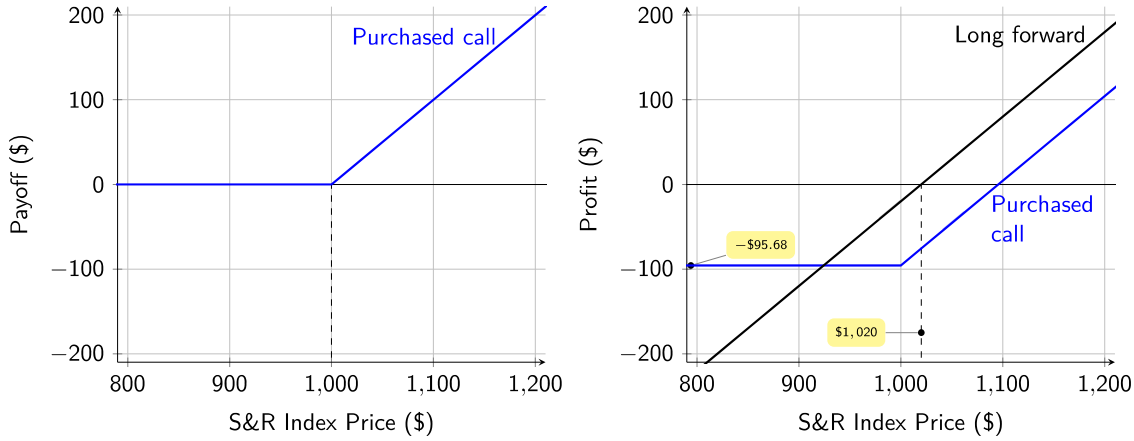


Figure 2: Long call: the left curve is $(S_T - K)^+$; the profit curve on the right subtracts the financed premium. Exercise begins at K; break-even is above K.

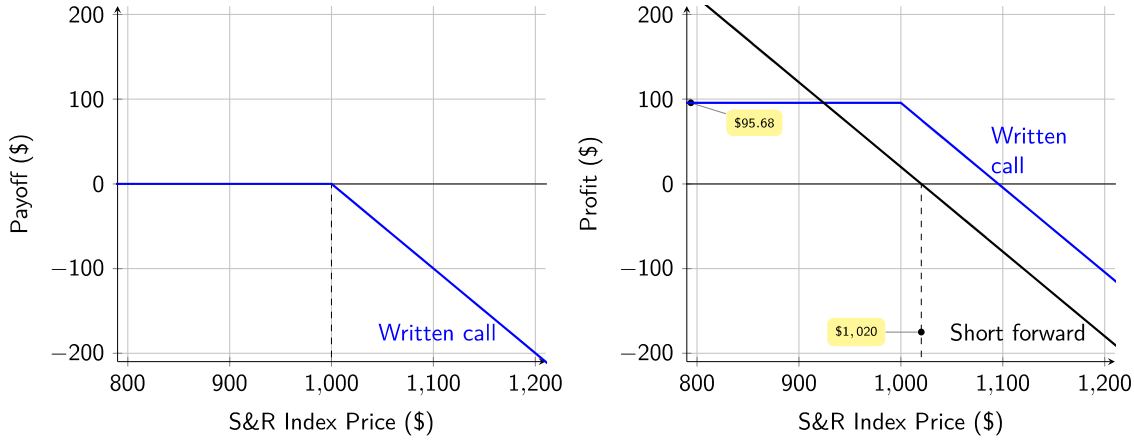


Figure 3: Short call: the premium creates a positive profit plateau below K, but losses are unbounded as S_T rises.

Worked example

Problem 5. A six-month S&R call has strike \$1,000 and future premium value \$95.68. Find the index price where the purchased-call profit diagram crosses the horizontal axis.

Solution. Solve $(S_T - 1000)^+ - 95.68 = 0$, obtaining $S_T = 1095.68$. At $S_T = 900$, exercise is declined: payoff is zero and profit is -\$95.68.

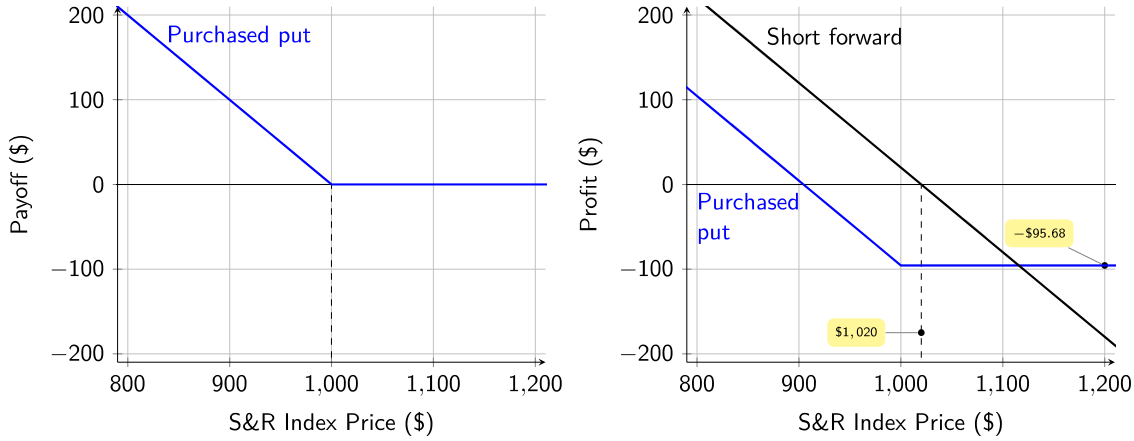


Figure 4: Long put: payoff decreases until $S_T = K$ and is then zero; financed premium shifts profit downward.

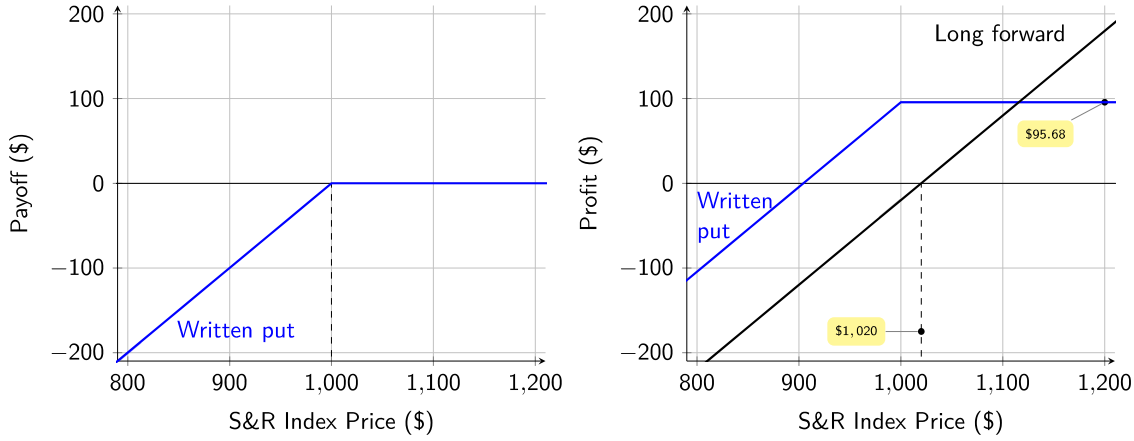


Figure 5: Short put: profit is capped at the financed premium; the downward-price risk lies below the strike.

Worked example

Problem 6. The stock price is \$40 and the effective annual interest rate is 8%. Draw payoff and profit for a 40-strike put costing \$3.26 with one year to maturity.

Solution. The financed premium is $3.26(1.08) = 3.5208$. Payoff is $(40 - S_T)^+$; profit is $(40 - S_T)^+ - 3.5208$. Break-even is $S_T = 36.4792$. The graph shifts the payoff down by the same financed premium at every terminal stock price.

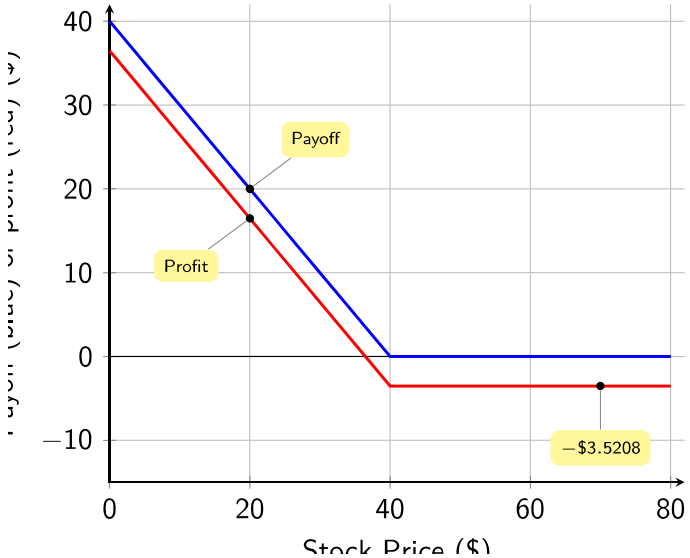


Figure 6: Put exercise: profit is payoff minus $3.26(1.08) = 3.5208$. The separation between the two curves is constant.

3.3 Put–call parity comes from equal terminal payoffs

Buy a European call and write a European put with the same stock, strike and maturity. Whatever S_T, the combined terminal payoff is

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K.$$

The same payoff arises from owning one share and borrowing the present value of K: the share is worth S_T and the loan repayment is K. If the stock pays no dividends before maturity and the frictionless assumptions apply, equal payoffs must have equal current prices:

$$c_0 - p_0 = S_0 - PV(K), \qquad PV(K) = Ke^{-rT}$$

for a continuously compounded rate r. Equivalently, $c_0 = p_0 + S_0 - PV(K)$ and $p_0 = c_0 - S_0 + PV(K)$. This is a replication identity, not a statement that calls and puts should have equal premiums. For an at-the-money, non-dividend-paying stock, $S_0 = K$, so $c_0 - p_0 = K - PV(K) > 0$ when interest is positive.

A synthetic call is a stock position financed by borrowing the discounted strike and insured by a put; a synthetic put insures a short stock position with a call and includes a bond paying the strike.

Worked example

Problem 7. The S&R index is \$1,000; a six-month call costs \$109.20 and a same-strike, same-maturity put costs \$60.18. The effective six-month interest rate is 2%. Find the strike.

Solution. In the lecture's six-month index exercise, $S_0 = \$1{,}000$, $c_0 = \$109.20$, $p_0 = \$60.18$, and the *effective six-month* interest rate is 2%. Hence $109.20 - 60.18 = 1000 - K/1.02$, so $K = \$970.00$. The 2% is already a six-month rate; it must not be compounded for another half-year.

When the stock pays known dividends before expiration, the share in the replication delivers those dividends while the options do not. Subtract their present value: $c_0 - p_0 = S_0 - PV(\text{Div}) - PV(K)$. These exact equalities are for European options under their stated assumptions; early exercise changes the American-option relationships.

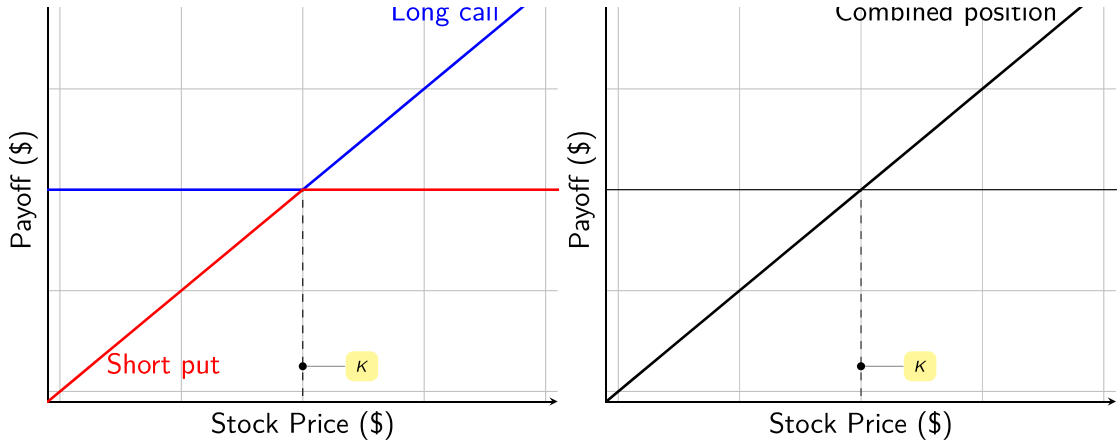


Figure 7: Long call plus short put: the two signed legs add to the linear payoff $S_T - K$, which explains put–call parity.

3.4 Bounds are quick arbitrage checks

A non-dividend-paying European call cannot cost more than its stock, and a European put cannot cost more than the discounted strike: $0 \le c_0 \le S_0$, $0 \le p_0 \le PV(K)$. Nonnegative option prices plus parity also imply

$$c_0 \ge \max\{0, S_0 - PV(K)\}, \qquad p_0 \ge \max\{0, PV(K) - S_0\}.$$

These are bounds, not valuation formulas. They can expose impossible quotes even before choosing a stochastic pricing model.

Worked example

Problem 8. A one-month European put on a non-dividend-paying stock costs 2.50, has strike 50 and spot 46, with an interest rate of 10%. Is there an arbitrage, and which positions implement it?

Solution. The lecture asks whether a one-month European put with $K = \$50$, $S_0 = \$46$, premium \$2.50, and 10% interest is too cheap. Interpreting the unspecified 10% as an annual continuously compounded rate, $PV(K) = 50e^{-0.10/12} \approx \$49.585$, so the lower bound is approximately \$3.585. The put is underpriced by about \$1.085. Buy the put and share, and borrow $PV(K)$: the initial net cash is positive, while the terminal package pays $\max(S_T, 50) - 50 \ge 0$. The conclusion is unchanged under an effective-annual 10% convention, but the exact arbitrage amount changes slightly; the source does not specify the convention.

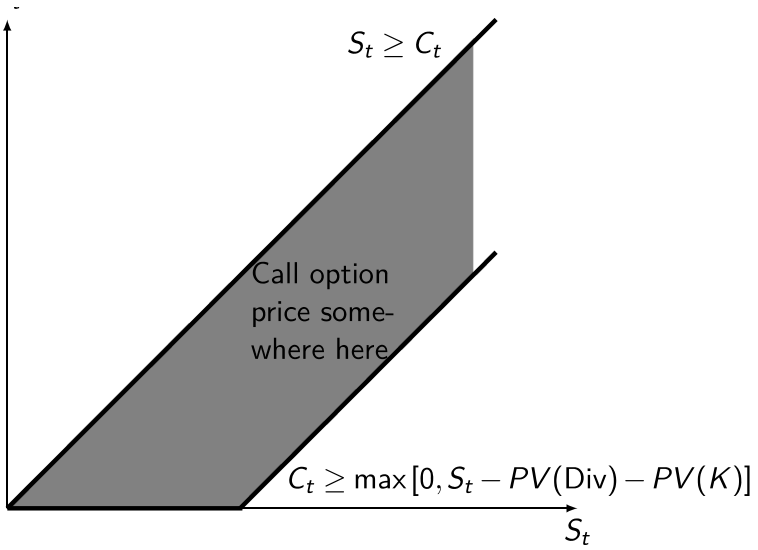


Figure 8: Call-price bounds: the shaded region lies below S_t and above $\max(0, S_t - PV(\text{Div}) - PV(K))$. The horizontal variable is today's stock price, not the terminal payoff variable.

Why it matters

The source drawing gives the loose upper bound $P_t \leq K$. Its printed dividend term in the lower bound has a minus sign; for a European put with known dividends, put–call parity gives the corrected lower bound $P_t \geq \max\{0, PV(K) - S_t + PV(\text{Div})\}$. This correction follows from $p_t = c_t - S_t + PV(\text{Div}) + PV(K)$ and $c_t \geq 0$. Its K upper bound is looser than the European discounted-strike bound under nonnegative interest rates. For European options with known dividends, the corresponding call lower bound is $C_t \geq \max\{0, S_t - PV(\text{Div}) - PV(K)\}$. Do not apply an exact European parity identity to American early-exercise rights.

3.5 Build the payoff you need

Options can reshape an existing exposure. A long stock plus a put has terminal value $S_T + (K - S_T)^+ = \max(S_T, K)$: the put creates a *floor* before premium and financing. A short stock plus a call has terminal value $-S_T + (S_T - K)^+ = -\min(S_T, K)$, limiting the damage from a price rise before costs. A principal-protected equity-linked deposit combines repayment of principal with an option-like participation in upside. The historical 1999 First Union National Bank example invests \$10,000 and, after 5.5 years, promises $10{,}000[1 + 0.7 \max(0, S_T/1300 - 1)]$; issuer credit risk and the opportunity cost of the locked-up principal are separate from this contractual payoff.

Common multi-option structures are specified by their signed legs in the table below. A strategy name alone is insufficient: write each leg's payoff and include the financed premium before discussing profit.

Worked example

Problem 9. Identify the call positions with strikes 90, 100 and 110 that produce the supplied payoff graph. All options mature in one year, today's stock price is \$100, the continuously compounded annual rate is 8%, and there are no dividends.

Solution. The lecture's payoff-identification exercise could not be solved from OCR alone, so its source-page figure was recovered. It is a *long call butterfly* with strikes 90, 100 and 110:

buy one 90-call, write two 100-calls, and buy one 110-call. The terminal payoff is

$$(S_T - 90)^+ - 2(S_T - 100)^+ + (S_T - 110)^+.$$

It is zero outside $[90, 110]$, rises with slope $+1$ from 90 to 100, reaches 10 at 100, then falls with slope -1 to 110. This identifies the positions from the graph, but it is a *payoff* only; option premiums are not supplied for a profit calculation.

The quoted 2006 option-market activity findings are historical observations, not a forecast about current customer behavior.

Why it matters

For every option strategy, separate three questions: What rights and obligations exist? What is the terminal *payoff* in each price region? What *profit* remains after bringing premiums to the same date? Parity and bounds check the answer before any model-specific valuation.

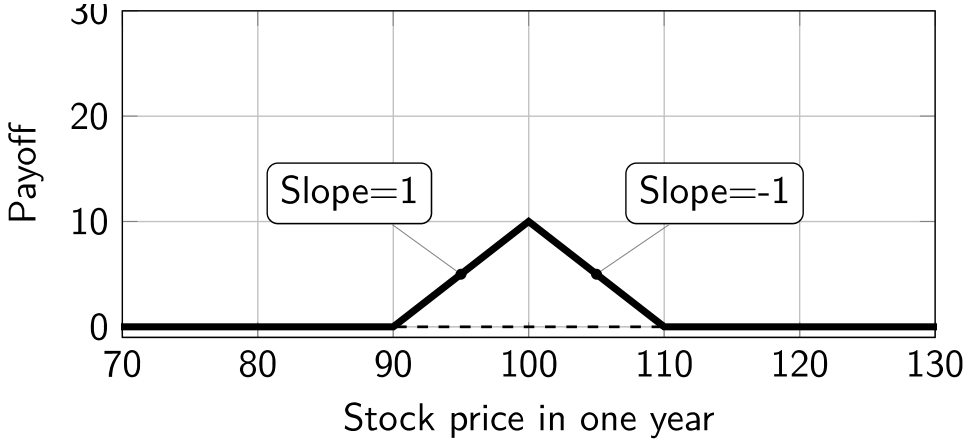


Figure 9: Identify the butterfly: slope $+1$ on $(90, 100)$, slope -1 on $(100, 110)$, peak 10 and zero payoff outside these strikes.

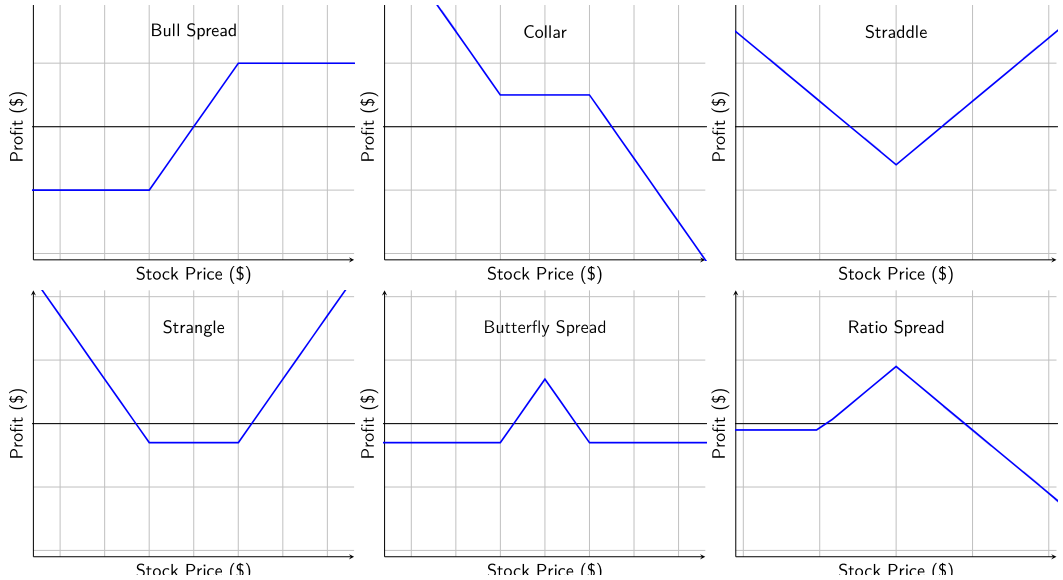


Figure 10: Strategy profit shapes. Read the signed option legs and the initial financed cost before using a strategy name; a zero premium does not remove tail risk.

The source strike-position grid is reconstructed below. A dash means no position; quantities refer to option contracts. The collar here contains only option legs: adding a long stock position gives a collared stock.

Strategy	Option	K_{low}	K_{ATM}	K_{high}
Bull spread	Call	–	Buy	Sell
	Put	–	–	–
Collar	Call	–	–	Sell
	Put	Buy	–	–
Straddle	Call	–	Buy	–
	Put	–	Buy	–
Strangle	Call	–	–	Buy
	Put	Buy	–	–
Butterfly	Call	Buy	Sell (2)	Buy
	Put	–	–	–
Ratio spread	Call	–	Buy	Sell (n)
	Put	–	–	–

The source notes that alternative combinations can produce the same results. When invoking put–call parity, include any accompanying cash/bond position: two bare call/put spreads need not have identical initial premiums or absolute terminal cash flows.

Why choose each structure? A bull call spread buys a call and writes a higher-strike call to reduce its premium while capping gains; reversing the exposure gives a bear spread. Put–call parity can relate call and put constructions, but the accompanying cash/bond position matters: buying a low-strike put and selling a high-strike put alone has a different payoff level from the call spread.

The option-only collar is a short-forward-like exposure between its strikes. Added to a long stock, it gives a floor and a cap; reversing the option legs gives a written collar. A straddle buys a call and put at the same strike to benefit from a large move either way, at the cost of two premiums. A strangle uses out-of-the-money strikes, usually costs less, and requires a larger move. Their written versions reverse both risk and premium.

A long call butterfly concentrates payoff near its middle strike. The lecture also motivates a different protected short-volatility structure: a written straddle protected by out-of-the-money put and call wings, commonly called a short iron butterfly; reversing its positions gives a long iron butterfly. The call-only butterfly in the table and the iron construction should be distinguished by their legs, initial cash flows and payoff levels.

A ratio spread buys one option and sells $n > 1$ more-out-of-the-money options of the same kind (calls or puts). Its ratio is short quantity divided by long quantity. A zero initial premium can be arranged in some quotes, but this does not remove the tail exposure. The dated strategy screens illustrate choosing, combining and valuing these legs. Lakonishok, Lee, Pearson, and Poteshman, Option Market Activity, The Review of Financial Studies, 2006

- Stylized facts about option trading
- Written option positions are more common than purchased positions
- About 4 times more purchased calls than puts
- Main driver of option market activity is speculating on the direction of underlying stock movements
- Option trading strategies
- Small fraction of volatility trading strategies (straddles and strangles)
- Large fraction of covered-call strategies
- Option market activity during the stock market bubble of the late 1990s
- Call buying and put writing increased dramatically (mostly on growth stocks)
- Purchased puts less common (little appetite for betting against the bubble)

4 Cross-hedging an imperfectly matched exposure

4.1 Why a related future can still reduce risk

A *cross hedge* uses a derivative on one asset to hedge a different, related asset. An airline buying jet fuel, for example, may use heating-oil futures when an exact jet-fuel contract is unavailable or illiquid. The hedge cannot generally cancel every price move because jet fuel and heating oil are not the same commodity. Let ΔS be the change in jet-fuel price per gallon and ΔF the change in the heating-oil futures price per gallon over the same horizon. Correlation measures how much of the exposure is shared; the volatility ratio tells us how many futures-price units offset one spot-price unit.

The *sign* of the position depends on the original exposure. Someone who owns an asset and fears a price fall normally sells futures. An airline that must *buy* fuel and fears a price rise normally buys futures, so futures gains offset a higher purchase cost. The addendum's generic expression $\Delta S + N\Delta F$ uses a signed N; the airline calculation below uses a positive number of long contracts and writes the residual cost as $\Delta S - h\Delta F$.

The source uses US jet-fuel hedging with crude-oil or heating-oil futures as its example of an unavailable or illiquid exact hedge.

4.2 Choose the hedge ratio by minimizing variance

For one unit of future fuel consumption, write the hedged change in cost as $X(h) = \Delta S - h\Delta F$. Expanding its variance and differentiating gives

$$\begin{aligned} \operatorname{Var}[X(h)] &= \sigma_S^2 + h^2\sigma_F^2 - 2h\operatorname{Cov}(\Delta S, \Delta F), \\ 0 &= 2h\sigma_F^2 - 2\operatorname{Cov}(\Delta S, \Delta F), \\ h^* &= \frac{\operatorname{Cov}(\Delta S, \Delta F)}{\sigma_F^2} = \rho\frac{\sigma_S}{\sigma_F}. \end{aligned}$$

This is also the slope in a least-squares regression of ΔS on ΔF with an intercept. With a single explanatory variable, the regression's $R^2 = \rho^2$. Under the same variance assumptions, the minimum residual variance is $\sigma_S^2(1-\rho^2)$. Here R^2 measures variance reduction in the historical sample, not a guaranteed reduction in every future month.

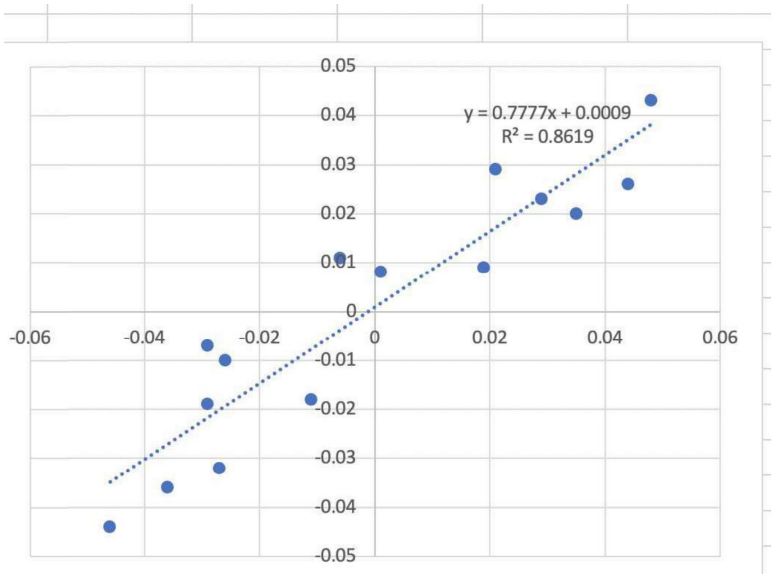
The airline expects to buy 2 million gallons in one month. In the regression notation,

$$\Delta S = a + h^*\Delta F + \epsilon.$$

Worked example

The source prints rounded inputs $\sigma_F = 0.0313$, $\sigma_S = 0.0263$ and $\rho = 0.9284$. The 15 monthly price-change pairs in the addendum give, before rounding, $\sigma_F \approx 0.0313434$, $\sigma_S \approx 0.0262548$, and $\rho \approx 0.9283723$. Thus $h^* \approx 0.77765$, reported in the slides as 0.7777, while $R^2 \approx 0.8619$. If one instead substitutes the *printed rounded* standard deviations 0.0313 and 0.0263 into the formula, one gets about 0.7801. This small discrepancy is rounding of the inputs, not a different hedge rule; the unrounded 15 observations reproduce the slide's result.

Month	ΔF	ΔS
1	0.021	0.029
2	0.035	0.02
3	-0.046	-0.044
4	0.001	0.008
5	0.044	0.026
6	-0.029	-0.019
7	-0.026	-0.01
8	-0.029	-0.007
9	0.048	0.043
10	-0.006	0.011
11	-0.036	-0.036
12	-0.011	-0.018
13	0.019	0.009
14	-0.027	-0.032
15	0.029	0.023



The scatter plot places ΔF horizontally and ΔS vertically. Its fitted line is $\Delta S = 0.0009 + 0.7777\Delta F$, with $R^2 = 0.8619$; the intercept is retained in the regression but does not enter the variance-minimising hedge ratio.

4.3 Turn the ratio into tradable contracts

If the exposure is Q_A units and one futures contract represents Q_F units, the continuous optimum is $N^* = h^* Q_A / Q_F$. The airline expects to purchase 2 million gallons, and the historical NYMEX heating-oil contract in the example covers 42,000 gallons. Therefore

$$N^* \approx \frac{0.7777(2{,}000{,}000)}{42{,}000} = 37.03,$$

or 37 *long* contracts after whole-contract rounding. Contract rounding, imperfect correlation, and differences in location, quality or delivery date all leave basis risk. A futures hedge also requires liquidity for mark-to-market cash flows even if it reduces the variance of final purchase cost.

The addendum mentions *tailing the hedge*: adjusting the contract count slightly for the timing of daily futures settlement. No explicit tailing formula is supplied in these slides, so 37 is the untailored classroom answer rather than a claimed exact implementation for all settlement schedules.

Why it matters

The minimum-variance hedge is about covariance, not just correlation: direction follows the exposure, magnitude follows $\rho\sigma_S/\sigma_F$, and the contract count also needs both physical quantities. A strong historical fit reduces expected variance but never makes two different commodities identical.

Further reading: Hull, eighth edition, Chapter 5; McDonald, third edition, Chapters 5 and 6.

5 State prices and risk-neutral valuation

5.1 A two-state example fixes the intuition

Consider one period and two possible states tomorrow. The source slide's price vector was flattened by OCR; checking the original page shows it is $S = (1, 100, c)^\top$. A risk-free asset costs 1 and pays 1.1 in either state. A risky asset costs 100 and pays 100 in state 1 or 150 in state 2. A call costs the unknown c and pays 0 or 50. In matrix form, with assets as rows and states as columns,

$$S = \begin{pmatrix} 1 \\ 100 \\ c \end{pmatrix}, \qquad Q = \begin{pmatrix} 1.1 & 1.1 \\ 100 & 150 \\ 0 & 50 \end{pmatrix}.$$

The question is not which state is objectively more likely. It is what price for the call is consistent with the two already traded assets.

Let ϕ_1, ϕ_2 be the current prices of claims paying one unit only in state 1 or state 2. The first two assets imply

$$1 = 1.1(\phi_1 + \phi_2), \qquad 100 = 100\phi_1 + 150\phi_2.$$

Solving gives $\phi_1 = 8/11 \approx 0.7273$ and $\phi_2 = 2/11 \approx 0.1818$. Thus $c = 0\phi_1 + 50\phi_2 = 100/11 \approx 9.09$. We can see the same price without naming state prices: one share of the risky asset minus a position that owes 100 tomorrow pays $(0, 50)$. The borrowing needed to owe 100 tomorrow brings in $100/1.1$ today, so the replicating strategy costs $100 - 100/1.1 = 100/11$.

Normalize the state prices by the gross risk-free return, 1.1: $p_1^* = 1.1\phi_1 = 0.8$ and $p_2^* = 1.1\phi_2 = 0.2$. These positive weights sum to one. The call price is the discounted expectation under them,

$$c = \frac{0(0.8) + 50(0.2)}{1.1} = \frac{100}{11}.$$

They are *risk-neutral probabilities*, pricing weights inferred from traded prices, not a claim that state 2 occurs in 20% of actual observations.

5.2 The general finite-state framework

Assume a one-period finite-state market with infinitely divisible securities, unrestricted short sales, no transaction costs and no taxes. Objective probabilities satisfy $\pi_s > 0$ and $\sum_{s=1}^{K} \pi_s = 1$; they will be needed for the SDF representation, not for finding replicating prices. The source takes its securities' current price vector to be strictly positive.

Suppose there are N traded securities and K mutually exclusive states. Let $S \in \mathbb{R}^N$ be today's price vector, and let $Q \in \mathbb{R}^{N \times K}$ contain their future payoffs, one asset per row. A portfolio $\theta \in \mathbb{R}^N$ costs $S^\top\theta$ and pays the state vector $Q^\top\theta$. This orientation matters: multiplying Q by state prices produces an N-vector of asset prices.

An arbitrage is a trading strategy with no possible loss, no positive initial investment, and a strict gain somewhere (either upfront or in at least one possible state). The lecture illustrates particularly clear cases: non-positive cost with strictly positive payoffs in every state, or strictly negative cost with non-negative payoffs everywhere. Writing $X = Q^\top\theta$, these two illustrative

types are $S^\top\theta \leq 0,\ X > 0$ in every state and $S^\top\theta < 0,\ X \geq 0$ in every state. The more general wording also rules out a zero-cost payoff that is positive in some states and zero in the rest.

In the finite-state, frictionless setting of the lecture, absence of arbitrage is equivalent to the existence of a strictly positive state-price vector $\phi \in \mathbb{R}^K_{++}$ satisfying

$$S = Q\phi, \qquad S_i = \sum_{s=1}^{K} \phi_s q_i(s).$$

The equation prices all traded payoff rows with one linear rule. If two portfolios deliver exactly the same payoff in every state but cost different amounts, buying the cheaper and selling the dearer creates an arbitrage. Positivity prevents a positive state-contingent payoff from having a non-positive price under the stated assumptions. State prices may not be unique in an incomplete market, but any payoff replicated by traded assets has a unique no-arbitrage price.

5.3 From Arrow–Debreu prices to probability weights

An Arrow–Debreu claim for state s pays one unit there and zero elsewhere; its price is exactly ϕ_s. With one claim for each state, their payoff matrix is $Q^{AD} = I_K$ and $S^{AD} = I_K\phi = \phi$. Buying one such claim for each state pays one unit for sure and costs $B_0 = \sum_s \phi_s$. Therefore its gross risk-free return is $1 + R = 1/B_0$. Define $p^*_s = (1 + R)\phi_s$, also written $\phi^\star(s)$ in the source; then $p^*_s > 0$ and $\sum_s p^*_s = 1$. The asset-pricing equation becomes

$$S_i = \frac{1}{1+R} \sum_{s=1}^{K} p^*_s q_i(s) = \frac{1}{1+R} \mathbb{E}^*[q_i].$$

This is not a separate model layered onto state prices: it is the same linear pricing rule written as a discount factor times an expectation. The objective probabilities π_s are not needed for this form. It applies directly to traded or replicated payoffs; an unspanned payoff in an incomplete market need not have a unique price.

5.4 The same rule under objective probabilities

If objective state probabilities $\pi_s > 0$ are given, define the *stochastic discount factor* $M_s = \phi_s/\pi_s$. Then

$$S_i = \sum_s \pi_s M_s q_i(s) = \mathbb{E}[M q_i].$$

Equivalently, the likelihood ratio $Z_s = p^*_s/\pi_s$ satisfies $M_s = Z_s/(1 + R)$, yielding $\mathbb{E}[Mq_i] = (1 + R)^{-1}\mathbb{E}^*[q_i]$. These formulas distinguish two roles: objective probabilities describe frequency beliefs, while state prices (or risk-neutral probabilities) encode time discounting and how the market values payoffs across states. A payoff in a bad state may command a high state price because it delivers when resources are especially valuable; the next chapter derives this link from marginal utility.

Why it matters

The practical sequence is: write the payoff matrix, find a positive state-price vector from traded prices, value any replicated claim, and only then normalize state prices into probabilities if the expectation form is useful. Never substitute empirical frequencies for risk-neutral weights without a pricing argument.

6 Utility, the Euler equation and equilibrium

6.1 No-arbitrage prices versus an economic explanation

The previous chapter showed that, under no arbitrage, positive state prices ϕ_s can price traded payoffs. It did not explain why a payoff in one state might be worth more than the same payoff in another. A one-period consumption-investment problem supplies an economic interpretation through *marginal utility*: an additional unit of consumption is especially useful when consumption is scarce.

The lecture writes the investor's objective without an explicit time-preference multiplier:

$$\max_{c_0,\{c_s\},\theta} U(c_0) + \sum_{s=1}^{K} \pi_s U(c_s),$$

Consumption is perishable. Here $\pi_s > 0$ are objective state probabilities and U is increasing and concave. This exact classroom specification is used below; adding a separate discount factor would alter the formula. Current endowment is e_0, future state endowment is e_s, and the investor chooses holdings θ_i of securities priced S_i today and paying $q_i(s)$ tomorrow.

6.2 Trace one extra unit of an asset through both budgets

The resource constraints are $c_0 + \sum_i \theta_i S_i \le e_0$ and $c_s - \sum_i \theta_i q_i(s) \le e_s$ for every state. With no waste and increasing utility, the budgets bind:

$$c_0 = e_0 - \sum_i \theta_i S_i, \qquad c_s = e_s + \sum_i \theta_i q_i(s).$$

Substitution leaves a portfolio-only problem:

$$\max_{\theta} \left\{ U\left(e_0 - \sum_i \theta_i S_i\right) + \sum_s \pi_s U\left(e_s + \sum_i \theta_i q_i(s)\right) \right\}.$$

Buying a little more of asset i reduces current consumption by S_i per unit and raises state-s future consumption by $q_i(s)$. For an interior, unconstrained optimum, differentiating expected utility with respect to θ_i yields

$$-S_i U'(c_0) + \sum_s \pi_s q_i(s) U'(c_s) = 0.$$

Rearrange and divide by positive current marginal utility:

$$S_i = \sum_s \pi_s \underbrace{\frac{U'(c_s)}{U'(c_0)}}_{M_s} q_i(s) = \mathbb{E}[M q_i].$$

This is the *asset-pricing Euler equation*. The random variable M_s is a stochastic discount factor (SDF), also denoted $\xi(s)$, equal to the investor's intertemporal marginal rate of substitution. Portfolio constraints, transaction costs or boundary choices can turn a first-order equality into a different condition; they are excluded from this derivation.

6.3 Connect the three pricing languages

Comparing the Euler equation to $S_i = \sum_s \phi_s q_i(s)$ identifies $\phi_s = \pi_s M_s$. If a one-unit risk-free payoff is traded, $\sum_s \phi_s = (1+R)^{-1}$, and risk-neutral probabilities are $p_s^* = (1+R)\phi_s$. Hence one and the same traded-asset price can be written in three equivalent ways:

$$\boxed{S_i = \mathbb{E}[M q_i] = \sum_s \phi_s q_i(s) = \frac{1}{1+R}\mathbb{E}^*[q_i]}.$$

The expectation on the left uses objective probabilities π_s; the one on the right uses risk-neutral probabilities p_s^*. The latter do *not* assert that investors are risk neutral. They move the risk adjustment from the discount factor into the probability weights.

With concave utility, low consumption means high marginal utility. Thus a bad consumption state can have high M_s, and its state price $\phi_s = \pi_s M_s$ can be high relative to its objective likelihood. This is why an asset paying in bad states can be valuable insurance. A high state price depends on both likelihood and marginal utility; one should not infer it from low consumption alone without considering π_s.

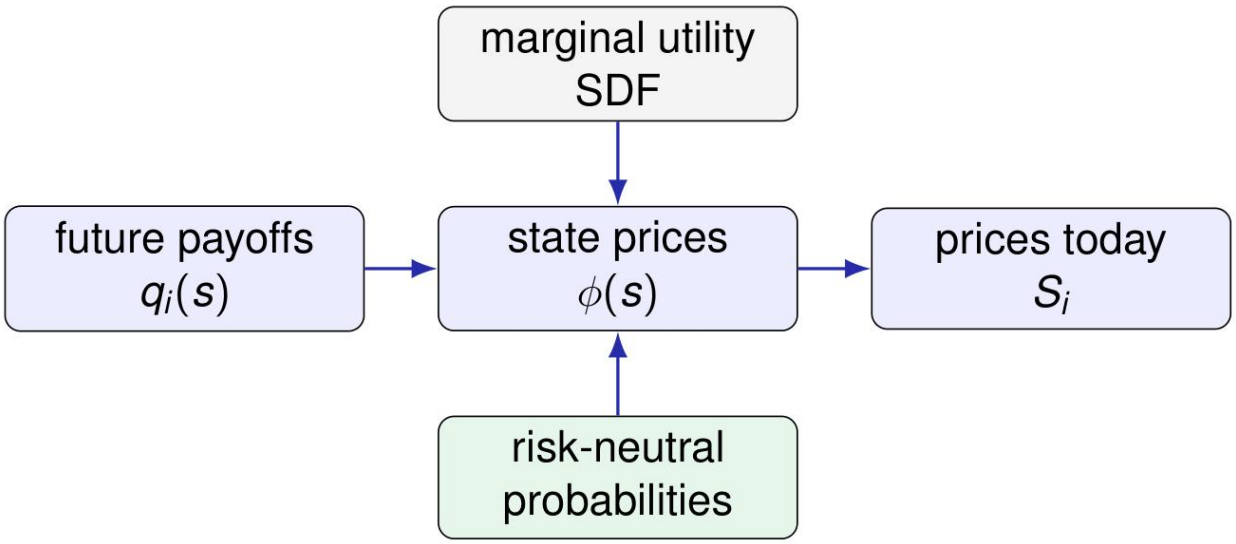
6.4 What equilibrium adds

Individual optimization supplies candidate pricing weights at the person's optimal consumption allocation. *Equilibrium* also requires market clearing: across investors, desired holdings must equal available securities, and consumption must be feasible given endowments and aggregate resources. Preferences, endowments, traded payoffs and supply jointly determine prices and allocations. The lecture's final diagrams summarize this path from fundamentals to state prices; they do not supply a separate numerical equilibrium model.

The hierarchy is useful when solving exercises. For a claim spanned by traded assets, replication and no arbitrage already fix a unique price. State prices and risk-neutral probabilities express that same price compactly. Utility and equilibrium explain why those pricing weights may take their observed values, but they are not required to compute the price of the two-state call in Section 5.

Why it matters

Ask two different questions separately: *What price avoids arbitrage given traded payoffs?* and *Why do investors value some states more?* The first is answered by replication and positive state prices; the second by marginal utility and market clearing.



7 Forwards and futures: exercises with worked answers

The twelve supplied problems below retain their complete statements and alternatives. The worked answers are newly generated explanations, not a supplied answer key. Rate conventions and missing trading assumptions are stated in each answer.

Derivatives QFI 2023-24 Exercises & Solutions Series I Topic: Forwards and Futures

Problem 1.1

Stock XYZ has a current price of 100. The forward price for delivery of this stock in 1 year is 110. Unless otherwise indicated, the stock pays no dividends, and the annual effective risk-free interest rate is 10%. Determine which of the following statements is FALSE.

(A)
The time-1 profit diagram and the time-1 payoff diagram for long positions in this forward contract are identical.

(B)
The time-1 profit for a long position in this forward contract is exactly opposite to the time-1 profit for the corresponding short forward position.

(C)
There is no comparative advantage to investing in the stock versus investing in the forward contract.

(D)
If the 10% interest rate was continuously compounded instead of annual effective, then it would be more beneficial to invest in the stock, rather than the forward contract.

(E)
If there was a dividend of \$3.00 paid 6 months from now, then it would be more beneficial to invest in the stock, rather than the forward contract.

Worked answer (generated). With a 10% *effective annual* rate, $100(1.10) = 110$, exactly the quoted one-year forward price. A long forward has zero initial premium, so its time-1 payoff and profit are both $S_T - 110$; a short forward has the opposite profit. Buying the stock with borrowed funds gives the same time-1 profit. If 10% were instead *continuously* compounded, the fair delivery price would be $100e^{0.10} \approx 110.52$, making a long forward at 110 more favorable, not the stock. A dividend paid on the stock but not the forward does benefit direct ownership, all else equal.

Problem 1.2

A market maker in stock index forward contracts observes a 6-month forward price of 112 on the index. The index spot price is 110 and the continuously compounded dividend yield on the index is 2%. The continuously compounded risk-free interest rate is 5%. Describe actions the market maker could take to exploit an arbitrage opportunity and calculate the resulting profit (per index unit).

(A)
Buy observed forward, sell synthetic forward, Profit = 0.34

(B)
Buy observed forward, sell synthetic forward, Profit = 0.78

(C)
Buy observed forward, sell synthetic forward, Profit = 1.35

(D)
Sell observed forward, buy synthetic forward, Profit = 0.78

(E)
Sell observed forward, buy synthetic forward, Profit = 0.34

Worked answer (generated). With $S_0 = 110$, $r = 5\%$, $\delta = 2\%$, and $T = 0.5$, fair carry is $110e^{(0.05-0.02)0.5} \approx 111.6624$. The quoted forward at 112 is high by \$0.3376. Sell that forward and buy the synthetic forward (the dividend-reinvested index position financed at the risk-free rate), locking in about \$0.34 at delivery under the frictionless assumptions.

Problem 1.3

Judy decides to take a short position in 20 contracts of S&P 500 futures. Each contract is for the delivery of 250 units of the index at a price of 1500 per unit, exactly one month from now. The initial margin is 5% of the notional value, and the maintenance margin is 90% of the initial margin. Judy earns a continuously compounded risk-free interest rate of 4% on her margin balance. The position is marked-to-market on a daily basis. On the day of the first marking-to-market, the value of the index drops to 1498. On the day of the second marking-to-market, the value of the index is X and Judy is not required to add anything to the margin account. Calculate the largest possible value of X.

(A)
1490.50
(B)
1492.50
(C)
1500.50
(D)
1505.50
(E)
1507.50

Worked answer (generated). Twenty contracts at a \$250 multiplier give 5,000 index-point dollars. Initial margin is $0.05(5{,}000)(1{,}500) = \$375{,}000$; maintenance is \$337,500. Assuming 365 days per year and interest credited daily, a short gains $5{,}000(1{,}500 - 1{,}498) = \$10{,}000$ on day 1. Its balance is $M_1 = 375{,}000e^{0.04/365} + 10{,}000$. No top-up on day 2 requires $M_1 e^{0.04/365} + 5{,}000(1{,}498 - X) \geq 337{,}500$, hence $X \leq 1507.5167$. The nearest listed maximum is 1507.50. A different day-count convention changes only the last decimals.

Problem 1.4

The current price of a medical company's stock is 75. The expected value of the stock price in three years is 90 per share. The stock pays no dividends. You are also given: i) The risk-free interest rate is positive. ii) There are no transaction costs. iii) Investors require compensation for risk. The price of a three-year forward on a share of this stock is X, and at this price an investor is willing to enter into the forward. Determine what can be concluded about X.

(A)
X < 75
(B)
X = 75
(C)
75 < X < 90
(D)
90 < X

Worked answer (generated). No dividends and a positive risk-free rate imply a fair forward above today's 75. If the stock's expected terminal price is 90 and investors demand a positive expected risk premium, the expected stock return exceeds the risk-free return; hence the fair delivery price lies below 90. Therefore $75 < X < 90$. The expectation of 90 alone does not price the forward.

Problem 1.5

The ask price for a share of ABC company is 100.50 and the bid price is 100. Suppose an investor can borrow at an annual effective rate of 3.05% and lend (i.e., save) at an annual effective rate of 3%. Assume there are no transaction costs and no dividends. Determine which of the following strategies does not create an arbitrage opportunity.

(A)
Short sell one share, and enter into a long one-year forward contract on one share with a forward price of 102.50.

(B)
Short sell one share, and enter into a long one-year forward contract on one share with a forward price of 102.75.

(C)
Short sell one share, and enter into a long one-year forward contract on one share with a forward price of 103.00.

(D)
Purchase one share with borrowed money, and enter into a short one year forward contract on one share with a forward price of 103.60.

(E)
Purchase one share with borrowed money, and enter into a short one year forward contract on one share with a forward price of 103.75.

Worked answer (generated). The bid is 100 and lending grows it to $100(1.03) = 103$; a forward below 103 allows short-stock/long-forward arbitrage. The ask is 100.50 and borrowing it costs $100.50(1.0305) = 103.56525$ in one year; a forward above that allows buy-stock/short-forward arbitrage. Thus the no-arbitrage band is $103 \leq F \leq 103.56525$. Of the listed *strategies*, the one with forward price 103.00 produces zero, not a strictly positive arbitrage payoff. Prices 102.50, 102.75, 103.60 and 103.75 lie outside the band in the indicated trade direction.

Problem 1.6

Investors in a certain stock demand to be compensated for risk. The current stock price is 100. The stock pays dividends at a rate proportional to its price. The dividend yield is 2%. The continuously compounded risk-free interest rate is 5%. Assume there are no transaction costs. Let X represent the expected value of the stock price 2 years from today. Assume it is known that X is a whole number. Determine which of the following statements is true about X.

(A)
The only possible value of X is 105.

(B)
The largest possible value of X is 106.

(C)
The smallest possible value of X is 107.

(D)
The largest possible value of X is 110.

(E)
The smallest possible value of X is 111.

Worked answer (generated). With continuously compounded $r = 5\%$ and dividend yield $\delta = 2\%$, the two-year forward price is $100e^{(0.05-0.02)2} \approx 106.18$. Positive compensation for stock risk means its expected *total* return exceeds the risk-free return. Under the proportional-dividend model used in the course, expected terminal stock value must therefore exceed the forward benchmark. If X is

an integer, its smallest possible value is 107. This conclusion uses the stated positive risk premium; the forward price itself does not forecast X.

Problem 1.7

An off-market forward contract is a foward where either you have to pay a premium or you receive a premium for entering into the contract. (With a standard forward contract, the premium is zero.) Suppose the effective annual interest rate is 10% and the S&R index is 1000. Consider 1-year forward contracts.

(a)
Verify that if the forward price is \$1100, the profit diagrams for the index and the 1-year forward are the same.

(b)
Suppose you are offered a long forward contract at a forward price of \$1200. How much would you need to be paid to enter into this contract?

(c)
Suppose you are offered a long forward contract at a forward price of \$1000. How much would you be willing to pay to enter into this forward contract?

Worked answer (generated). At the 10% effective rate, a \$1,000 index investment financed for one year costs \$1,100 at maturity, matching a zero-premium long forward at $K = \$1{,}100$; both have time-1 profit $S_T - 1{,}100$. A long forward at $K = \$1{,}200$ is worth $1000 - 1200/1.1 = -\$90.91$ today, so the entrant must be *paid* \$90.91. At $K = \$1{,}000$, it is worth $1000 - 1000/1.1 = +\$90.91$, the most the entrant would pay under the model. Carrying those entry premiums forward reconciles the time-1 profit diagrams.

Problem 1.8

Suppose the stock price is \$35 and the continuously compounded interest rate is 5%.

(a)
What is the 6-month forward price, assuming dividends are zero?

(b)
If the 6-month forward price is \$35.50, what is the annualized forward premium?

(c)
If the forward price is \$35.50, what is the annualized continuous dividend yield?

Worked answer (generated). (a) The six-month no-dividend forward is $35e^{0.05(0.5)} \approx \$35.8860$. (b) For observed $F = \$35.50$, the annualized *continuously compounded* forward premium is $\ln(35.50/35)/0.5 \approx 2.8369\%$; if "forward premium" means a simple annualized percentage instead, it is $[(35.50/35) - 1]/0.5 \approx 2.8571\%$. (c) The continuous dividend yield implied by the observed forward is $\delta = 0.05 - \ln(35.50/35)/0.5 \approx 2.1631\%$. Distinguishing the two premium conventions prevents a silent mismatch.

Problem 1.9

The S&R spot price is 1100, the risk-free rate is 5%, and the dividend yield on the index is 0.

(a)
Suppose you observe a 6-month forward price of 1135. What arbitrage would you undertake?

(b)
Suppose you observe a 6-month forward price of 1115. What arbitrage would you undertake?

Worked answer (generated). Assuming the stated 5% rate is continuously compounded, the no-dividend six-month forward is $1100e^{0.05(0.5)} \approx 1127.85$. At 1135 the observed forward is high: short it and buy the stock/index financed with borrowing, yielding about \$7.15 per unit at maturity. At 1115 it is low: buy it and short the underlying, investing the short-sale proceeds, yielding about \$12.85. If 5% were effective annual instead, the exact gaps change but both trade directions remain the same.

Problem 1.10

The S&R index spot price is 1100 and the continuously compounded risk-free rate is 5%. You observe a 9-month forward price of 1129.257.

(a)
What dividend yield is implied by this forward price?

(b)
Suppose you believe the dividend yield over the next 9 months will be only 0.5%. What arbitrage would you undertake?

(c)
Suppose you believe the dividend yield will be 3% over the next 9 months. What arbitrage would you undertake?

Worked answer (generated). For $S_0 = 1100$, $F = 1129.257$, $r = 5\%$ continuous and $T = 0.75$, $\delta = r - \ln(F/S_0)/T \approx 1.5000\%$. If the future dividend yield were *known* to be 0.5%, fair carry would be about 1137.76: buy the underpriced observed forward and short its synthetic counterpart. At a known 3% yield, fair carry would be about 1116.62: sell the overpriced observed forward and buy the synthetic. The question says "believe" rather than "know". A forecast about uncertain dividends makes these directional trades, *not* riskless arbitrages, unless the dividend stream can be locked in and the replication assumptions hold.

Problem 1.11

Suppose the S&P 500 index is currently 950 and the initial margin is 10%. You wish to enter into 10 S&P 500 futures contracts.

(a)
What is the notional value of your position? What is the margin?

(b)
Suppose you earn a continuously compounded rate of 6% on your margin balance, your position is marked to market weekly, and the maintenance margin is 80% of the initial margin. What is the greatest S&P 500 index futures price 1 week from today at which you will receive a margin call?

Worked answer (generated). Using the lecture's historical \$250-per-index-point contract multiplier and interpreting "enter into 10 contracts" as a *long* position (the direction implied by asking for a low-price margin call), notional is $10(250)(950) = \$2{,}375{,}000$. Initial margin is \$237,500, maintenance is $0.8(237{,}500) = \$190{,}000$. After one week the account is $237{,}500e^{0.06/52} + 2500(F_1 - 950)$; it hits maintenance at $F_1 \approx 930.89$. A short position would instead face a call after a *rise*; the problem does not explicitly state the direction or multiplier, so both assumptions are recorded here.

Problem 1.12

A one-year forward contract on a stock has a price of \$75. The stock is expected to pay a dividend of \$1.50 at two future times, six months from now and one year from now, and the annual effective risk-free interest rate is 6%. Calculate the current stock price.

(A)
70.75
(B)
73.63
(C)
75.81
(D)
77.87
(E)
78.04

Worked answer (generated). With an effective annual 6% rate and \$1.50 dividends at 0.5 and 1 year, delivery price $F = 75$ implies $S_0 = 75/1.06 + 1.50/(1.06)^{1/2} + 1.50/1.06 \approx \$73.63$. This is the stock's current value: the stock price equals the present value of its future delivery price *plus* dividends received along the way.

8 Binomial pricing: replication before probabilities

8.1 Why a new pricing model is needed

Payoff diagrams, put–call parity and no-arbitrage bounds tell us important things about an option without determining its unique price from the underlying alone. Parity can price a call from a put or conversely because the other option is already an input. A binomial model supplies a simple description of future stock prices and traded assets with which to replicate the option directly.

Over a period of length h, a stock currently worth S_0 has two possible ex-dividend prices, uS_0 and dS_0, with $u > d > 0$. "Binomial" refers to these two movements, not to an assumption that real prices have only two possible values for all time. Repeated steps expand the possible terminal values. The method's central operation is to work backwards from a known terminal payoff.

8.2 The one-period example, including the financing leg

The source example has a non-dividend-paying stock priced at 41, a European call with $K = 40$, one year to maturity and continuously compounded $r = 8\%$. The stock will be 59.954 or 32.903, corresponding to $u = 1.4623$ and $d = 0.8025$. The call therefore pays

$$C_u = (59.954 - 40)^+ = 19.954, \qquad C_d = (32.903 - 40)^+ = 0.$$

Choose Δ shares and an amount B in the bank account at time zero. A negative B is borrowing. Replication requires the two equations

$$\begin{cases} 59.954\Delta + e^{0.08}B = 19.954, \\ 32.903\Delta + e^{0.08}B = 0. \end{cases}$$

Subtracting the equations first isolates the stock holding. The source's rounded solution is $\Delta = 0.738$ and $B = -22.405$. The portfolio costs

$$C_0 = \Delta S_0 + B = 0.738(41) - 22.405 = 7.839.$$

It has the same payoff as the call in both states, so the law of one price gives the call the same price. The shares cost money now, while borrowing finances part of that purchase; B is not the repayment at maturity. That repayment is approximately $22.405e^{0.08} = 24.271$.

Worked example

Mispricing creates an explicit trade. If the call sells for 8 instead of 7.839, write the call and buy its replicating stock-and-bank portfolio. Receive $8 - 7.839 = 0.161$ now, and the net risky payoff at maturity is zero:

Position at maturity	Down stock 32.903	Up stock 59.954
Written call	0	−19.954
Purchased 0.738 shares	24.271	44.225
Loan repayment	−24.271	−24.271
Net payoff	0	0

If the call instead costs 7.50, buy it, short the replication and receive $7.839 - 7.50 = 0.339$ now. The reverse terminal legs cancel in each state. These rounded tables retain the course example; small rounding differences are not new economic profits.

8.3 Dividend yield changes the replication, not the principle

Suppose the stock pays a continuously compounded dividend yield δ, with dividends reinvested. An initial share holding of Δ grows to $\Delta e^{\delta h}$ shares over the period, while uS_0 and dS_0 remain the *ex-dividend* end-of-period prices. Consequently,

$$\begin{cases} \Delta S_0 u e^{\delta h} + B e^{rh} = C_u, \\ \Delta S_0 d e^{\delta h} + B e^{rh} = C_d. \end{cases}$$

The solution is

$$\Delta = e^{-\delta h}\frac{C_u - C_d}{S_0(u-d)}, \qquad B = e^{-rh}\frac{uC_d - dC_u}{u-d} = e^{-rh}\left(C_u - \Delta S_0 u e^{\delta h}\right).$$

The option price can therefore be written without computing the hedge separately:

$$C_0 = \Delta S_0 + B = e^{-rh}\left[\frac{e^{(r-\delta)h} - d}{u-d}C_u + \frac{u - e^{(r-\delta)h}}{u-d}C_d\right].$$

These expressions apply to a general payoff, not only a call, when C_u, C_d are replaced by that payoff's state values.

Why it matters

No probability of an up move was needed to solve the two replication equations. In a complete two-state market, the option price is pinned down by the traded stock and bank account, so changing an investor's belief about up/down probabilities does not change the no-arbitrage replication cost. Beliefs remain relevant to investment decisions; they are not a missing input to this replicated price.

No-arbitrage requires

$$d < e^{(r-\delta)h} < u.$$

The stock's financed, dividend-adjusted growth must lie between its two possible returns. If it were on or outside the interval, a combination of the stock and bank account would generate an

arbitrage or a degenerate boundary inconsistent with the strictly positive state probabilities used here.

8.4 Delta is a holding and a local sensitivity

Δ is the number of shares needed in the hedge, adjusted for reinvested dividends in the convention above. It is also the sensitivity of the model option value to a small change in the underlying price. The stock price and the node-specific hedge change over time, so replication over several periods is dynamic. The source's Bloomberg delta screenshot supplies no additional derivation; its teaching point is expressed here without an unexplained terminal image.

For the source's special case $\delta = 0, h = 1$,

$$\Delta = \frac{C_u - C_d}{S_0(u-d)}, \quad B = e^{-r}(C_u - \Delta S_0 u), \quad C_0 = e^{-r}\left[\frac{e^r - d}{u-d}C_u + \frac{u - e^r}{u-d}C_d\right].$$

Worked example

Lecture Problem 1, preserved. Let $S_0 = 100, K = 105, r = 8\%$ continuously compounded, $T = 0.5$, $\delta = 0, u = 1.3, d = 0.8$ and one binomial period. Find the call premium, Δ and B. Here the period is $h = T = 0.5$, not one year. Terminal payoffs are 25 and 0; $\Delta = 25/(130 - 80) = 0.5$, $B = -40e^{-0.04} = -38.4316$ and $C_0 = 50 - 38.4316 = 11.5684$.

8.5 Two periods: calculate terminal payoffs, then step back

Return to $S_0 = 41, K = 40, r = 0.08, \delta = 0$, now with two one-year periods and the same u, d. The intermediate prices are 59.954 and 32.903, and the terminal recombining prices are approximately 87.669, 48.114, 26.405. The call terminal payoffs are 47.669, 8.114, 0.

The up-then-down path and down-then-up path reach the same stock price because $ud = du$; a terminal payoff depending only on that price is consequently the same at the shared node. Starting at time 1, replicate the next period's option values separately at each node:

Node	Call value	Shares Δ	Bank investment B
Time 1, up	23.029	1	−36.925
Time 1, down	3.187	0.374	−9.111
Time 0	10.737	0.734	−19.337

For example, the first row is $23.029 = 59.954 - 36.925$. Rolling the two time-1 values back gives the initial call value 10.737. The source's complete labelled tree is retained as a diagram below because it connects prices, payoffs and the hedge at each node, rather than merely showing a screen.

8.6 Three periods and self-financing rebalancing

Adding a third period produces terminal stock prices approximately 128.198, 70.356, 38.612, 21.191 and call payoffs 88.198, 30.356, 0, 0. Backward induction yields these source node values:

Node	Call value	Shares	Bank investment
Time 2, up-up	50.745	1	−36.925
Time 2, mixed	11.925	0.956	−34.086
Time 2, down-down	0	0	0
Time 1, up	26.258	0.981	−32.580
Time 1, down	4.685	0.549	−13.390
Time 0	12.799	0.798	−19.899

Figure 11: The arrows connect each current hedge to the next state. The middle terminal price is shared by up–down and down–up paths.

These are rounded readings of the source tree; the precise values follow from the same recursion. At a node, the arriving hedge has the resources required to buy the new hedge: one rebalances the stock and bank positions without an external cash injection. This is the *self-financing* property. Keeping the original time-zero delta unchanged for all future nodes would not be the replication strategy described by the tree.

9 Puts, path dependence and the information a tree must retain

9.1 The put uses the same stock tree

For a European put, change the terminal condition to $(K - S_T)^+$; the stock tree and rollback rule stay the same. In the two-year $S_0 = 41, K = 40$ example, terminal put payoffs are $0, 0, 13.595$. At the up node the put is zero with a zero hedge. At the down node its value is 7.209, with approximately $\Delta = -0.626, B = 27.814$. At time zero the put costs 3.823, with $\Delta = -0.266, B = 14.749$.

For a non-dividend-paying stock, put–call parity applies at every node:

$$C_t - P_t = S_t - Ke^{-r(T-t)}.$$

Thus the 10.737 two-year call price and the corresponding stock and discounted strike imply the same 3.823 put price. Applying parity at intermediate nodes is a useful check on the entire calculation, not merely a shortcut at the initial date.

Worked example

Lecture Problem 2, preserved. With $S_0 = 100, K = 105, r = 8\%$ continuously compounded, $T = 0.5, \delta = 0, u = 1.3, d = 0.8$ and one period, find the put premium, Δ and B. Put payoffs are 0 and 25; $\Delta = (0 - 25)/(130 - 80) = -0.5$, $B = 65e^{-0.04} = 62.4513$ and $P_0 = -50 + 62.4513 = 12.4513$. The source's call-minus-put identity provides an independent check.

9.2 The three supplied interview questions

1. How many nodes are there in a recombining binomial tree with N time steps?
2. How many nodes are there in a non-recombining binomial tree with N time steps?
3. Assume no dividends, stock price 100, annual continuously compounded risk-free rate 5%, and a one-year European call struck at the money. If volatility is zero, what is the call worth?

Generated explanations. At terminal time there are $N + 1$ recombining nodes and 2^N non-recombining nodes. If "nodes in a tree" means all dates including the root, the counts are $(N + 1)(N + 2)/2$ and $2^{N+1} - 1$ respectively; state the interpretation rather than confusing the terminal count with the whole-tree count. With zero volatility and no dividends, $S_T = 100e^{0.05}$ is deterministic, so the call price is $e^{-0.05}(100e^{0.05} - 100) = 100(1 - e^{-0.05}) \simeq 4.8771$. Zero volatility does not make an at-the-money call worthless when deterministic growth moves the terminal stock above the strike.

9.3 An Asian payoff cannot forget the path

For an Asian option, an average of observed prices enters the payoff. Two paths can have the same terminal stock price but different intermediate prices and hence different averages. Recombining solely on the terminal stock price would merge economically different states. A full non-recombining tree retains the path but grows exponentially, with 2^n states at date n instead of $n + 1$. Other state representations may compress information, but must still retain what the payoff needs.

The arithmetic-average source example uses $S_0 = 41, K = 40, u = 1.4623, d = 0.8025, r = 0.08, T = 2, h = 1$ and observes all three dates, including time zero:

$$G_T = \left(\frac{1}{3} \sum_{k=0}^{2} S_k - 40 \right)^+ .$$

The source spreadsheet's economically meaningful information is reconstructed as a table, rather than left as a screenshot:

Path	Terminal stock	Arithmetic average	Call payoff
up-up	87.6712	62.8752	22.8752
up-down	48.1133	49.6892	9.6892
down-up	48.1133	40.6719	0.6719
down-down	26.4043	33.4356	0

The intermediate up-node has stock 59.9543, current average 50.4772, option value 14.1243, $\Delta = 0.3333$ and $B = -5.8605$. The down-node has stock 32.9025, average 36.9513, option value 0.2640, $\Delta = 0.0310$ and $B = -0.7544$. At the root, stock and current average both equal 41, the option is worth 5.6886 and the hedge has approximately $\Delta = 0.5124, B = -15.3182$. In particular, the up-down and down-up rows cannot be merged even though the terminal stock prices match.

The source also sets a geometric-average exercise:

$$G_T = \left[\left(\prod_{k=0}^{2} S_k \right)^{1/3} - 40 \right]^+ ,$$

and refers to McDonald, Chapter 14.2. Preserve the three observation dates and replace the arithmetic average by the geometric average; do not price a geometric-average contract using the arithmetic-average terminal values. The reference is optional further reading on this specific topic rather than an unexplained external link.

10 Volatility and risk-neutral weights

10.1 Square-root-of-time scaling and the forward tree

The source measures uncertainty by the annualized standard deviation σ of continuously compounded stock returns. If period returns are uncorrelated in the stated model, a period of length h has standard deviation

$$\sigma_h = \sigma\sqrt{h}.$$

With no uncertainty, stock growth must satisfy $S_{t+h} = S_t e^{(r-\delta)h}$. Uncertainty is introduced with

$$u = e^{(r-\delta)h+\sigma\sqrt{h}}, \qquad d = e^{(r-\delta)h-\sigma\sqrt{h}}.$$

The deterministic drift and the two opposite shocks have different roles. Setting $\sigma = 0$ fixes the future price in advance, but does not make prices constant through time. This *forward tree* construction places $e^{(r-\delta)h}$ between d and u when $\sigma > 0$. In the original one-year example, $r = 0.08, \delta = 0, \sigma = 0.3$ gives $u = e^{0.38} = 1.4623$ and $d = e^{-0.22} = 0.8025$.

Worked example

Lecture Problem 3, preserved. A stock index has $S_0 = 100, \sigma = 30\%, r = 5\%, \delta = 3\%, T = 3$ and three periods. Find European call and put prices at strike 95. Using $h = 1$ and the forward-tree parameters, the source reports call 24.0058 and put 14.3799. The dividend-adjusted parity check is $C_0 - P_0 = 100e^{-0.03(3)} - 95e^{-0.05(3)}$. This checks both the dividend convention and the maturity.

10.2 The pricing weights are risk-neutral probabilities

Under the strict no-arbitrage inequality, both coefficients inside the replication formula are positive and add to one. Define

$$p^* = \frac{e^{(r-\delta)h} - d}{u - d}, \qquad 1 - p^* = \frac{u - e^{(r-\delta)h}}{u - d}.$$

Then the one-step value is

$$V_t = e^{-rh}\left[p^* V^u_{t+h} + (1 - p^*)V^d_{t+h}\right].$$

This is a discounted expectation under the risk-neutral pricing measure. It is an interpretation of the replicating price, not an estimate of the empirical frequency of up moves. A risk-neutral investor is indifferent between a sure payment and a risky payment with the same expected amount; a risk-averse investor prefers the sure payment at that same real-world expectation. Changing to pricing probabilities does not assert that actual investors become risk-neutral.

For the one-year 41 stock example, $p^* = 0.4256$ and $1 - p^* = 0.5744$, so $C_0 = e^{-0.08}[0.4256(19.954) + 0.5744(0)] = 7.839$. For two periods, the terminal call payoffs and probabilities are

State	Call payoff	Probability
up-up	47.669	$(p^*)^2 = 0.1811$
one up, one down	8.114	$2p^*(1 - p^*) = 0.4889$
down-down	0	$(1 - p^*)^2 = 0.3300$

Their discounted weighted sum is 10.737. The factor two counts two different sequences reaching the shared middle stock node. For three periods, the source payoffs $88.198, 30.356, 0, 0$ have probabilities approximately $0.0771, 0.3121, 0.4213, 0.1896$ and give the price 12.799. These probabilities sum to one; their products count paths rather than expressing a new price model.

10.3 The n-period European formula

With constant u, d, p^* and a payoff depending only on terminal stock price,

$$C_0 = e^{-rT} \sum_{k=0}^{n} \binom{n}{k} (p^*)^k (1-p^*)^{n-k} \left(S_0 u^k d^{n-k} - K\right)^+, \quad T = nh.$$

For a put, replace the final positive part by $(K - S_0 u^k d^{n-k})^+$. This one-step terminal summation matches backward induction for European terminal-price payoffs. It does not eliminate the need to retain path information for an Asian option or to compare early exercise for an American option. As the discretization is refined in the specified diffusion limit, the binomial method connects to Black–Scholes; a finite tree is an approximation, not the continuous model itself.

Worked example

Lecture Problem 4, preserved. Let $S_0 = 100, \sigma = 0.3, r = 0.08, T = 1, \delta = 0$ and $n = 3$. Use the terminal formula and plot the risk-neutral distribution of the year-1 stock price. The source gives $u = 1.2212, d = 0.8637, p^* = 0.4568$ and this distribution:

Up moves	Stock price	Probability
3	182.14	0.0953
2	128.81	0.3400
1	91.10	0.4044
0	64.43	0.1603

The probabilities are masses on four discrete prices, not the heights of a continuous probability density. The supplied exercise does not identify a strike for a unique numerical call price; the formula above describes how to price a chosen terminal payoff.

11 American exercise and real investment options

11.1 Exercise value versus continuation value

An American option can be exercised before maturity and must be at least as valuable as the corresponding European option:

$$C_{\text{Am}} \geq C_{\text{Eur}}, \qquad P_{\text{Am}} \geq P_{\text{Eur}}.$$

The source combines this with the loose bounds $S \geq C_{\text{Am}} \geq C_{\text{Eur}} \geq \max(0, S - PV(\text{Div}) - PV(K))$ and $K \geq P_{\text{Am}} \geq P_{\text{Eur}} \geq \max(0, PV(K) - S + PV(\text{Div}))$. Their applicability depends on the dividend and rate assumptions developed in the bounds chapter.

At each permitted exercise node, compute discounted continuation and compare it with immediate exercise:

$$V_t^{\text{Am}} = \max\left\{ g(S_t), e^{-rh}\left[p^* V_{t+h}^u + (1-p^*) V_{t+h}^d\right] \right\}.$$

Use $g(S) = (S-K)^+$ for a call and $(K-S)^+$ for a put. The terminal values are the same European-style intrinsic payoffs; the extra maximization occurs before expiry.

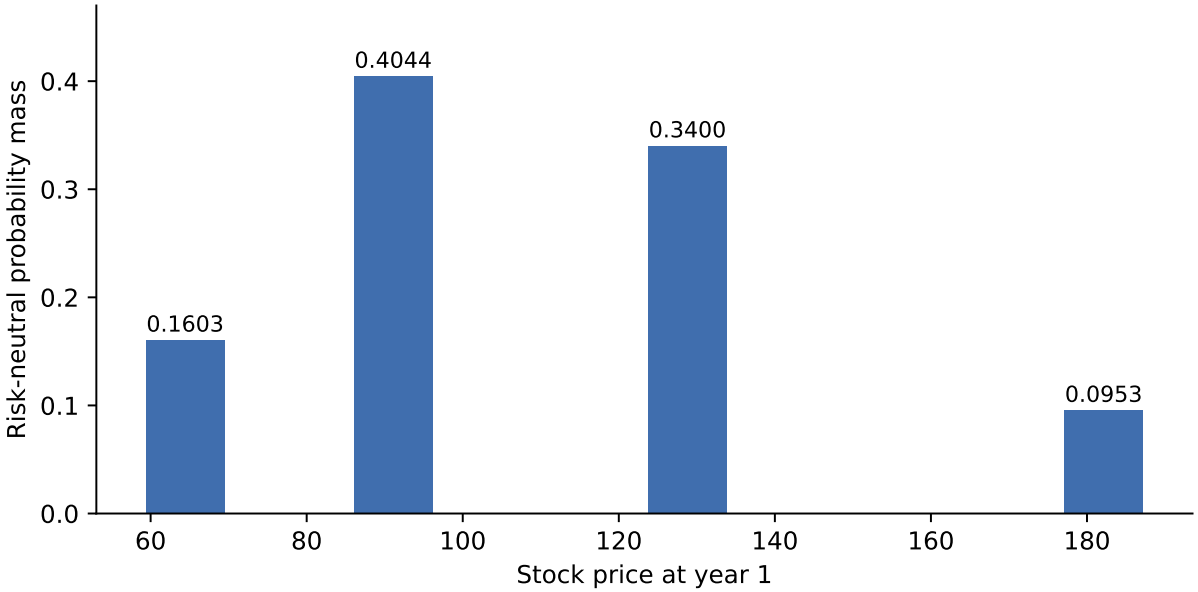


Figure 12: Discrete risk-neutral probability masses for Lecture Problem 4. The labels correspond to the table, not observed empirical frequencies.

For a non-dividend-paying stock under the course's nonnegative-rate assumptions, early exercise of an American call is not optimal. The holder would pay the strike earlier and surrender the remaining insurance/time value without gaining dividends. With dividends, exercise can become optimal: receiving the stock gives access to dividends that an unexercised call does not receive. The source modifies its original example to a continuously compounded yield of 6.5% and illustrates a call exercise node at stock price 76.982. Its labelled exercise/continuation comparison is retained in a graph rather than a whole slide.

An American put may optimally be exercised early even with no dividends. If the stock approaches zero, the European put is approximately worth $PV(K)$ while immediate exercise pays K; at a positive rate, earlier receipt of the strike can be worth more than the lost protection. The source's dividend-yield example identifies an early-exercise node at stock 23.187. The graph distinguishes the value if exercised, P_{EX}, from continuation, P_{NO}.

Worked example

Lecture Problem 5, preserved. For $S_0 = 100, \sigma = 30\%, r = 5\%, \delta = 3\%, T = 3, n = 3$, price the American call and put struck at 95. The source reports call 24.1650 and put 15.2593, both above their European counterparts 24.0058 and 14.3799. The call also benefits from exercise flexibility here because the example has dividends; the non-dividend call result is not applicable.

The costs and benefits of early exercise are asymmetric. A call holder gains the stock/dividends, but pays the strike sooner and loses insurance against a later fall. A put holder receives the strike sooner, but loses dividends on a stock position surrendered through exercise and loses insurance against a later rise. Compare values at each node; "in the money" alone does not determine optimal early exercise.

11.2 A real option is a right to take a business decision

A real option applies contingent-decision logic to an investment project rather than a traded option contract. Paying an investment cost to receive a project's future cash flows resembles paying a

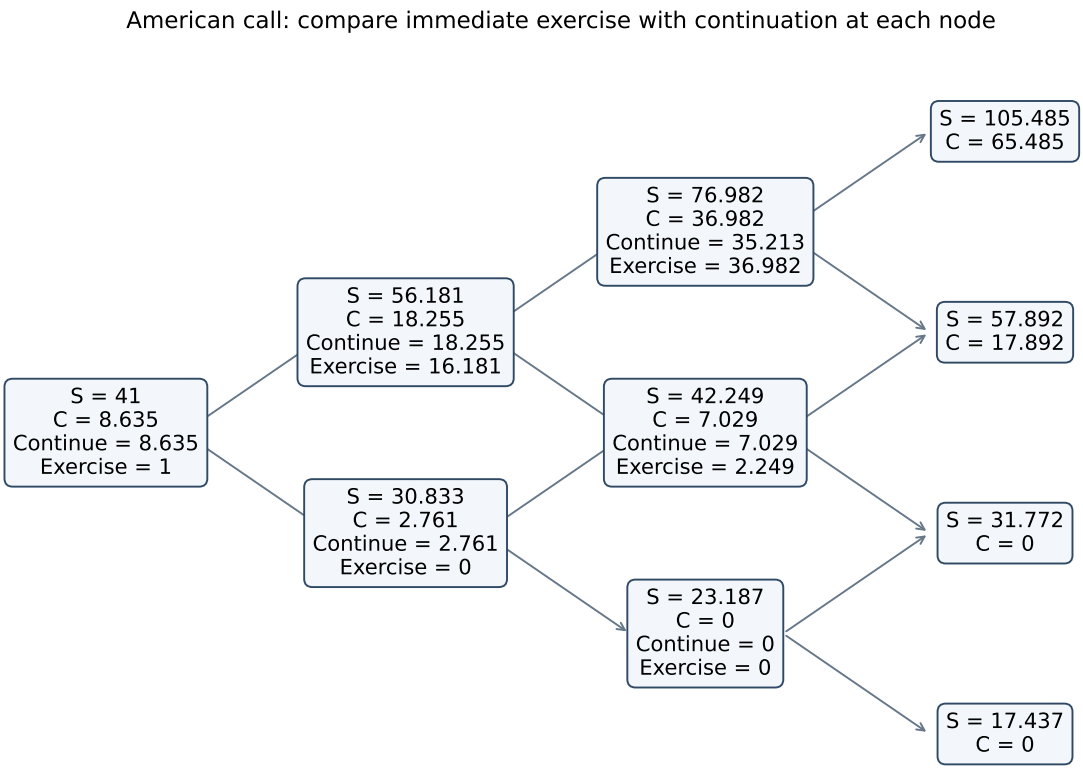


Figure 13: Source call tree reconstructed as a diagram: at $S = 76.982$, exercise 36.982 exceeds continuation 35.213.

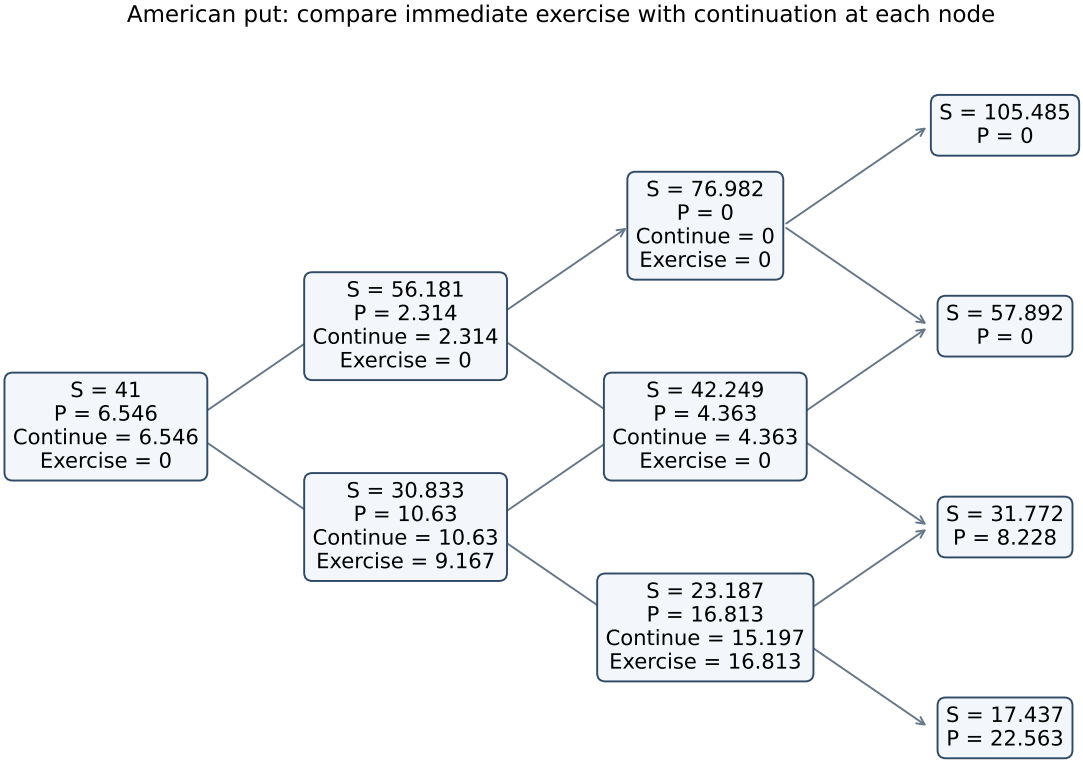


Figure 14: Source put tree reconstructed: at $S = 23.187$, exercise 16.813 exceeds continuation 15.197. Other nodes retain the option when continuation is higher.

call strike to receive an asset. The mapping is investment cost $\leftrightarrow$ strike, present value of project $\leftrightarrow$ underlying asset value. Waiting, expansion, shutdown and abandonment change the feasible future decisions.

The source project requires an initial investment of 100 and is expected to produce a perpetual cash flow of 18 in year 1, growing at 3% annually thereafter. With the stated 15% discount rate for the growing perpetuity, $PV = 18/(0.15 - 0.03) = 150$ and the static NPV is $150 - 100 = 50$. Its cash-flow yield is 18/150, represented continuously as $\delta = \log(1 + 18/150) = 0.1133$. These project-discounting inputs are not the risk-free rate subsequently used in the option-pricing tree.

The source uses continuously compounded risk-free $r = 6.766\%$ and annual project-return volatility 50%. A forward tree with $u = 1.5751, d = 0.5795$ produces the following cash flows and project values:

Node	Project cash flow	Present value of project
Initial	18.00	150.00
Year 1, up	28.35	236.27
Year 1, down	10.43	86.92
Year 2, up-up	44.66	372.15
Year 2, mixed	16.43	136.91
Year 2, down-down	6.04	50.37

The risk-neutral probability is approximately 0.3775 at each step. Investment can occur at time 0, 1 or 2. At the final date, the exercise values are 272.15, 36.91 and 0. At the year-1 up node, continuation is 117.50 but exercise is $236.27 - 100 = 136.27$, so invest. At the down node, continuation is 13.02 while immediate exercise has value zero, so wait. At time zero, continuation is 55.66 versus immediate investment NPV of 50.

The ability to wait adds value by allowing rejection of bad future investments without giving up the opportunity to invest in good states. Investing immediately abandons the remaining exercise flexibility, not merely a five-unit accounting item. The source also asks the reader to repeat the calculation with 30% volatility and discuss the change; that exercise is retained as an invitation to compare the whole decision tree, not answered by simply assuming that static NPV changes with volatility.

11.3 The supplied expansion exercise

Desautels Industries considers a plant costing 140 million initially. Cash flows at year 2 are 200 million after two good years, 150 million after one good and one bad year, and 100 million after two bad years. After year 1 the company can double capacity by investing another 140 million. The annual risk-free rate is 6% and the risk-neutral probability of a good year is 0.53. Compute the project value with and without the option to expand. These exact supplied inputs are the exercise; the timing and incremental cash flows must be used consistently.

Generated solution, for the stated two-year terminal cash-flow interpretation. Write $q = 0.53$ and use the source's 6% annual risk-free rate. Without expansion,

$$NPV_0 = \frac{q^2(200) + 2q(1-q)(150) + (1-q)^2(100)}{(1.06)^2} - 140.$$

At year 1 after a good year, the extra capacity has incremental discounted expected cash flow $[q(200) + (1-q)(150)]/1.06$ and costs 140. After a bad year, substitute $[q(150) + (1-q)(100)]/1.06$. Take the positive part at each node and discount the risk-neutral weighted incremental exercise values to time zero. The option value is added to the no-expansion NPV. This generated calculation assumes effective annual compounding for the source's unqualified "6% per year"; a continuous interpretation would use different discount factors and must be stated explicitly.

Numerically, the no-expansion NPV is -3.8305 million. The incremental year-1 NPVs are 26.5094 after a good year and -20.6604 after a bad year, so expand only after a good year. The capacity option is worth 13.2547 million initially; the project NPV including expansion is 9.4242 million.

11.4 Other real options and limitations of the model

The source examples classify when to invest as a call option; shutting down, restarting or permanently abandoning a project as put-like rights; an investment that generates later opportunities as a compound option; and flexibility to switch inputs, outputs or technologies as a rainbow option. The economic payoff and permitted decision times must be specified before using these names.

The final slide cautions that the model's assumptions simplify reality. Volatility need not be constant; stock prices may jump; returns may exhibit short-run momentum and longer-run negative correlation rather than time independence. Dividends are often discrete quarterly or annual payments, and their amount can be uncertain, unlike a known continuous yield. The source notes that a binomial tree can be adapted for discrete dividends and refers to McDonald, Chapter 11.4. Model assumptions are conditions for a calculation, not properties guaranteed by a fitted tree.

12 Options: supplied exercise series II

Problem 2.1

Happy Jalapenos, LLC has an exclusive contract to supply jalapeno peppers to the organizers of the annual jalapeno eating contest. The contract states that the contest organizers will take delivery of 10,000 jalapenos in one year at the market price. It will cost Happy Jalapenos 1,000 to provide 10,000 jalapenos and today's market price is 0.12 for one jalapeno. The continuously compounded risk-free interest rate is 6%.

Happy Jalapenos has decided to hedge as follows: Buy 10,000 0.12-strike put options for 84.30 and sell 10,000 0.14-stike call options for 74.80. Both op- tions are one-year European. Happy Jalapenos believes the market price in one year will be somewhere between 0.10 and 0.15 per jalapeno.

Determine which of the following intervals represents the range of possible profit one year from now for Happy Jalapenos.

(A) - 200 to 100
(B) -110 to 190
(C) -100 to 200
(D) 190 to 390
(E) 200 to 400

Worked example

Generated solution. At expiry, the collar replaces the selling price by $\min(0.14, \max(S_T, 0.12))$. The net premium paid now is $84.30 - 74.80 = 9.50$. Profit at the common terminal date is

$$\Pi_T = 10000 \min(0.14, \max(S_T, 0.12)) - 1000 - 9.50e^{0.06}.$$

For the supplied price range, profit runs from 189.91 to 389.91, rounded to 190–390. Answer D. The financed premium must be subtracted even when the collar payoff is zero.

Problem 2.2

The PS index has the following characteristics:

1. One share of the PS index currently sells for 1,000.
2. The PS index does not pay dividends.

Sam wants to lock in the ability to buy this index in one year for a price of 1,025. He can do this by buying or selling European put and call options with a strike price of 1,025. The annual effective risk-free interest rate is 5%.

Determine which of the following gives the hedging strategy that will achieve Sam's objective and also gives the cost today of establishing this position.

(A) Buy the put and sell the call, receive 23.81
(B) Buy the put and sell the call, spend 23.81
(C) Buy the put and sell the call, no cost
(D) Buy the call and sell the put, receive 23.81
(E) Buy the call and sell the put, spend 23.81

Worked example

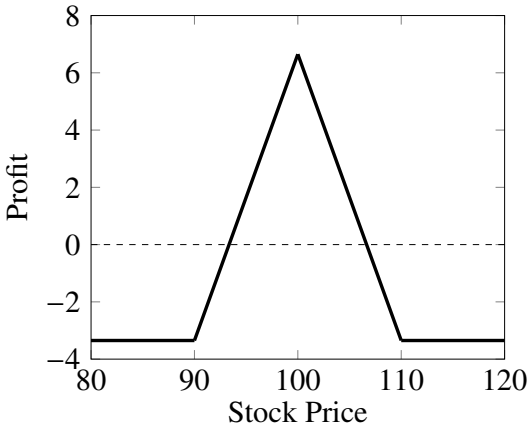
Generated solution. Buy the call and sell the put at strike 1025. The combined terminal payoff is $S_T - 1025$, so together with payment of 1025 it delivers the index. Put–call parity gives $C_0 - P_0 = 1000 - 1025/1.05 = 23.8095$. The position costs 23.81 now. Answer E.

Problem 2.3

Stock ABC has the following characteristics:
1. The current price to buy one share is 100.
2. The stock does not pay dividends.
3. European options on one share expiring in one year have the following prices:

Strike price	Call option price	Put option price
90	14.63	0.24
100	6.80	1.93
110	2.17	6.81

A butterfly spread on this stock has the following profit diagram.
The continuously compounded risk-free interest rate is 5%.
Determine which of the following will NOT produce this profit diagram.
(A) Buy a 90 put, buy a 110 put, sell two 100 puts
(B) Buy a 90 call, buy a 110 call, sell two 100 calls
(C) Buy a 90 put, sell a 100 put, sell a 100 call, buy a 110 call
(D) Buy one share of the stock, buy a 90 call, buy a 110 put, sell two 100 puts
(E) Buy one share of the stock, buy a 90 put, buy a 110 call, sell two 100 calls.



The source profit curve has flat negative tails and a peak around $S_T = 100$.

Worked example

Generated solution. Answer D. The call butterfly in B has payoff $(S_T - 90)^+ - 2(S_T - 100)^+ + (S_T - 110)^+$, a tent of height 10; A has the same payoff. C has that payoff minus 10; E has that payoff plus 90. Their corresponding financed costs offset these constant differences (up to rounding in the quoted premiums). D instead has payoff $2S_T - 90$ below 90 and above 110, and so cannot produce a flat tail. For B the premium is $14.63 + 2.17 - 2(6.80) = 3.20$, and the tails of its profit diagram are $-3.20e^{0.05} \simeq -3.364$. Always compare profit with profit, accounting for the initial stock position as well as option premiums.

Problem 2.4

A trader shorts one share of a stock index for 50 and buys a 60 strike European call option on that stock that expires in 2 years for 10. Assume the annual effective risk-free interest rate is 3%. The stock index increases to 75 after 2 years. Calculate the profit on your combined position, and determine an alternative name for this combined position.

(A) Profit: -22.64 Name: Floor
(B) Profit: -17.56 Name: Floor
(C) Profit: -22.64 Name: Cap
(D) Profit: -17.56 Name: Cap
(E) Profit: -22.64 Name: "Written" covered call

Worked example

Generated solution. Short stock plus long call is a cap on the short-stock exposure. The initial receipt is $50 - 10 = 40$. At expiry the position pays $-75 + (75 - 60) = -60$, hence $\Pi_T = 40(1.03)^2 - 60 = -17.564$. Answer D: profit -17.56, cap.

Problem 2.5

The current price of a non-dividend paying stock is 40 and the continuously compounded risk-free interest rate is 8%. You are given that the price of a 35-strike call option is 3.35 higher than the price of a 40 strike call option, where both options expire in 3 months.

Calculate the amount by which the price of an otherwise equivalent 40-strike put option exceeds the price of an otherwise equivalent 35-strike put option.

(A) 1.55
(B) 1.65
(C) 1.75
(D) 3.25
(E) 3.75

Worked example

Generated solution. Subtract put–call parity at the two strikes:

$$P_{40} - P_{35} = C_{40} - C_{35} + 5e^{-0.08(3/12)} = -3.35 + 4.90099 = 1.55099.$$

Answer A.

Problem 2.6

The current price of a non-dividend paying stock is 40 and the continuously compounded risk-free interest rate is 8%. You enter into a short position on 3 call options, each with 3 months to maturity, a strike price of 35, and an option premium of 6.13. Simultaneously, you enter into a long position on 5 call options, each with 3 months to maturity, a strike price of 40, and an option premium of 2.78. All 8 options are held until maturity.

Calculate the maximum possible profit and the maximum possible loss for the entire option portfolio.

(A) Maximum profit: 3.42 Maximum loss: 4.58
(B) Maximum profit: 4.58 Maximum loss: 10.42
(C) Maximum profit: Unlimited Maximum loss: 10.42
(D) Maximum profit: 4.58 Maximum loss: Unlimited
(E) Maximum profit: Unlimited Maximum loss: Unlimited

Worked example

Generated solution. The initial receipt is $3(6.13) - 5(2.78) = 4.49$, whose future value is $4.49e^{0.02} = 4.5807$. The payoff is $-3(S_T - 35)^+ + 5(S_T - 40)^+$: zero below 35; $-3(S_T - 35)$ between 35 and 40; $2S_T - 95$ above 40. Its minimum is -15 at 40. Maximum loss is therefore $15 - 4.5807 = 10.4193$; the upper tail is unbounded. Answer C.

Problem 2.7

An investor is analyzing the costs of two-year, European options for aluminum and zinc at a particular strike price. For each ton of aluminum, the two-year forward price is 1400, a call option costs 700, and a put option costs 550. For each ton of zinc, the two-year forward price is 1600 and a put option costs 550. The annual effective risk-free interest rate is 6%.

Calculate the cost of a call option per ton of zinc.

(A) 522
(B) 800
(C) 878
(D) 900
(E) 1231

Worked example

Generated solution. Use forward-form parity $C - P = (F - K)/(1.06)^2$. For aluminium, $700 - 550 = 150$, giving $K = 1400 - 150(1.06)^2 = 1231.46$. Zinc then has $C = 550 + (1600 - 1231.46)/(1.06)^2 = 877.999$. Answer C: 878.

Problem 2.8

You are given:

i) An investor short-sells a non-dividend paying stock that has a current price of 44 per share.
ii) This investor also writes a collar on this stock consisting of a 40-strike European put option and a 50-strike European call option. Both options expire in one year.
iii) The prices of the options on this stock are:

Strike	Call	Put
40	8.42	2.47
50	3.86	7.42

iv) The continuously compounded risk-free interest rate is 5%.
v) Assume there are no transaction costs.

Calculate the maximum profit for the overall position at expiration.

(A) 2.61
(B) 3.37
(C) 4.79
(D) 5.21
(E) 7.39

Worked example

Generated solution. Writing the collar means short the 40-put and long the 50-call, combined with short stock. The initial cash receipt is $44 + 2.47 - 3.86 = 42.61$. Terminal payoff is $-S_T - (40 - S_T)^+ + (S_T - 50)^+$: -40 below 40, $-S_T$ between 40 and 50, and -50 above 50. Maximum profit is $42.61e^{0.05} - 40 = 4.7947$. Answer C.

Problem 2.9

An insurance company sells single premium deferred annuity contracts with return linked to a stock index, the time-t value of one unit of which is denoted by S (t). The contracts offer a minimum guarantee return rate of g%. At time 0, a single premium of amount π is paid by the policyholder, and $\pi \times$ y% is deducted by the insurance company. Thus, at the contract maturity date, T, the insurance company will pay the policyholder

$$\pi(1 - y\%) \max\left[\frac{S(T)}{S(0)}, (1 + g\%)^T\right].$$

You are given the following information:

(i) The contract will mature in one year.
(ii) The minimum guarantee rate of return, g%, is 3%.
(iii) Dividends are incorporated in the stock index. That is, the stock index is constructed with all stock dividends reinvested.
(iv) S (0) = 100
(v) The price of a one-year European put option, with strike price \$103, on the stock index is \$15.21.

Determine y%, so that the insurance company does not make or lose money on this contract.

(A) 12.8%
(B) 13.0%
(C) 13.2%
(D) 13.4%
(E) 13.6%

Worked example

Generated solution. For $T = 1$, the payoff is $\pi(1 - y)\max(S_T, 103)/100$, where y is the deduction as a decimal fraction. Since $\max(S_T, 103) = S_T + (103 - S_T)^+$, the replicating stock-plus-put costs $100 + 15.21 = 115.21$ per 100 of participation. Fairness requires $(1 - y)115.21/100 = 1$, so $y = 15.21/115.21 = 0.132019$. Answer C: 13.2%. Dividends are reinvested in the index, as specified in the question.

Problem 2.10

Consider a chooser option (also known as an as-you-like-it op- tion) on a nondividend-paying stock. At time 1, its holder will choose whether it becomes a European call option or a European

put option, each of which will expire at time 3 with a strike price of $100. The chooser option price is $20 at time t = 0. The stock price is $95 at time t = 0. Let C(T) denote the price of a European call option at time t = 0 on the stock expiring at time T, T > 0, with a strike price of $100.

You are given:

(i) The risk-free interest rate is 0.

(ii) C(1) = $4.

Determine C(3)

(A) $9

(B) $11

(C) $13

(D) $15

(E) $17

Worked example

Generated solution. At time 1, maturity-3 parity with $r = 0$ gives $P_{1,3} = C_{1,3} + 100 - S_1$. Thus the chooser is worth $C_{1,3} + (100 - S_1)^+$, replicated by a maturity-3 call plus a maturity-1 put of strike 100. At time zero the latter put costs $P(1) = C(1) + 100 - S_0 = 4 + 5 = 9$. Hence $20 = C(3) + 9$ and $C(3) = 11$. Answer B.

Problem 2.11

The current price of a non-dividend paying stock is 40 and the continuously compounded risk-free interest rate is 8%. The following table shows call and put option premiums for three-month European of various ex- ercise prices:

Strike	Call	Put
35	6.13	0.44
40	2.78	1.99
45	0.97	5.08

A trader interested in speculating on volatility in the stock price is consider- ing two investment strategies. The first is a 40-strike straddle. The second is a strangle consisting of a 35-strike put and a 45-strike call.

Determine the range of stock prices in 3 months for which the strangle out- performs the straddle.

(A) The strangle never outperforms the straddle

(B) 33.56 < S T < 46.44

(C) 35.13 < S T < 44.87

(D) 36.57 < S T < 43.43

(E) The strangle always outperforms the straddle.

Worked example

Generated solution. The straddle costs $2.78 + 1.99 = 4.77$; the strangle costs $0.44 + 0.97 = 1.41$. The strangle saves $3.36e^{0.02} = 3.42788$ at expiry. For $35 < S_T < 40$, its profit advantage is $S_T - 40 + 3.42788$; for $40 < S_T < 45$, it is $40 - S_T + 3.42788$. Outside these intervals it loses five units of payoff relative to the straddle. The positive-advantage interval is $36.5721 < S_T < 43.4279$. Answer D.

Problem 2.12

For each ton of a certain type of rice commodity, the four-year forward price is 300. A four-year 400-strike European call option costs 110. The continuously compounded risk-free interest rate is 6.5%.

Calculate the cost of a four-year 400-strike European put option for this rice commodity

(A) 10.00
(B) 32.89
(C) 118.42
(D) 187.11
(E) 210.00

Worked example

Generated solution. Forward-form parity gives $P = C + e^{-rT}(K - F) = 110 + 100e^{-0.065(4)} = 187.1052$. Answer D: 187.11.

Problem 2.13

Suppose call and put prices are given by

Strike	50	55
Call premium	9	10
Put premium	7	6

What no-arbitrage property is violated? What spread position would you use to effect arbitrage? Demonstrate that the spread position is an arbitrage.

Worked example

Generated solution. Calls must be nonincreasing and puts nondecreasing in strike for equal maturities and otherwise identical contracts. Both inequalities are violated. Buy the 50-call for 9 and sell the 55-call for 10: receive 1 immediately and obtain $(S_T - 50)^+ - (S_T - 55)^+ \geq 0$ later. Alternatively buy the 55-put for 6 and sell the 50-put for 7: receive 1 immediately and obtain $(55 - S_T)^+ - (50 - S_T)^+ \geq 0$. For the call spread the terminal payoff is 0, $S_T - 50$, or 5 in the regions below 50, between 50 and 55, or above 55. For the put spread it is 5, $55 - S_T$, or 0 in those regions. Neither strategy requires a forecast.

13 Binomial model: complete exercise set

The supplied exercise statements are retained below. The worked solutions are **generated study explanations**, independently calculated from their inputs. Unless explicit up/down multipliers are given, the course uses the forward-tree convention

$$h = T/n, \quad u = e^{(r-\delta)h+\sigma\sqrt{h}}, \quad d = e^{(r-\delta)h-\sigma\sqrt{h}}, \quad p^* = \frac{e^{(r-\delta)h} - d}{u - d}.$$

These u, d differ from the drift-free CRR convention; changing conventions changes finite-step numerical prices. American values are obtained by comparing continuation with immediate exercise at every node.

13.1 Problem 3.1: replication versus immediate exercise

Statement. A stock is initially 50 and after one year is either 125 or 50. The annual *effective* risk-free rate is 50%. Consider a European put with strike 150. (a) What are the payoffs? (b) Which stock and risk-free-bond portfolio replicates the option? (c) What is its price? (d) Verify using risk-neutral probabilities. (e) Would an American put have the same price?

Generated solution. The up/down payoffs are 25 and 100. With B the current bank-account investment,

$$125\Delta + 1.5B = 25, \quad 50\Delta + 1.5B = 100 \quad \Longrightarrow \quad \Delta = -1,\ B = 100.$$

The replicating portfolio costs $-50 + 100 = 50$. The risk-neutral up probability is $p^* = (1.5 - 1)/(2.5 - 1) = 1/3$, so $(25/3 + 200/3)/1.5 = 50$. Immediate exercise gives $150 - 50 = 100$. The American put is worth 100 and is exercised now. An effective 50% rate means a growth factor 1.5, not $e^{0.5}$.

13.2 Problem 3.2: American call in a two-period tree

Statement. Each period is one year. A non-dividend-paying stock has $S_0 = 20$, $u = 1.2840$ and $d = 0.8607$ (gross up/down capital-gain factors). The continuously compounded risk-free rate is 5%. Calculate a two-period American call with strike 22. Choices: (A) 0; (B) 1; (C) 2; (D) 3; (E) 4.

Generated solution. $p^* = (e^{.05} - .8607)/(1.2840 - .8607) = .4502034$. Terminal payoffs in down-down, mixed and up-up states are $0, .102776, 10.973120$. The one-period continuation values are .0440135 and 4.7529527, and the root is 2.0584547. Exercise never improves the call's value: choose **C**, the nearest listed value.

13.3 Problem 3.3: call and put hedge quantities

Statement. $S_0 = 100, K = 105, r = 8\%, T = .5, \delta = 0, u = 1.3, d = .8, n = 1$. Find (a) the premium, Δ and B for a European call; (b) the corresponding quantities for a European put. Rates are continuously compounded.

Generated solution. The call payoffs are 25 and 0, the put payoffs 0 and 25. Therefore

$$\Delta_C = .5,\ B_C = -40e^{-.04} = -38.431578,\ C_0 = 11.568422;$$

$$\Delta_P = -.5,\ B_P = 65e^{-.04} = 62.451314,\ P_0 = 12.451314.$$

The negative call bank position is borrowing. Check $C_0 - P_0 = 100 - 105e^{-.04}$. Here B denotes the amount invested today, rather than its terminal repayment.

13.4 Problem 3.4: dividends and early exercise

Statement. A one-period model has $h = 1, S_0 = 100, r = 0, \sigma = 30\%, \delta = .08$. Compute American calls for strikes 70, 80, 90 and 100. (a) Which strikes lead to early exercise? (b) Use put–call parity to explain why early exercise does not occur at the higher strikes and occurs for all strikes lower than those identified in (a).

Generated solution. $u = e^{.22} = 1.2460767$, $d = e^{-.38} = .6838614$ and $p^* = .4255575$. Compare each European continuation value with $100 - K$:

K	Continuation	Immediate exercise	American price
70	23.2387	30	30
80	18.9831	20	20
90	14.7276	10	14.7276
100	10.4720	0	10.4720

Exercise occurs at 70 and 80. European parity with $r = 0$ gives $C_E - (S_0 - K) = P_E - S_0(1 - e^{-\delta})$. The lost dividend value is fixed at 7.688365, while the European put increases with K. At 80 the put is still below that amount, so every lower strike also favours exercise; at 90 and 100 it exceeds that amount. This is an application of *European* parity to the continuation value; American prices do not obey the same equality.

13.5 Problem 3.5: three-year American put

Statement. A stock pays a continuous dividend yield of 6.5%. Each period is one year and the continuously compounded risk-free rate is 10%. The three-period stock tree is

Time	Stock prices (highest to lowest)
0	300
1	375, 210
2	468.75, 262.5, 147
3	585.9375, 328.125, 183.75, 102.9

Find a three-year at-the-money American put. Choices: (A) 15.86; (B) 27.40; (C) 32.60; (D) 39.73; (E) 57.49.

Generated solution. $K = 300, u = 1.25, d = .7$ and $p^* = (e^{.035} - .7)/.55 = .6102177$. Terminal payoffs from highest to lowest are $0, 0, 116.25, 197.10$. American values at time 2 are $0, 41.000172, 153$ and at time 1 are $14.460336, 90$. Early exercise occurs at $S = 147$ at time 2 and at $S = 210$ at time 1. The root is 39.726311: choose **D**.

13.6 Problem 3.6: the first integer strike that should be exercised

Statement. Use the usual one-period tree for a non-dividend-paying stock. The period is 3 months, $S_0 = 100$, volatility 30%, and continuously compounded risk-free rate 4%. An investor owns an American put expiring at the period's end. Find the smallest integer strike for which exercise at the beginning is optimal. Choices: (A) 114; (B) 115; (C) 116; (D) 117; (E) 118.

Generated solution. $h = .25, u = e^{.16} = 1.1735109, d = e^{-.14} = .8693582, p^* = .4625702$. In the relevant strike range $Sd < K < Su$, only the down payoff is positive. Immediate exercise dominates when

$$K - S_0 \geq e^{-rh}(1 - p^*)(K - S_0 d) \quad \Longleftrightarrow \quad K \geq \frac{S_0[1 - e^{-rh}(1 - p^*)d]}{1 - e^{-rh}(1 - p^*)}.$$

At 114 continuation is $14.400370 > 14$; at 115 it is approximately $14.932452 < 15$. The first integer is **115 (B)**.

13.7 Problem 3.7: full node values and parity

Statement. The annual effective riskless rate is 25% per period. A stock initially worth 100 doubles or halves each period: prices at time 1 are 200 and 50, and at time 2 are 400, 100 and 25. Consider European and American options with $K = 75$ and maturity two periods away. (a) Find risk-neutral probabilities. (b) Find European put values at each node. (c) Find American put values at each node. (d) Find the European call using parity.

Generated solution. $p^* = (1.25 - .5)/(2 - .5) = .5$, so up and down are equally weighted. Put payoffs at 400, 100, 25 are $0, 0, 50$. European values at 200 and 50 are 0 and 20, giving $P_E = 8$. The American values at those nodes are 0 and 25, since exercise at 50 gives 25; thus $P_A = 10$. Terminal values coincide for both exercise styles. The European call is

$$C_E = P_E + 100 - 75/(1.25)^2 = 8 + 100 - 48 = 60.$$

13.8 Problem 3.8: two-step stock tree

Statement. $S_0 = 40, K = 40, r = 8\%$ continuously compounded, $\sigma = 30\%, \delta = 0, T = .5, n = 2$. (a) Construct the stock tree and give u, d. (b) Show the European call is 4.110. (c) Compute European and American puts.

Generated solution. $h = .25$, $u = e^{.17} = 1.1853049$, $d = e^{-.13} = .8780954$, $p^* = .4625702$. Prices at time 1 are 47.412194, 35.123817, and terminal prices are 56.197904, 41.632431, 30.842063. Backward induction gives $C_E = 4.109801$, $P_E = 2.541379$, $P_A = 2.568715$. At the down node the American put exercises for $40 - 35.123817 = 4.876183$, versus continuation 4.824291. European parity checks $C_E - P_E = 40 - 40e^{-.04}$.

13.9 Problem 3.9: three-step values and the exercise decision

Statement. $S_0 = 100, K = 95, r = 8\%$ continuously compounded, $\sigma = 30\%, \delta = 0, T = 1, n = 3$. (a) Verify the American call is 18.283, and that it never exercises early so the European call is equal. (b) Show the European put is 5.979 and verify parity. (c) Verify the American put is 6.678.

Generated solution. $h = 1/3$, $u = 1.2212461, d = .8636926, p^* = .4568067$. The call continuation values always dominate exercise at every pre-expiry node, giving $C_A = C_E = 18.282552$. The European put is 5.978605 and

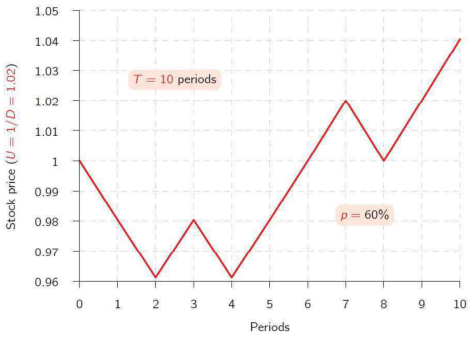
$$18.282552 - 5.978605 = 100 - 95e^{-.08} = 12.303947.$$

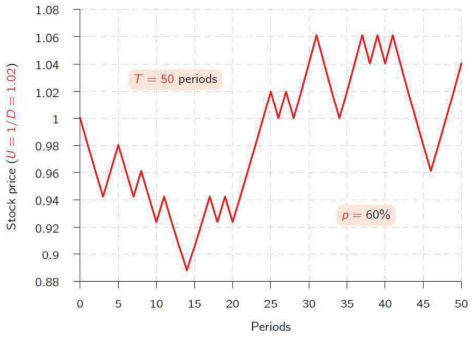
The American put is 6.677901. At the lowest time-2 node $S = 74.596483$, exercise pays 20.403517, exceeding European continuation 17.903663. This benefit is propagated backwards to the root.

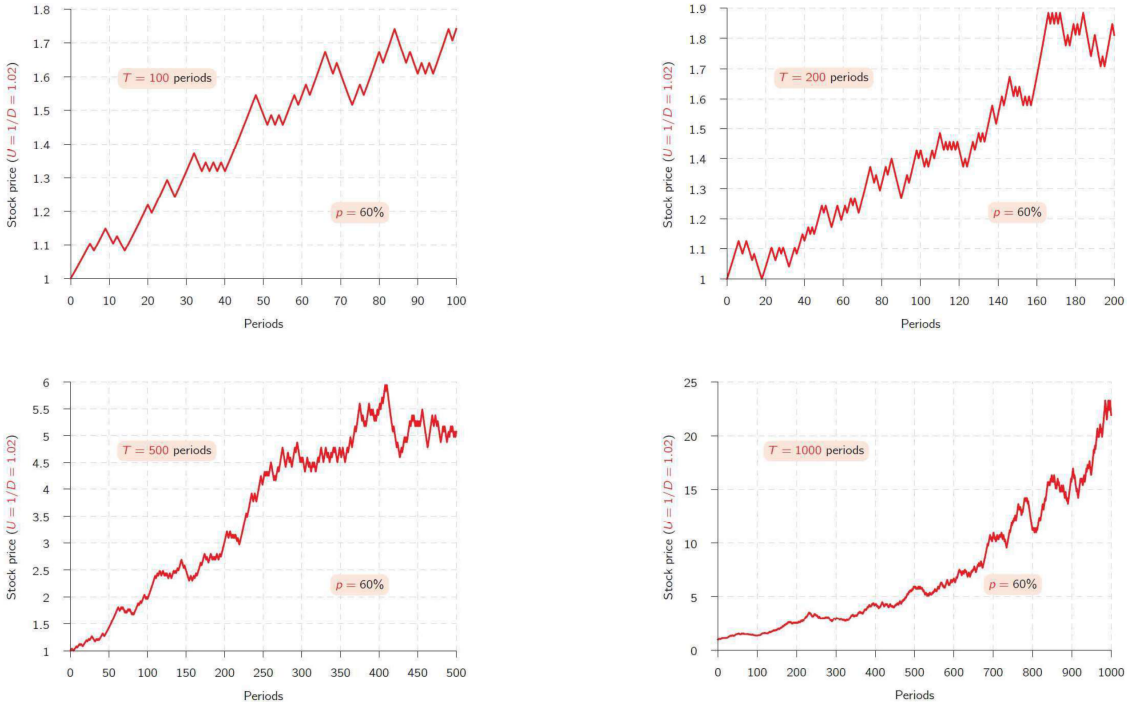
14 From binomial trees to Black–Scholes

14.1 Discrete steps and the continuous-time limit

A binomial model evolves at distinct dates. Continuous time is the limit in which the interval between those dates becomes arbitrarily small. This changes the model's time structure: it does not mean that a finite tree is literally a continuous path. The successive simulated stock paths show what happens when 10, 50, 100, 200, 500 and 1000 increments are used; the illustrated up-move probability is 60%. These paths demonstrate increasing time resolution, rather than supplying a risk-neutral probability estimate.







The axes identify the number of periods and normalised stock price. Increasing the number of draws changes the realised path too, so these examples should not be read as successive observations of the same underlying path.

14.2 Refining the tree changes both the outcomes and their weights

A coarse tree gives only a few terminal prices. Divide the time to maturity T into n periods, $h = T/n$, and recompute

$$u = e^{(r-\delta)h+\sigma\sqrt{h}}, \quad d = e^{(r-\delta)h-\sigma\sqrt{h}}, \quad p^* = \frac{e^{(r-\delta)h} - d}{u - d}.$$

For the original one-year European call $S_0 = 41, K = 40, r = .08, \sigma = .3, \delta = 0$, three periods give $h = 1/3, u = 1.2212, d = .8637, p^* = .4568$, rather than the one-period multipliers 1.4623, .8025. The terminal payoffs and probabilities, with k the number of up moves, are

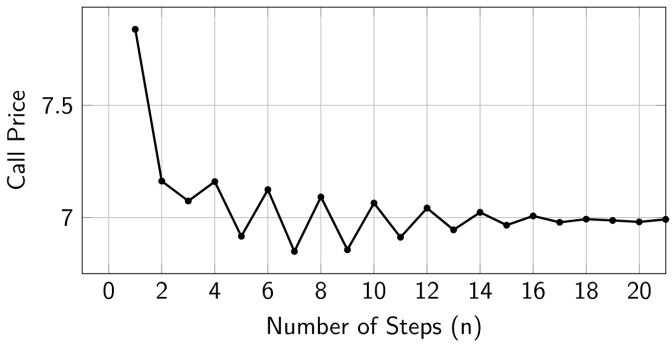
k	Payoff	Probability
3	34.678	$(p^*)^3 = .0953$
2	12.814	$3(p^*)^2(1-p^*) = .3400$
1	0	$3p^*(1-p^*)^2 = .4044$
0	0	$(1-p^*)^3 = .1603$

Thus $C_0 = e^{-.08}[.0953(34.678) + .3400(12.814)] \simeq 7.0739$. For general n,

$$C_0 = e^{-rT} \sum_{k=0}^{n} \binom{n}{k} (p^*)^k (1-p^*)^{n-k} \max\{S_0 u^k d^{n-k} - K, 0\}.$$

The binomial coefficient counts paths reaching the same recombining terminal state. Holding u, d, p^* fixed while increasing n would change the experiment rather than refine its time discretisation.

Number of steps	Call price
1	7.839
2	7.162
3	7.074
4	7.160
10	7.065
50	6.969
100	6.966
500	6.960
∞	6.961

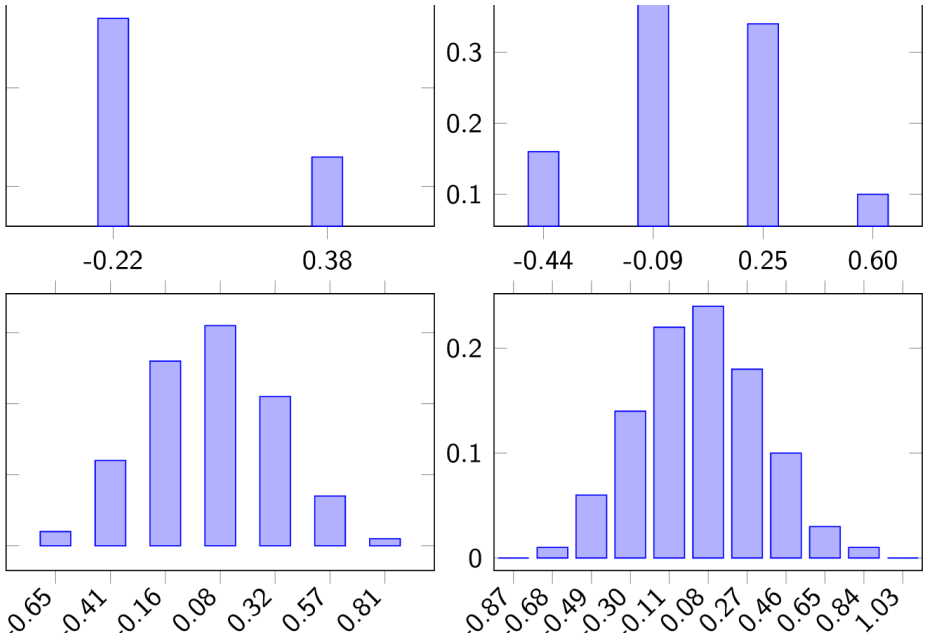


The oscillation around the limiting value is caused by the finite tree's terminal grid relative to the strike. Convergence is not monotone.

Problem 12.2, retained with its results. Use the binomial function for a call with $S_0 = 41, K = 40, \sigma = .3, r = .08, T = .25$ (three months), $\delta = 0$. Calculate $n = 8, 9, 10, 11, 12$ and describe the approximation's behaviour. The respective prices are $3.464, 3.361, 3.454, 3.348, 3.446$. They slowly approach the Black–Scholes price 3.399, with odd/even oscillations. These values were independently recomputed with the course's forward-tree convention.

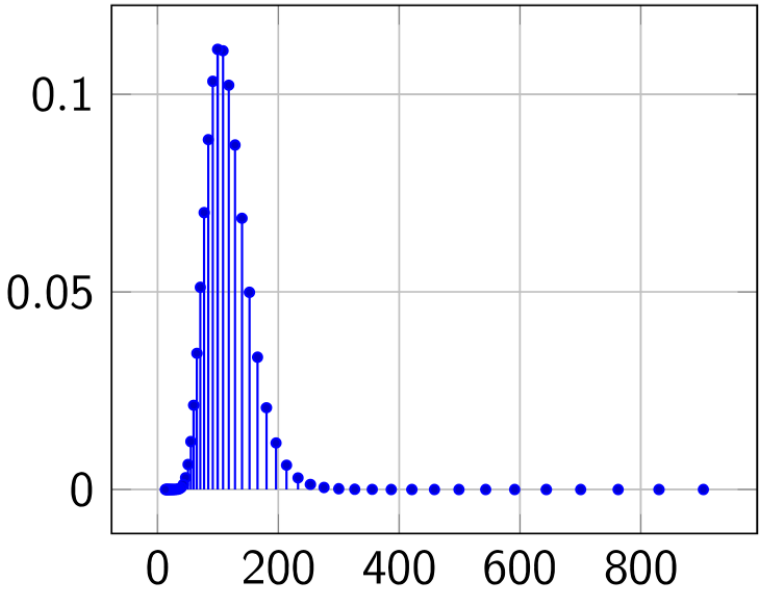
14.3 Normal log returns and lognormal stock prices

The lognormal distribution arises when *continuously compounded returns* are normally distributed. As the tree is refined, the distribution of accumulated log returns approaches a normal distribution. The four panels compare $n = 2, 3, 6, 10$; ten steps already give a recognisable bell-shaped log-return distribution.



Stock prices themselves are exponentials of log returns, so their limiting distribution is lognormal, with positive support and a right tail. A normal stock-price model and a normal log-return model are different assumptions.

Problem 11.13, retained. Let $S_0 = 100, \sigma = .30, r = .08, T = 1, \delta = 0$. Use the terminal binomial formula to calculate each terminal price and its probability, plot the year-1 risk-neutral stock distribution for $n = 50$, and find $\Pr(S_1 < 80)$ and $\Pr(S_1 > 120)$. With $u = 1.0450, d = .9600, p^* = .4894$, each state $S_0 u^k d^{50-k}$ has probability $\binom{50}{k}(p^*)^k(1-p^*)^{50-k}$. Sum the states satisfying the strict inequalities: $\Pr(S_1 < 80) = .2006$ and $\Pr(S_1 > 120) = .2829$.



These are pricing-measure probabilities, not historical forecasts. The unequal tails result from the distribution and the chosen thresholds, even though both thresholds are 20 currency units from the initial price.

14.4 Model assumptions and inputs

- Continuously compounded stock returns are normal and independent over disjoint time intervals; volatility is known and constant.
- Future dividends are known, either as cash amounts in an appropriate treatment or as a fixed yield. The displayed formulas below specifically use a continuous proportional dividend yield δ.
- The risk-free interest rate is known and constant.
- There are no transaction costs or taxes, short selling is costless, and borrowing is possible at the risk-free rate.

The six Black–Scholes inputs are current stock price S_0, strike K, volatility σ, continuously compounded risk-free rate r, maturity T, and dividend yield δ. The finite binomial model has one extra numerical input: number of periods n.

14.5 The limiting derivation: two binomial tails

An option is in the money when $S_0 u^k d^{n-k} > K$. Writing $h = T/n$, this means $k > \alpha$, where

$$\alpha = \frac{n}{2} - \frac{\log(S_0/K) + (r-\delta)T}{2\sigma\sqrt{T/n}}.$$

Separate the payoff into its stock and strike components:

$$\begin{aligned} C_0 &= e^{-rT} \sum_{k>\alpha} \binom{n}{k} (p^*)^k (1-p^*)^{n-k} (S_0 u^k d^{n-k} - K) \\ &= e^{-rT}(S_0 U_1 - K U_2), \\ U_2 &= \sum_{k>\alpha} \binom{n}{k} (p^*)^k (1-p^*)^{n-k}, \\ U_1 &= \sum_{k>\alpha} \binom{n}{k} (p^* u)^k [(1-p^*)d]^{n-k}. \end{aligned}$$

U_2 is the risk-neutral in-the-money probability. The binomial normal-tail approximation is $U_2 \simeq N((np^* - \alpha)/\sqrt{np^*(1-p^*)})$. Substitution of the time-step parameters gives the limit

$$U_2 \longrightarrow N\left(\frac{\log(S_0/K) + (r - \delta - \sigma^2/2)T}{\sigma\sqrt{T}}\right) = N(d_2).$$

For U_1, use the stock-weighted probability

$$q = \frac{p^* u}{p^* u + (1-p^*)d} = \frac{p^* u}{e^{(r-\delta)T/n}}, \qquad 1 - q = \frac{(1-p^*)d}{e^{(r-\delta)T/n}}.$$

Then

$$U_1 = e^{(r-\delta)T} \sum_{k>\alpha} \binom{n}{k} q^k (1-q)^{n-k} \longrightarrow e^{(r-\delta)T} N(d_1),$$

where

$$d_1 = \frac{\log(S_0/K) + (r - \delta + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

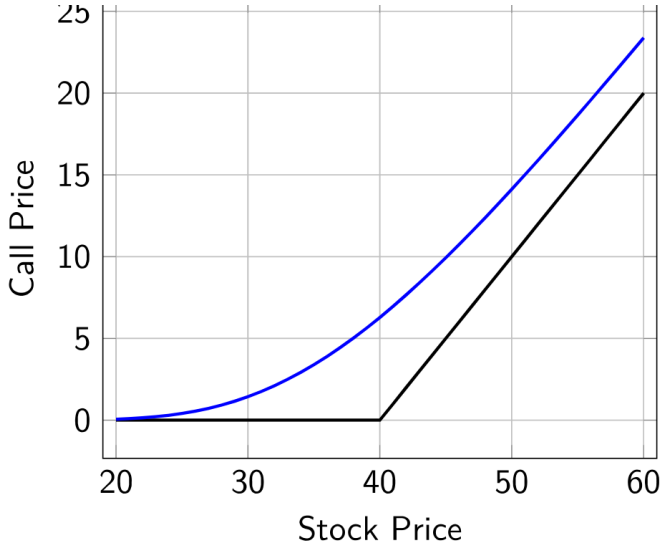
The stock-tail approximation is $U_1 \simeq e^{(r-\delta)T} N((nq - \alpha)/\sqrt{nq(1-q)})$. The factor $e^{(r-\delta)T}$ follows by factoring $[p^* u + (1-p^*)d]^n$. Combining both limiting tails gives

$$\boxed{C_0 = S_0 e^{-\delta T} N(d_1) - K e^{-rT} N(d_2).}$$

$N(x) = \Pr(Z \le x)$ for a standard normal Z. The risk-neutral exercise probability is $N(d_2)$; $N(d_1)$ comes from the stock-weighted tail and is not the same probability under the same measure.

14.6 European call, put, replication and beta

For the original inputs $41, 40, .3, .08, 1, 0$, $d_1 = .4998, d_2 = .1998$, $N(d_1) = .6911, N(d_2) = .5789$ and $C_0 = 6.961$. The stock-price plot compares the smooth pre-expiry call value with its terminal payoff.



The historical expected return on the stock does not enter the formula. The hedge is

$$C_0 = \Delta S_0 + B, \quad \Delta = e^{-\delta T} N(d_1), \quad B = -Ke^{-rT} N(d_2).$$

If the stock has beta β_S and the financing asset has zero beta, the option's beta is

$$\beta_{\text{option}} = \frac{\Delta S_0}{\Delta S_0 + B}\, \beta_S.$$

A larger stock beta raises the option's expected payoff and its required expected-return discount rate. In this idealised replication setting the effects offset: beta is irrelevant to the arbitrage price. A higher historical expected return alone does not justify substituting that return for r in the pricing formula.

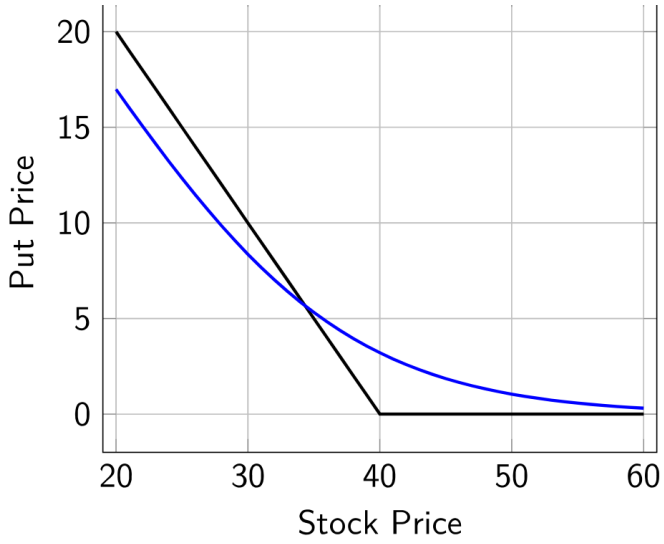
The European put is

$$\boxed{P_0 = -S_0 e^{-\delta T} N(-d_1) + Ke^{-rT} N(-d_2).}$$

Since $N(-x) = 1 - N(x)$,

$$P_0 = C_0 + Ke^{-rT} - S_0 e^{-\delta T}.$$

For the same 41 stock example, $P_0 = 2.886$; a 500-step tree gives approximately 2.885. Parity independently gives $6.961 + 40e^{-.08} - 41 = 2.886$.



Problem 12.20, retained. (a) Price a call with $S_0 = 100, K = 90, \sigma = 30\%, r = 8\%, \delta = 5\%, T = 1$. (b) Price a put with $S_0 = 90, K = 100, \sigma = 30\%, r = 5\%, \delta = 8\%, T = 1$. The supplied spreadsheet outputs are both 17.70; independent evaluation gives 17.698814 for each. Swapping S_0, K and r, δ relates the two formulas; the equality is not evidence that calls and puts with identical inputs generally cost the same.

14.7 Precursors and the replication breakthrough

Before organised options markets developed, many option-valuations concerned warrants and OTC claims. The historical introduction dates modern listed options to April 1973 and mentions a Genesis narrative as an early option analogy. It names Castelli (1877), Bachelier (1900), Sprenkle (1961), Ayres (1965), Boness (1964), Samuelson (1965), Baumol, Malkiel and Quandt (1966), and Chen (1970). The subsequent comparison also includes Samuelson and Merton (1969) and Thorp and Kassouf (1967). Their limitations show why selecting a suitable expected-return discount rate was difficult; the shared missing ingredient is a self-financing riskless hedge.

Bachelier. The printed historical expression, with the strike denoted by K, is

$$C(S,T) = S\Phi\left(\frac{S-K}{\sigma\sqrt{T}}\right) - K\Phi\left(\frac{S-K}{\sigma\sqrt{T}}\right) - \sigma\sqrt{T}\left(\frac{K-S}{\sigma\sqrt{T}}\right).$$

Typography/formula issue. The last term is printed without a normal-density symbol and with the sign shown above. It cannot be silently treated as a verified pricing identity: at $S = K$ the displayed expression gives zero despite positive volatility. The accompanying prose says that negative underlying and option prices are possible and that the time value of money is omitted; the claim about negative *option* prices does not hold for the expectation of a nonnegative call payoff.

Generated clarification. If one specifically assumes $S_T = S + \sigma\sqrt{T}Z$ with Z standard normal and no discounting, directly integrating $(S_T - K)^+$ gives

$$C = S\,\Phi\left(\frac{S-K}{\sigma\sqrt{T}}\right) - K\,\Phi\left(\frac{S-K}{\sigma\sqrt{T}}\right) + \sigma\sqrt{T}\,\phi\left(\frac{K-S}{\sigma\sqrt{T}}\right).$$

Here σ is a price-change volatility, Φ the standard normal CDF and ϕ its density. This generated expectation is nonnegative; it is not a claim that the ambiguous printed last term has been transcribed correctly by replacing it. Normal price levels can be negative. The volatility convention differs from Black–Scholes return volatility, and discounting has been omitted. The handwritten comment credits Bachelier's ideas as ahead of their time and names his *Théorie de la spéculation*.

Sprenkle. The lecture presents

$$C(S,T) = e^{\rho T} S\Phi(d_1) - (1-A)K\Phi(d_2), \quad d_1 = \frac{\log(e^{\rho T}S/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

ρ is expected share growth and A represents risk aversion. Riskless discounting is not included in this expression; the resulting premium is described as too large.

Boness and Samuelson. Boness writes $C = S\Phi(d_1) - Ke^{-\rho T}\Phi(d_2)$ with $d_1 = [\log(S/K) + (\rho + \sigma^2/2)T]/(\sigma\sqrt{T})$. This accounts for time value but still needs the stock's expected return. Samuelson distinguishes stock expected return ρ and the option's expected return α:

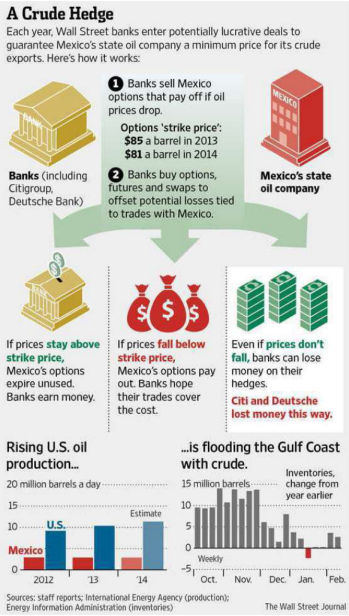
$$C = Se^{(\rho-\alpha)T}\Phi(d_1) - Ke^{-\alpha T}\Phi(d_2), \quad d_1 = \frac{\log(S/K) + (\rho + \sigma^2/2)T}{\sigma\sqrt{T}}.$$

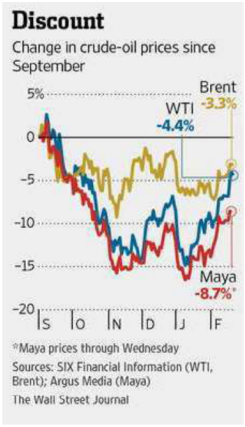
The printed historical expression places an additional minus sign before $(\rho - \alpha)T$. The displayed version here is a **generated algebraic clarification**: expected stock growth $e^{\rho T}$ discounted at $e^{-\alpha T}$ gives $e^{(\rho-\alpha)T}$. The original printed sign is retained in the faithful Notes and flagged rather than silently accepted as a verified identity. The two expected-return inputs remain unknown. Samuelson and Merton derive an option return as a function of stock price using a representative investor's utility, with many conditional assumptions. Thorp and Kassouf approximate warrant prices using linear regression for an exponent; the lecture notes that the hedged position's no-arbitrage expected return should be the riskless rate and that their approach lacked a direct volatility measure.

Black, Scholes and Merton. The breakthrough identifies the continuously rebalanced stock-and-bond replicating portfolio, whose riskless hedged return must match the riskless rate. Merton emphasises the *self-financing* condition and the arbitrage argument. The resulting value no longer depends on the stock's expected return. The lecture writes the familiar no-dividend form $C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2)$ with $d_1 = [\log(S/K) + (r + \sigma^2/2)T]/(\sigma\sqrt{T})$. The historical commentary distinguishes the original two-security hedge from Merton's three-security portfolio involving the option, stock and bank account, $w_C C + w_S S + w_B B$, with self-financing explicitly imposed. It credits Merton's help in the original paper's footnote and a direct comment from Robert Merton to the authors of this historical discussion. Scholes and Merton later received the economics prize; the printed year 1996 is a typo: the official Nobel record gives 1997. Black had died before the award.

14.8 Historical oil hedges: basis risk and feedback

The supplied 2014 news examples discuss bank hedges of Mexican oil-company put protection, crude-price spreads, and the difference between the client's exposure and the futures used to hedge it. The two visuals preserve the contract/hedge chain and the Brent–WTI discount comparison.





The Reuters example describes producers buying put protection with strikes around \$75–85. As oil prices fell toward those strikes, banks sold futures to reduce the changing exposure of their written puts. Traders cited this delta-hedging flow as one factor behind a rapid price fall. The teaching points are the sign and changing size of the hedge, basis risk between different oils, and potential feedback when many hedgers trade together. These historical observations do not establish that hedge trades were the sole cause of the price move.

15 A measure-change derivation of Black–Scholes

15.1 Historical dynamics and the discounted stock

On a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0\le t\le T}, \mathbb{P})$ carrying a standard Brownian motion $W^{\mathbb{P}}$, suppose the non-dividend-paying stock satisfies

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t^{\mathbb{P}}, \qquad \sigma > 0.$$

For constant risk-free rate r, the discount factor is $\beta(t) = e^{-rt}$, so $d\beta(t) = -r\beta(t)dt$. Define $Y_t = \beta(t)S_t$, the stock price in discounted units. The product rule gives

$$\begin{aligned} dY_t &= d(\beta_t S_t) = \beta_t\,dS_t + S_t\,d\beta_t + d[\beta, S]_t \\ &= \beta_t(\mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}) - r\beta_t S_t dt \\ &= (\mu - r)Y_t dt + \sigma Y_t dW_t^{\mathbb{P}}. \end{aligned}$$

The quadratic-covariation term vanishes because β has finite variation. Under $\mathbb{P}$, discounted stock still has drift $\mu - r$.

15.2 Eliminating the drift changes the measure

Set $W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \int_0^t \gamma_s ds$. Under the historical measure this shifted process is generally *not* a martingale; adding a drift does not by itself create another Brownian motion. Substitution gives

$$\begin{aligned} Y_t &= Y_0 + \int_0^t (\mu - r)Y_s ds + \int_0^t \sigma Y_s dW_s^{\mathbb{P}} \\ &= Y_0 + \int_0^t (\mu - r - \sigma\gamma_s)Y_s ds + \int_0^t \sigma Y_s dW_s^{\mathbb{Q}}. \end{aligned}$$

Choosing $\gamma_s = (\mu - r)/\sigma$ removes the drift. We must also find an equivalent measure under which $W^{\mathbb{Q}}$ is a standard Brownian motion. The measure change makes the stochastic-integral representation a martingale representation under the appropriate integrability conditions.

15.3 Radon–Nikodym and Cameron–Martin–Girsanov

For equivalent measures $\mathbb{P}, \mathbb{Q}$, the Radon–Nikodym theorem provides a positive random variable $Z = d\mathbb{Q}/d\mathbb{P}$ with

$$\mathbb{Q}(A) = \mathbb{E}^{\mathbb{P}}[Z\mathbf{1}_A], \qquad \mathbb{E}^{\mathbb{P}}[Z] = 1.$$

Equivalent measures agree on which events have probability zero; they need not assign the same positive probability to other events.

Let γ be predictable and satisfy the sufficient Novikov condition

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2}\int_0^T \gamma_t^2 dt\right)\right] < \infty.$$

Then the exponential density

$$Z_T = \exp\left[-\int_0^T \gamma_t \, dW_t^{\mathbb{P}} - \frac{1}{2}\int_0^T \gamma_t^2 dt\right]$$

defines a measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ on $\mathcal{F}_T$, and

$$\widetilde{W}_t = W_t^{\mathbb{P}} + \int_0^t \gamma_s ds$$

is a Brownian motion under $(\mathcal{F}_t, \mathbb{Q})$. The sign in the density and the sign of the Brownian drift shift must match this convention. For constant $\gamma = (\mu - r)/\sigma$ over a finite horizon, Novikov is satisfied. Consequently

$$dY_t = \sigma Y_t \, dW_t^{\mathbb{Q}}, \qquad dS_t = rS_t \, dt + \sigma S_t \, dW_t^{\mathbb{Q}}.$$

The stock's risk-neutral drift is r in this *no-dividend* derivation. With a continuous dividend yield the ex-dividend drift is $r - \delta$ instead.

15.4 Computing the call expectation explicitly

Under $\mathbb{Q}$, the solution is

$$S_T = S_0 \exp\left[(r - \tfrac{1}{2}\sigma^2)T + \sigma W_T^{\mathbb{Q}}\right], \qquad W_T^{\mathbb{Q}} \sim \mathcal{N}(0, T).$$

Therefore

$$\begin{aligned} C_0 &= \mathbb{E}^{\mathbb{Q}}[e^{-rT}(S_T - K)^+] \\ &= \mathbb{E}^{\mathbb{Q}}\left[\left(S_0 e^{-\sigma^2 T/2 + \sigma W_T^{\mathbb{Q}}} - Ke^{-rT}\right)^+\right] \\ &= \int_{-\infty}^{\infty} \left(S_0 e^{-\sigma^2 T/2 + \sigma z} - Ke^{-rT}\right)^+ f_T(z)\, dz, \end{aligned}$$

where $f_T(z) = (2\pi T)^{-1/2} e^{-z^2/(2T)}$. The positive-payoff region satisfies

$$\begin{aligned} S_0 e^{-\sigma^2 T/2 + \sigma z} > Ke^{-rT} &\iff \sigma z > \log K - \log S_0 - rT + \tfrac{1}{2}\sigma^2 T \\ &\iff z > -d_2\sqrt{T}, \end{aligned}$$

with $d_2 = [\log(S_0/K) + (r - \sigma^2/2)T]/(\sigma\sqrt{T})$. Thus

$$C_0 = S_0 \int_{-d_2\sqrt{T}}^{\infty} \frac{1}{\sqrt{2\pi T}} \exp\left[-\frac{z^2 - 2T\sigma z + \sigma^2 T^2}{2T}\right] dz$$
$$- Ke^{-rT} \int_{-d_2\sqrt{T}}^{\infty} \frac{1}{\sqrt{2\pi T}} e^{-z^2/(2T)} dz.$$

Complete the square in the first integrand: $z^2 - 2T\sigma z + \sigma^2 T^2 = (z - \sigma T)^2$. Set $u = (z - \sigma T)/\sqrt{T}$ in the first integral and $v = z/\sqrt{T}$ in the second:

$$\begin{aligned} C_0 &= S_0 \int_{-d_2 - \sigma\sqrt{T}}^{\infty} \phi(u)du - Ke^{-rT} \int_{-d_2}^{\infty} \phi(v)dv \\ &= S_0[1 - N(-d_2 - \sigma\sqrt{T})] - Ke^{-rT}[1 - N(-d_2)] \\ &= S_0 N(d_1) - Ke^{-rT} N(d_2), \qquad d_1 = d_2 + \sigma\sqrt{T}. \end{aligned}$$

This retains the density theorem, integrability condition, drift-removal calculation, payoff threshold and both changes of integration variable. It also explains why the historical drift μ disappears from the option price.

16 Greeks, market-making and dynamic hedging

16.1 An unhedged dealer has an arbitrary position

A market-maker sells to buyers and buys from sellers. The position generated by customer orders can have uncontrolled risk, so the dealer trades to hedge it. Consider selling 100 European calls with $S_0 = 41, K = 40, \sigma = .3, r = .08, \delta = 0$ and 91 days to expiry, initially $\tau = 91/365 = .249$. With no hedge, the stock path and daily option-position changes are:

Day	Stock	Marked value of 100 short calls	Daily profit
0	41	-339.47	—
1	42.5	-441.04	-101.57
2	39.5	-246.31	194.73
3	37	-129.49	116.82
4	40	-271.04	-141.55
5	40	-269.27	1.77

Profit is measured by marking the position to market, as if it were liquidated each day. It is not the terminal exercise payoff. A price increase hurts the short-call seller and a fall helps; the sign follows the short position.

16.2 Delta neutrality and daily rebalancing

Initially one call has delta about .645, so a one-dollar stock move changes its value by about .645. The short 100-call position is offset by approximately 64.54 shares. The delta changes as the stock and remaining life change; the hedge is recalculated and rebalanced each day.

Day	Stock	Call delta	Shares	Call profit	Share profit	Total
0	41	.645	64.54	—	—	—
1	42.5	.730	73.03	-101.57	96.81	-4.77
2	39.5	.548	54.81	194.73	-219.10	-24.38
3	37	.373	37.27	116.82	-137.02	-20.20
4	40	.581	58.06	-141.55	111.82	-29.73
5	40	.580	58.01	1.77	0.00	1.77

The share profit over a day uses the *previous day's* hedge: for example $64.54(42.5 - 41) = 96.81$. Rebalancing at day 1 then sets the shares held over the next day. The displayed totals use underlying unrounded values. Interest expenses are ignored in this illustrative table, as in the supplied example.

Delta hedging offsets the first-order response to *small* stock moves. For larger changes, the change in delta matters. The remaining losses are consistent with a short-gamma position and discrete rebalancing; the table is not a proof of exact continuous-time replication.

16.3 Definitions, signs and reporting units

Greeks are partial derivatives of an option price: vary one input while holding the others fixed. They quantify exposures of market-makers, index-option portfolios and individual investors. Let $\tau = T - t$, and denote a price by V.

Input	Greek	Meaning and convention
S	$\Delta = \partial_S V$	Currency change for a unit stock-price increase.
S	$\Gamma = \partial_{SS} V$	Change in delta for a unit stock-price increase.
S	$\Omega = (S/V)\Delta$	Percentage price change per percentage stock-price change.
σ	Vega $= \partial_\sigma V$	Divide by 100 to report a one percentage point volatility increase.
t	$\Theta = \partial_t V$	Calendar-time derivative with maturity fixed; divide by 365 for per-day theta.
r	$\rho = \partial_r V$	Divide by 100 for a percentage point, by 10,000 for a basis point.
δ	$\Psi = \partial_\delta V$	Divide by 100 for a percentage point dividend-yield increase.

The lecture's table uses daily theta and per-percentage-point vega, rho and psi. Check a terminal or spreadsheet's units before multiplying a Greek by an input change; a decimal-rate derivative and a displayed percentage-point sensitivity differ by a factor of 100.

For a standard European call and put with $\sigma > 0, \tau > 0$:

Input change	Call response	Put response
S rises	Increases, $\Delta_C > 0$	Decreases, $\Delta_P < 0$
S changes	$\Gamma_C > 0$	$\Gamma_P > 0$
S percentage rises	$\Omega_C \geq 1$ for usual r, δ inputs	$\Omega_P \leq 0$
σ rises	Increases, Vega > 0	Increases, Vega > 0
t rises (τ falls)	Generally decreases	Sign can be either
r rises	Increases, $\rho_C > 0$	Decreases, $\rho_P < 0$
δ rises	Decreases, $\Psi_C < 0$	Increases, $\Psi_P > 0$

Higher volatility raises the value of the option's convex payoff. A stock-price increase makes a call more likely to finish in the money and a put less likely. Higher r reduces the discounted strike paid by the call holder and received by the put holder. A call owner receives no stock dividends before exercise, so increasing the dividend yield reduces call value and raises put value. Time decay is *not universal*: a deep-in-the-money call on a stock with a high dividend yield can have positive theta, and a deep-in-the-money put can also gain from the passage of time. The lecture's signs are qualitative for ordinary inputs, rather than a claim that every option always loses value as calendar time advances.

16.4 Analytic formulas

Let

$$d_1 = \frac{\log(S/K) + (r - \delta + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}, \quad \phi(x) = N'(x).$$

$$\Delta_C = e^{-\delta\tau}N(d_1), \qquad \Delta_P = -e^{-\delta\tau}N(-d_1),$$
$$\Gamma_C = \frac{e^{-\delta\tau}\phi(d_1)}{S\sigma\sqrt{\tau}}, \qquad \Gamma_P = \Gamma_C,$$
$$\Omega_C = \frac{S\Delta_C}{C}, \qquad \Omega_P = \frac{S\Delta_P}{P},$$
$$\text{Vega}_C = Se^{-\delta\tau}\phi(d_1)\sqrt{\tau}, \qquad \text{Vega}_P = \text{Vega}_C,$$
$$\rho_C = K\tau e^{-r\tau}N(d_2), \qquad \rho_P = -K\tau e^{-r\tau}N(-d_2),$$
$$\Psi_C = -\tau Se^{-\delta\tau}N(d_1), \qquad \Psi_P = \tau Se^{-\delta\tau}N(-d_1).$$

The theta expressions, *before* division by 365, are

$$\Theta_C = \delta Se^{-\delta\tau}N(d_1) - rKe^{-r\tau}N(d_2) - \frac{Se^{-\delta\tau}\phi(d_1)\sigma}{2\sqrt{\tau}},$$
$$\Theta_P = \Theta_C + rKe^{-r\tau} - \delta Se^{-\delta\tau}.$$

The last equality follows by differentiating put–call parity with respect to calendar time while keeping T fixed. Using time *to maturity* instead would reverse the derivative's sign. Elasticity requires a nonzero option price. The lecture writes the diffusion term in theta as $-Ke^{-r\tau}\phi(d_2)\sigma/(2\sqrt{\tau})$. This is the same term because $Se^{-\delta\tau}\phi(d_1) = Ke^{-r\tau}\phi(d_2)$; both forms retain the same dividend and rate assumptions.

16.5 The complete numerical Greek example

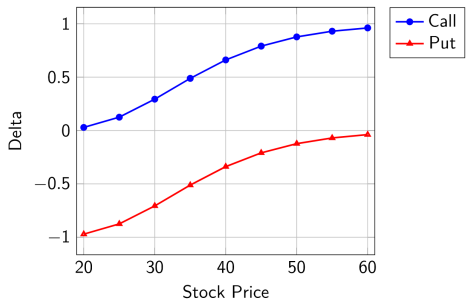
For $S = 41, K = 40, \sigma = .3, r = .08, \delta = 0, \tau = 1$, the displayed example reports:

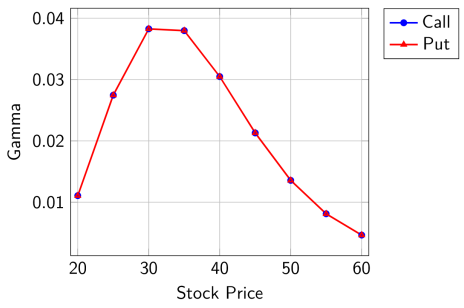
Greek and display unit	Call	Put
Delta	.691	-.309
Gamma	.029	.029
Elasticity	4.071	-4.389
Vega, per percentage point	.144	.144
Theta, per day	-.011	-.003
Rho, per percentage point	.214	-.156
Psi, per percentage point	-.283	.127

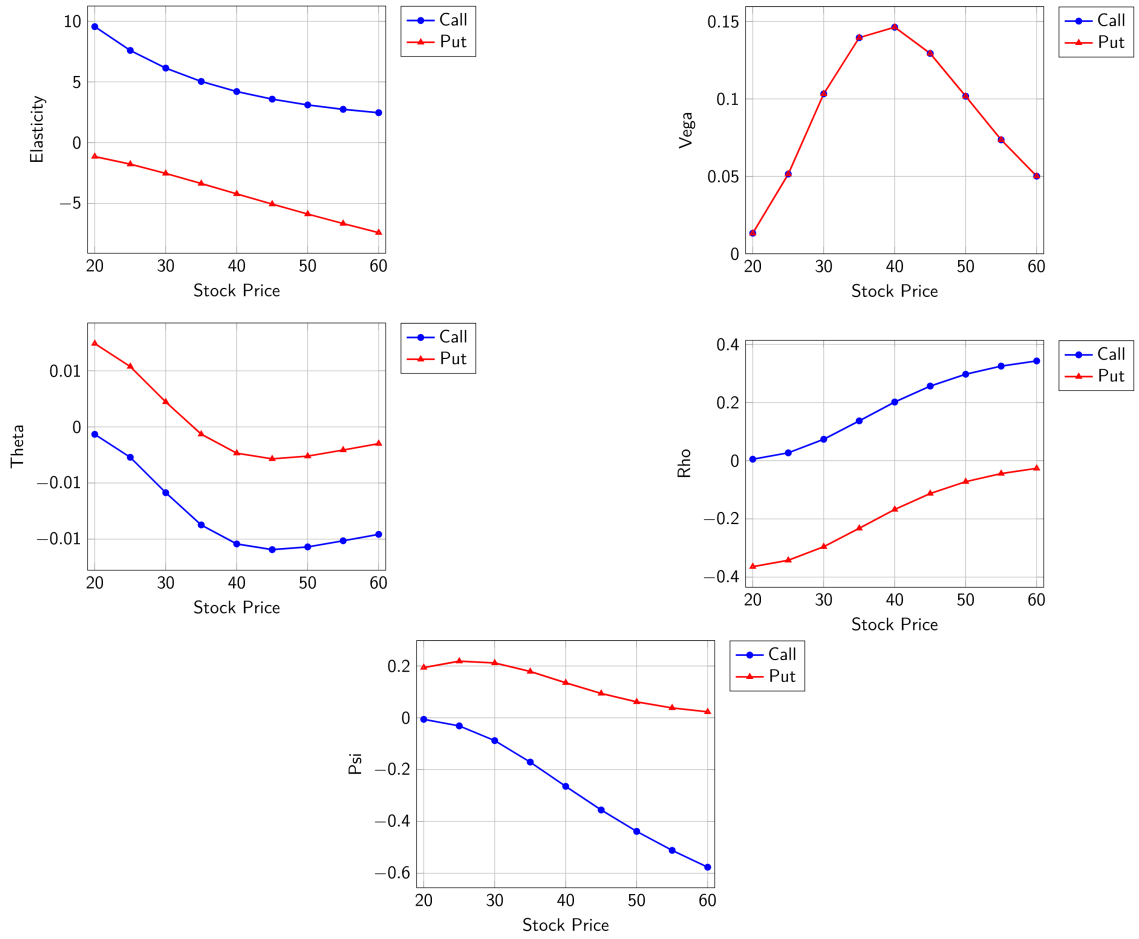
These values preserve the lecture's units and rounding. In particular, Δ is not gamma, and vega is a derivative with respect to *volatility*, not variance.

16.6 How the Greek curves change with moneyness

The seven original plots below compare call and put sensitivities as stock price runs from 20 to 60, with the other inputs fixed. Delta rises with S; gamma and vega are largest near the central region; put elasticity is negative; theta can change sign; rho and psi have opposite call/put signs.







The figures' scales are the display conventions explained above. The Bloomberg scenario illustrations distinguish unhedged, delta-hedged and delta–gamma-hedged exposures; the formulas and tables make their economic content explicit here. No additional terminal screenshot is needed to calculate the hedge.

16.7 Gamma neutrality requires another option

The gamma of a stock holding is zero because its delta is constant at one. Thus shares alone cannot neutralise option gamma. Return to the 91-day, 40-strike written call and choose a second, 120-day call with strike 45. The gamma ratio is

$$\frac{\Gamma_{40,91}}{\Gamma_{45,120}} \simeq 1.1213.$$

Buy approximately 1.1213 of the 45-strike calls for each 40-strike call sold. This ratio is obtained from unrounded gammas; the printed rounded components .0606/.0540 do not reproduce its last digits exactly.

	40-strike call	45-strike call	Combined option position
Price	3.395	1.707	-1.481
Delta	.645	.381	-.218
Gamma	.061	.054	0
Theta per day	-.018	-.014	.002

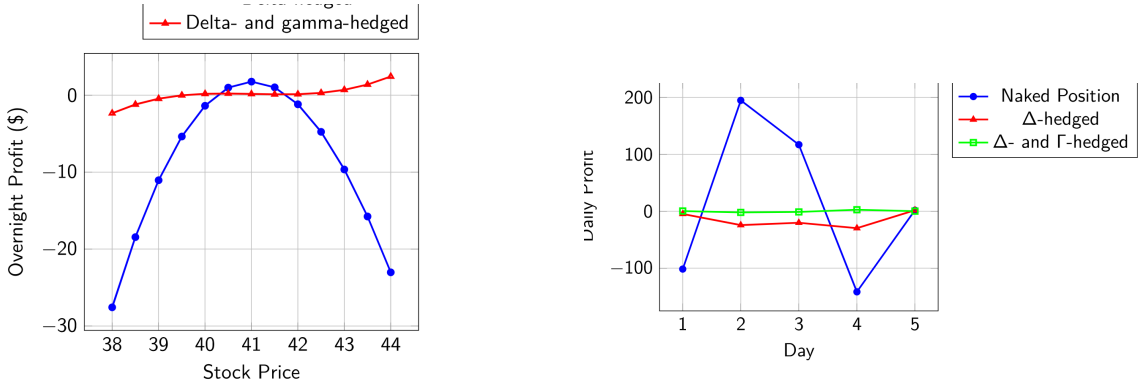
The combined position is one short 40-strike call and 1.1213 long 45-strike calls. Its residual delta is about $-.218$, so add .218 shares per original call, or 21.8 shares for 100 calls, to make both delta

and gamma zero at the current inputs.

$$\Gamma_{\text{portfolio}} = -\Gamma_{40} + w\Gamma_{45} = 0, \qquad \Delta_{\text{portfolio}} = -\Delta_{40} + w\Delta_{45} + x = 0.$$

Local Taylor expansion explains the improvement:

$$\Delta V \simeq \Delta_{\text{portfolio}}\Delta S + \tfrac{1}{2}\Gamma_{\text{portfolio}}(\Delta S)^2 + \Theta_{\text{portfolio}}\Delta t.$$

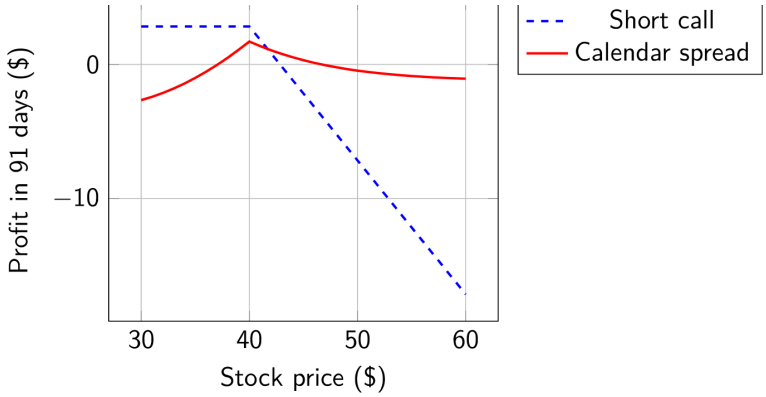


The first graph compares one-day profit as S changes; the second compares profits along the five-day stock path. The delta–gamma hedge is less sensitive to large moves than the delta-only hedge. The final three-curve comparison contains the entire unhedged and delta-only paths, so their earlier progressive plots are combined here. Its zero sensitivities are *local*; gamma neutrality does not eliminate every finite-move, volatility, funding or model risk. Further rebalancing is still necessary as inputs change.

16.8 Calendar spreads and unequal time decay

To protect against a stock-price increase when selling a call, one may buy another call with the same strike but a later expiry. The short near-expiry call generally loses time value faster than the longer-lived call; with no stock move this difference can favour the spread.

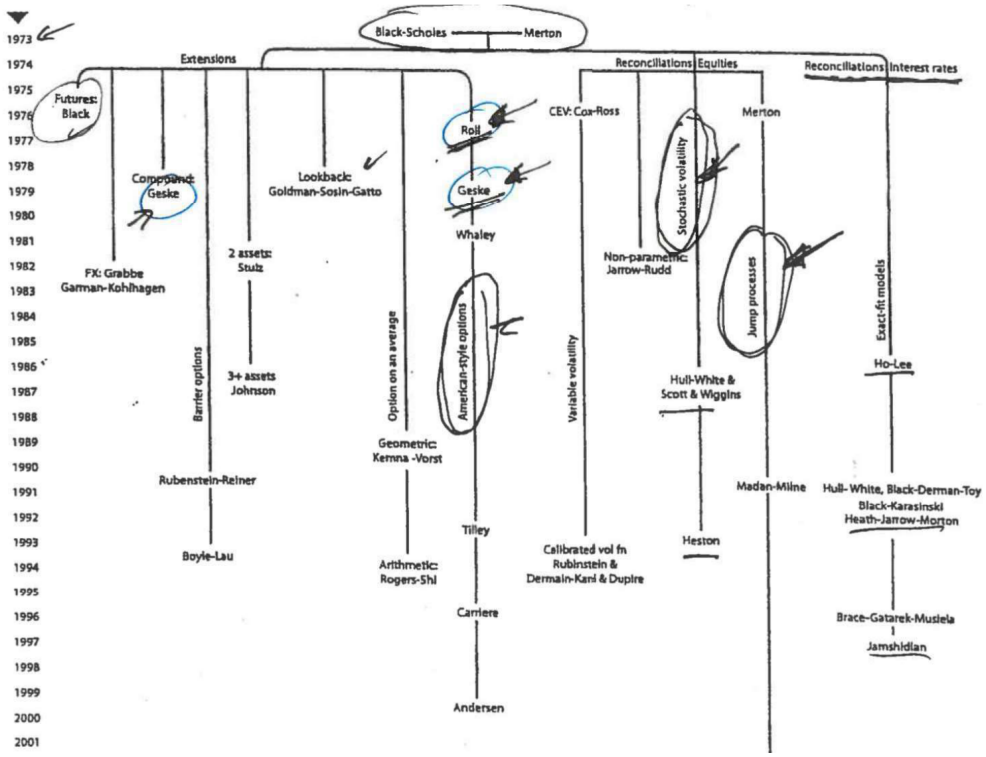
The supplied example sells a 40-strike call with 91 days left and buys a 40-strike one-year call. At $S = 40, r = 8\%, \sigma = 30\%, \delta = 0$, premiums are 2.78 and 6.28, so the initial net debit is about 3.50. Daily thetas are $-.0173$ and $-.0104$ respectively. The spread theta is $+.0069$ per share per day at these inputs: if the stock does not move over one day, the short option loses more value than the long option.



The graph evaluates profit over 91 days as a function of the stock price then. The short call expires at that horizon, but the long call still has remaining life and must be *valued*, not replaced by its final payoff. The "profitable if the stock does not move" explanation concerns time decay at the illustrated inputs; it does not make every calendar spread a guaranteed-profit position.

16.9 Developments after Black–Scholes

The historical diagram places Black–Scholes and Merton alongside later work on stochastic volatility, interest-rate models, jump models and more complex instruments. It is a map of modelling extensions rather than an extra pricing formula.



The original model's constant volatility, frictionless continuous hedge and simple rate/dividend assumptions are the reference point. Extensions change those assumptions and therefore change the hedge and valuation problem.

17 Discrete hedging, transaction costs and the charged premium

The Greek definitions and the local neutrality equations were developed in the preceding hedging chapter. This section adds the cash-account dynamics which turn those exposures into a complete trading strategy. It asks what changes when rebalancing is discrete and transactions are costly. The customer pays premium P for one call with strike K, maturity T; the dealer is short that call. Divide $[0, T]$ into n steps of length $h = T/n$, with stock observations S_j, hedge positions Δ_j and cash balance b_j immediately after trading at observation j.

Initially the premium finances the stock hedge and leaves

$$b_0 = P - \Delta_0 S_0.$$

At each intermediate date $j = 1, \ldots, n-1$, the previous cash balance earns bank interest and the dealer buys $\Delta_j - \Delta_{j-1}$ shares. A negative purchase is a sale:

$$b_j = e^{rh} b_{j-1} - (\Delta_j - \Delta_{j-1}) S_j.$$

At maturity, sell the last stock position and pay the customer's option payoff:

$$\boxed{b_n = e^{rh} b_{n-1} + \Delta_{n-1} S_n - (S_n - K)^+.}$$

The final b_n is the profit and loss of the complete funded strategy. It is not just the stock-and-option mark-to-market change. The hedge over $(t_{j-1}, t_j]$ uses the previous position; updating it after observing S_j cannot retroactively hedge the price move which has just occurred.

To study its distribution, simulate independent paths of $dS = \mu S dt + \sigma S dW^{\mathbb{P}}$ and apply the recurrence along every path. The physical drift is appropriate when studying realized risk, while the hedge deltas can still come from the specified Black–Scholes pricing model. Record means, standard deviations and tail quantiles of the final balances. Under the ideal continuously traded frictionless model, charging the replication price makes the hedging error converge to zero as the grid is refined; it is stronger than merely saying expected profit is zero. Finite trading dates leave residual risk, and the limiting statement does not justify ignoring transaction costs or model misspecification.

17.1 A quantile-based charge and its sign

Hold the simulated stock paths and hedge rule fixed and first set $P = 0$. Since the only effect of a charged premium is its accumulation in the cash account,

$$b_n(P) = b_n(0) + e^{rT}P.$$

Let $q_{.05}$ be the fifth percentile of $b_n(0)$. Translating every outcome by a constant translates the percentile by the same constant. Thus a charge which brings that percentile to zero is

$$q_{.05} + e^{rT}P = 0, \qquad \boxed{P = -e^{-rT}q_{.05}.}$$

The minus sign is essential when a negative profit quantile is used. A formula $P = e^{-rT}q_{.05}$ would charge a negative amount for a usual negative loss-tail profit quantile and move the balance in the wrong direction. If VaR is instead defined as the *positive loss amount* $\mathrm{VaR}_{.95} = -q_{.05}$, the same equation reads $P = e^{-rT}\mathrm{VaR}_{.95}$. For a continuous simulated distribution this puts approximately 5% of outcomes below zero; with atoms or estimated empirical quantiles, the exact strict-loss frequency depends on the quantile convention and sampling error. This risk-based charge is a management rule, not a unique arbitrage price for discrete hedging risk.

17.2 Proportional transaction costs

If each stock transaction costs a fraction ϵ of its absolute traded value,

$$c_0 = \epsilon|\Delta_0|S_0, \qquad c_j = \epsilon|\Delta_j - \Delta_{j-1}|S_j,$$

the balances become $b_0 = P - \Delta_0 S_0 - c_0$ and

$$b_j = e^{rh}b_{j-1} - (\Delta_j - \Delta_{j-1})S_j - c_j.$$

If final liquidation is charged too, the last recurrence also subtracts $\epsilon|\Delta_{n-1}|S_n$. The supplied simulator omits terminal liquidation costs; its reported balances should be interpreted with that convention. More trading reduces some discretization error but accumulates more cost. The effect on terminal risk and the low profit quantile therefore need not be monotone in trading frequency. A finer grid is not automatically an economically superior strategy once costs are included.

17.3 Funding the additional gamma-hedging option

For one written option, let q_j^2 be the quantity of a second call with strike $\widetilde{K} \neq K$ and maturity $\widetilde{T} \geq T$, and q_j^S the stock quantity. The local neutrality conditions already established are

$$-\Gamma_j^1 + q_j^2\Gamma_j^2 = 0, \qquad -\Delta_j^1 + q_j^2\Delta_j^2 + q_j^S = 0.$$

When $\Gamma_j^2 \neq 0$ this gives $q_j^2 = \Gamma_j^1/\Gamma_j^2$ and $q_j^S = \Delta_j^1 - q_j^2\Delta_j^2$. Setting the second-option holding to zero instead gives ordinary delta hedging. A very small denominator can demand an impractical

option position, especially near expiry or for a far out-of-the-money option. The implementation clips the ratio to $[-100, 100]$; once clipped, exact gamma neutrality is generally lost and should not still be claimed.

Let C_j^2 be the second option's marked price. With costs on both traded instruments, initialization and rebalancing are

$$\begin{aligned} b_0 &= P - q_0^S S_0 - q_0^2 C_0^2 - \epsilon(|q_0^S| S_0 + |q_0^2| C_0^2), \\ b_j &= e^{rh} b_{j-1} - (q_j^S - q_{j-1}^S) S_j - (q_j^2 - q_{j-1}^2) C_j^2 \\ &\quad - \epsilon \left(|q_j^S - q_{j-1}^S| S_j + |q_j^2 - q_{j-1}^2| C_j^2 \right). \end{aligned}$$

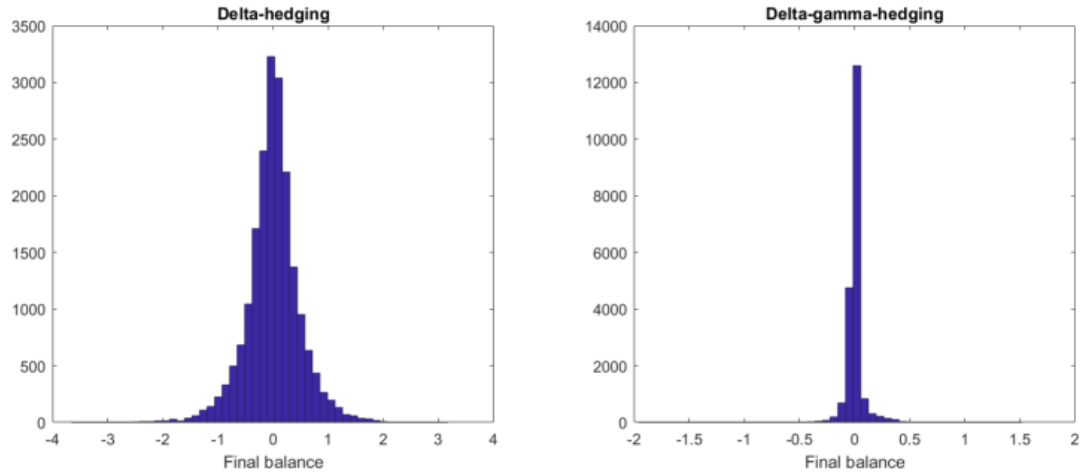
At T, liquidate the stock, sell the remaining second option at its value for remaining life $\widetilde{T} - T$, and pay the first call payoff. If $\widetilde{T} > T$, selling the second option at its *payoff* would be wrong: it still has time value. If maturities coincide, a terminal-payoff branch is required; a positive-time Black–Scholes routine cannot be evaluated at zero time by dividing by $\sqrt{0}$.

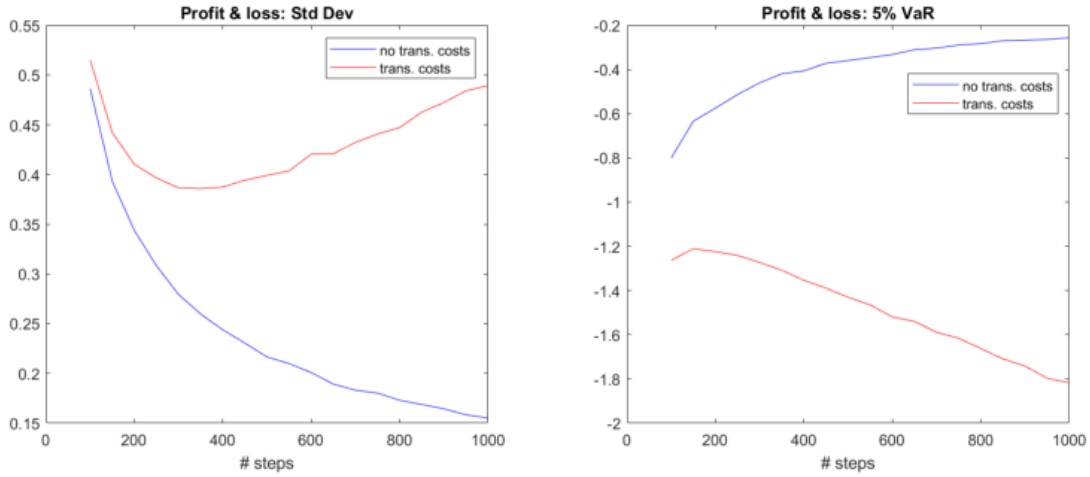
17.4 The supplied experiment and its interpretation

The illustrative inputs are $S_0 = 100$, $K = 110$, $T = .5$, $\sigma = .2$, physical drift $\mu = .05$, bank rate $r = .03$, second strike 115 and second maturity 1. The customer pays the Black–Scholes call price. With 20,000 paths and 100 steps, the reported output is

Hedge	Mean balance	Standard deviation	Fifth profit percentile
Delta	.0038	.4817	-.7863
Delta–gamma	.0003	.0765	-.0845

Both means are close to zero in the original simulation, while adding the second option substantially reduces dispersion in that run. They are original stochastic outputs, not deterministic values every new seed must reproduce. Their fifth-percentile column is a signed profit quantile even though the program labels it VaR.





The first comparison shows the narrower delta–gamma balance distribution. The second varies the number of steps from 100 to 1000 in increments of 50 and compares no costs with $\epsilon = .001$ costs. Its vertical axes are the balance standard deviation and the signed fifth profit percentile. The two diagnostics must be read together: lower standard deviation does not imply a less negative profit tail if cumulative costs shift the whole distribution down. Part II preserves the supplied vectorized Matlab implementation and the test script without printing a second copy of this theoretical account.

18 Measuring volatility, implied quotes and variance exposure

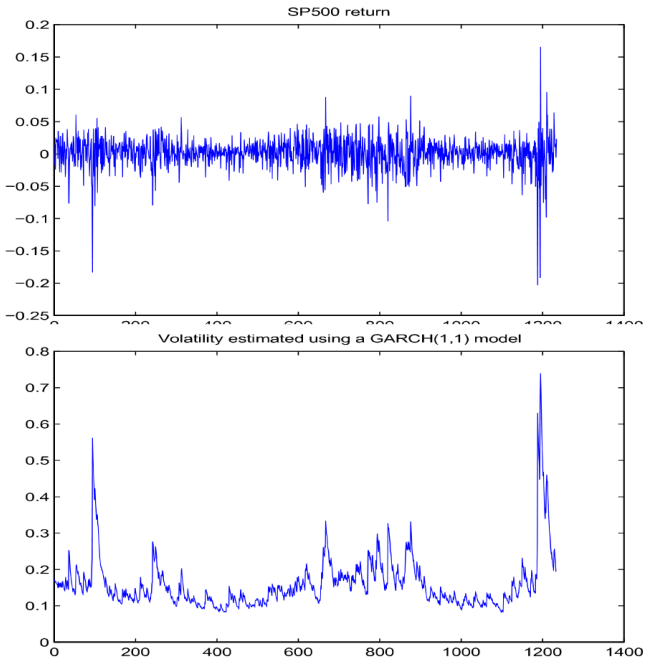
Volatility enters return forecasting, derivative pricing, the asset-allocation trade-off between return and risk, and portfolio risk management. A single number estimated over a long history and the conditional risk faced today answer different questions.

18.1 Unconditional dispersion and conditional volatility

For T observed log returns r_t with sample mean $\bar{r}$, the course uses

$$\widehat{\sigma} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2}.$$

This is the empirical root-mean-square deviation with divisor T; an unbiased sample-variance convention uses $T-1$ instead. The observation frequency must be stated before annualizing. The estimate summarizes unconditional dispersion over the whole window, whereas a time-varying σ_t conditions on the information available at date t.



The historical weekly S&P 500 return series in the upper panel shows clusters of large absolute movements separated by quieter periods; it integrates the same return illustration used to introduce clustering. The lower panel shows volatility estimated by a GARCH(1,1) model on the corresponding observation index. A useful added explanation is $h_t = \omega + \alpha\varepsilon_{t-1}^2 + \beta h_{t-1}$: recent squared shocks and past conditional variance both affect current conditional variance. With $\omega > 0$, $\alpha, \beta \geq 0$ and $\alpha + \beta < 1$, its stationary variance level is $\omega/(1-\alpha-\beta)$. This explains the mechanism shown; it is not a supplied parameter estimate. The plots do not justify substituting one unconditional history estimate for every current option's volatility.

18.2 Implied volatility is an inversion of a price

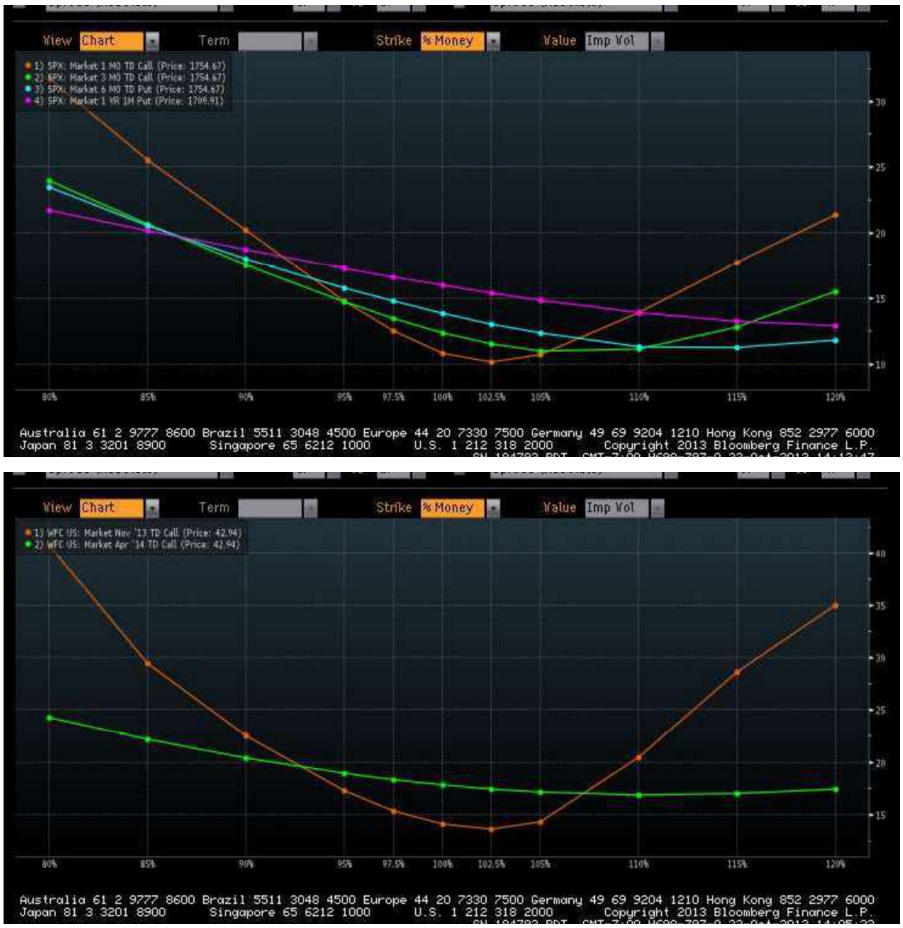
Volatility is not directly observed as a quoted stock price is. A historical estimate may be informative, but past volatility need not equal future volatility. Given an option quote, implied volatility is the σ which reproduces that quote in a specified pricing formula. It describes the quote through a model coordinate, rather than independently establishing that the quote is correct.

The example has $S = 100$, $K = 90$, $r = .08$, dividend yield $\delta = .05$, $T = 1$, and observed call price 18.25. Solve

$$100e^{-.05}N(d_1) - 90e^{-.08}N(d_2) = 18.25, \quad d_1 = \frac{\log(100/90) + (.08 - .05 + \sigma^2/2)}{\sigma}, \quad d_2 = d_1 - \sigma.$$

For a positive-time call with positive vega, the left-hand side increases with volatility; bracketed root finding gives approximately $\sigma = .3173$, or 31.73%. Substitution recovers the quoted price to its rounding. The spreadsheet illustration performs that same inversion, so its formula and inputs are integrated here instead of repeating a screenshot of a calculation already typeset.

Quotes at different strikes and maturities exhibit a systematic skew or smile. It is not a distinction between the names "put" and "call": at a fixed strike and maturity, put–call parity under consistent conventions makes the corresponding implied volatilities agree. The historical index and individual-equity terminal examples demonstrate that the pattern can differ across underlyings and expiries.



Each displayed curve is a different expiry's implied-volatility profile across moneyness/strike. These visuals retain their original legends and scales; their operational terminal menu is not a new mathematical definition. The index example has several downward-sloping curves, while the individual-equity illustration includes a pronounced curved profile. Both are evidence against one strike-independent volatility for all those quotes.

An implied-volatility surface can interpolate prices for unquoted contracts consistently with quoted options, and it provides a compact way to report the level of option prices. Its shape is also a diagnostic for a model's limitations. Option markets therefore reveal information about volatility and provide instruments with which to trade volatility exposure. Using a surface remains a modelling/interpolation decision; it does not guarantee that any arbitrary interpolation is free of arbitrage.

18.3 From a log payoff to variance

Consider a non-dividend-paying asset and a claim paying $\log(S_T/K)$ at T. Under constant-volatility risk-neutral dynamics,

$$L_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}[\log(S_T/K)] = e^{-rT}\left[\log(S_0/K) + (r - \tfrac{1}{2}\sigma^2)T\right].$$

Choose $K = F_{0,T} = S_0 e^{rT}$. Then $\log(S_0/K) = -rT$, so the rate terms cancel and

$$L_0 = -\tfrac{1}{2}e^{-rT}T\sigma^2, \qquad \boxed{\sigma^2 = \frac{2e^{rT}}{T}(-L_0).}$$

The negative expected log payoff is thus linked to variance. The argument does not require assuming a separately traded log contract exists: a strip of options can reproduce its nonlinear exposure.

For any terminal $s > 0$, static replication around the forward anchor gives

$$-\log(s/F) = -\frac{s-F}{F} + \int_0^F \frac{(K-s)^+}{K^2}\,dK + \int_F^\infty \frac{(s-K)^+}{K^2}\,dK.$$

This follows from the general twice-differentiable payoff expansion using the second derivative of $-\log s$, namely $1/s^2$. Under the risk-neutral forward condition $\mathbb{E}[S_T] = F$, the discounted expectation of the linear term is zero. Therefore

$$-L_0 = \int_0^F \frac{P_0(K)}{K^2}\,dK + \int_F^\infty \frac{C_0(K)}{K^2}\,dK,$$

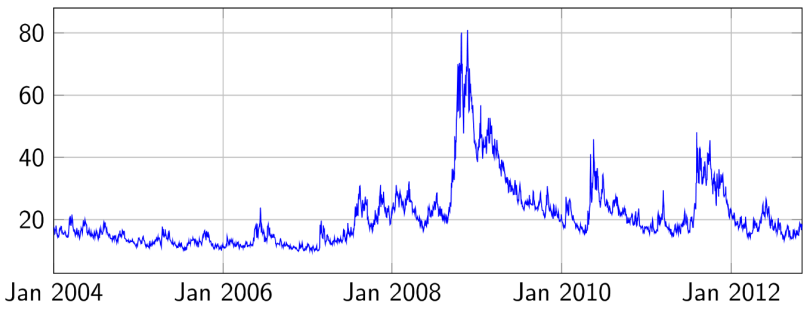
and

$$\boxed{\sigma^2 = \frac{2e^{rT}}{T}\left[\int_0^F \frac{P_0(K)}{K^2}\,dK + \int_F^\infty \frac{C_0(K)}{K^2}\,dK\right].}$$

Out-of-the-money means relative to the *forward*: puts below F, calls above F. The payoff identity is model-independent with suitable integrability. Identifying it with constant σ^2 used the constant-volatility model. For continuous stochastic volatility it relates to expected integrated variance; jumps require extra care, so it should not be advertised as an exact continuous-path realized-variance identity for every discontinuous model.

18.4 The volatility index and its discrete option strip

The historical introduction distinguishes the original 1993 index based on at-the-money S&P 100 options from the 2003 approach based on a broad S&P 500 put/call strip. Its target is an annualized 30-day volatility measure inferred from option prices. The term "fear gauge" describes the frequent association of high implied volatility with market stress, rather than a probability of an index decline or a guaranteed stock hedge.



The historical plot spans 2004–2012 and includes the large 2008 spike. The accompanying terminal description identifies the instrument as implied volatility rather than an asset-price return. Its dated descriptive statistics are historical snapshots; they do not provide a current trading signal. VIX futures and options can express or hedge volatility exposure, but their behaviour depends on maturity and pricing; the historically negative association with stock returns is not a contractual hedge ratio or a universal correlation.

For a single option maturity T, the discrete strip approximation is

$$\widehat{\sigma}^2(T) = \frac{2e^{rT}}{T}\sum_i \frac{\Delta K_i}{K_i^2} Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2.$$

Here K_0 is the largest listed strike no greater than the forward; Q selects the put below K_0 and call above it, with the put/call midpoint at K_0. Interior widths are $\Delta K_i = (K_{i+1} - K_{i-1})/2$, while endpoint widths use the adjacent strike difference. A broad, finite set of actual listed strikes

replaces the ideal continuous integrals, and the last term adjusts for an anchor not exactly equal to F. The squared correction is present in the original formula and must be retained. At the anchor the teaching note uses a simplified put-only convention; the midpoint convention above follows the Cboe methodology.

The single-maturity variance is only one ingredient of the target-horizon index: the calculation combines suitable maturities in variance-time units and reports volatility as 100 times the square root of the resulting annualized variance. Do not take an arithmetic average of volatilities when interpolating total variance. These implementation details distinguish the index from simply inverting one at-the-money Black–Scholes call.

18.5 References for the variance-replication argument

The log-contract connection is attributed to A. Neuberger, *The Log Contract* (1994), Journal of Portfolio Management 20. Static option replication is developed by P. Carr and D. Madan, *Towards a Theory of Volatility Trading* (1998), and K. Demeterfi, E. Derman, M. Kamal and J. Zou, *A Guide to Volatility and Variance Swaps* (1999), Journal of Derivatives 6(4), 9–32. E. Derman's *My Life as a Quant* (2004), Wiley, is the biographical reference mentioned alongside the argument. These references identify the source literature; the derivation required to use the formulas is included above.

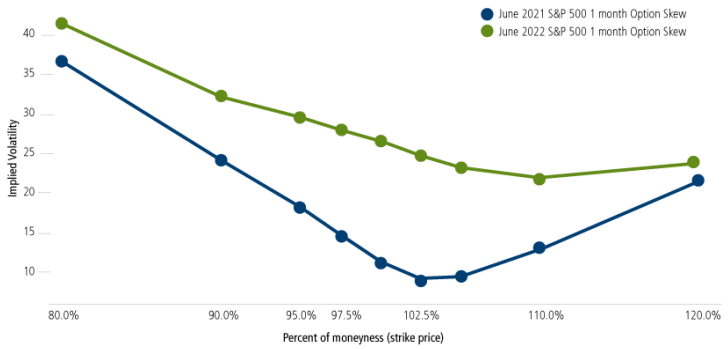
19 Beyond Black–Scholes: stochastic variance and jumps

19.1 What changes, and what remains from the benchmark

The Black–Scholes framework and its replication argument were developed earlier. Here the starting point is its distributional implication: under the physical measure, a geometric Brownian motion satisfies

$$S_t = S_0 \exp\{(\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\}, \qquad \log(S_t/S_0) \sim N((\mu - \tfrac{1}{2}\sigma^2)t, \sigma^2 t).$$

Constant σ makes the Black–Scholes implied volatility independent of strike and maturity when the model itself generates the prices. Market option quotes instead exhibit a *smile* or a decreasing, asymmetric *smirk*, and their shape changes with maturity. Return distributions also exhibit greater tail mass, asymmetry and changes through time compared with this constant-parameter Gaussian log-return benchmark. Large movements associated with rare, economically important news motivate models that go beyond a constant diffusion coefficient.



The two curves show implied volatility against the percentage moneyness of put options at different observation dates. Both their non-flat shape and the difference between the curves matter: the challenge is not merely to choose a different single constant volatility. The figure is a historical illustration, not a claim about a current volatility surface.

Two extensions address different mechanisms. Heston makes the instantaneous *variance* stochastic while keeping continuous stock paths. Merton adds a compound-Poisson jump component, allowing a discontinuous change over an arbitrarily short interval. Both retain risk-neutral discounted-payoff valuation under a specified pricing measure. Their additional parameters make calibration more flexible, but do not by themselves guarantee a perfect fit or exact hedging in an incomplete market.

19.2 Heston: two state variables and correlated shocks

Under the risk-neutral measure, with constant risk-free rate and no dividend yield in this example, write

$$\begin{aligned} dS_t &= rS_t\,dt + \sqrt{v_t}S_t\,dW_t^S, \\ dv_t &= \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v, \qquad\qquad d\langle W^S, W^v\rangle_t = \rho\,dt. \end{aligned}$$

Here $S_0 > 0$, $v_0 > 0$, $\kappa, \theta, \xi > 0$, and $-1 \le \rho \le 1$. The notation ξ denotes the volatility-of-variance coefficient called σ in the model presentation; v_t denotes its ν_t. Distinguishing these from the Black–Scholes constant volatility avoids confusing variance with its square root.

The variance process is a square-root, mean-reverting CIR process. Its conditional mean is

$$\mathbb{E}[v_{t+h} \mid v_t] = \theta + (v_t - \theta)e^{-\kappa h}.$$

Thus θ is the long-run variance level and κ the speed of adjustment. The coefficient ξ controls the randomness of variance. The exact CIR process is nonnegative; the sufficient Feller condition stated in the course, $2\kappa\theta > \xi^2$, makes zero unattainable from a strictly positive initial value. This property of the continuous process must not be transferred automatically to a discretized approximation.

The parameters have distinct economic roles. A larger long-run variance raises the variance level towards which the process reverts and ordinarily raises option values in the comparison considered. Larger κ brings variance back to that level more rapidly. Negative ρ connects a negative stock shock with a positive variance shock, creating the leverage effect and a heavier left tail; it helps reproduce a downward-sloping implied-volatility skew. Larger ξ, particularly when mean reversion is not too fast, allows wider variation in variance and helps change tail thickness and smile curvature. These are comparative interpretations with other inputs held fixed, not a substitute for calibrating the parameters jointly.

Vanilla prices can be used to calibrate the model; the calibrated dynamics can then price claims for which the model offers no simple closed-form payoff formula, including exotic claims. The benefit of matching liquid vanillas does not eliminate simulation, numerical or model risk for a different payoff.

19.2.1 Pricing decomposition and its probability interpretation

Let $\tau = T - t$, $x = \log S_t$, and condition throughout on the current (S_t, v_t). The European call price is

$$C(S_t, v_t, \tau) = e^{-r\tau}\mathbb{E}_t^{\mathbb{Q}}[(S_T - K)^+] = S_t P_1 - Ke^{-r\tau}P_2.$$

To understand the two terms, expand the payoff before using a Fourier representation:

$$C = e^{-r\tau}\mathbb{E}_t^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T > K\}}] - Ke^{-r\tau}\mathbb{Q}_t(S_T > K).$$

Therefore $P_2 = \mathbb{Q}_t(S_T > K)$ is the exercise probability under the money-market pricing measure. Provided the discounted stock is a true martingale, changing to the stock numeraire makes the first expectation $S_t\mathbb{Q}_t^S(S_T > K)$, so P_1 is the corresponding exercise probability under the stock measure. It is not the same probability as P_2. Because S_T scales linearly with the initial stock price when variance is held fixed, differentiation of the payoff expectation gives $\partial C/\partial S_t = P_1$ under the usual differentiation assumptions. A vanilla call thus has a delta interpretable through this first probability. Also, $S_t > 0$ means $x \in \mathbb{R}$; it does *not* mean that the log price must be positive.

19.2.2 Characteristic functions and inverse Fourier integration

The affine form used for each probability is

$$f_j(x, v, \tau; \phi) = \exp\{C_j(\tau; \phi) + D_j(\tau; \phi)v + i\phi x\}, \qquad j = 1, 2.$$

Using the risk-neutral variance parameters above, set

$$a = \kappa\theta, \quad b_1 = \kappa - \rho\xi, \quad b_2 = \kappa, \qquad u_1 = \tfrac{1}{2}, \quad u_2 = -\tfrac{1}{2},$$

and, for each j, define

$$\begin{aligned}
d_j &= \sqrt{(\rho\xi i\phi - b_j)^2 - \xi^2(2u_j i\phi - \phi^2)}, \\
g_j &= \frac{b_j - \rho\xi i\phi + d_j}{b_j - \rho\xi i\phi - d_j}, \\
C_j &= ri\phi\tau + \frac{a}{\xi^2}\left\{(b_j - \rho\xi i\phi + d_j)\tau - 2\log\left(\frac{1 - g_j e^{d_j\tau}}{1 - g_j}\right)\right\}, \\
D_j &= \frac{b_j - \rho\xi i\phi + d_j}{\xi^2}\frac{1 - e^{d_j\tau}}{1 - g_j e^{d_j\tau}}.
\end{aligned}$$

The probabilities are recovered by

$$P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-i\phi\log K} f_j(x, v, \tau; \phi)}{i\phi}\right] d\phi.$$

These expressions give a semi-analytic price: a numerical one-dimensional integral remains. Complex square roots and logarithms require mutually consistent branches. The representation with $e^{d_j\tau}$ can suffer overflow or discontinuities if implemented naively; algebraically equivalent stable forms and appropriate integration controls are used in practice. Reproducing the symbolic expression alone is not a numerical-stability check. The functions f_j do not depend on K, so the expensive characteristic-function values can be reused for different strikes with the same maturity; only the strike-dependent Fourier factor changes.

Parameter-convention clarification. The alternative notation in the pricing formula uses $a = k\theta_P$, $b_1 = k + \lambda - \rho\xi$ and $b_2 = k + \lambda$. This is consistent when k, θ_P denote physical variance parameters and the chosen variance-risk premium produces $\kappa_Q = k + \lambda$ and $\kappa_Q\theta_Q = k\theta_P$. If k, θ already describe the *risk-neutral* dynamics written above, one must use them directly rather than add another λ. The course's two parametrizations cannot be mixed without this identification. In particular, setting $\lambda = 0$ recovers the displayed risk-neutral coefficients with $k = \kappa$.

19.2.3 Euler simulation and the positivity problem

The integrated equations over a step h are

$$\begin{aligned}
S_{t+h} &= S_t + \int_t^{t+h} rS_u\, du + \int_t^{t+h} \sqrt{v_u} S_u\, dW_u^S, \\
v_{t+h} &= v_t + \int_t^{t+h} \kappa(\theta - v_u)\, du + \int_t^{t+h} \xi\sqrt{v_u}\, dW_u^v.
\end{aligned}$$

Freezing coefficients at the start of the interval gives the Euler approximation

$$\begin{aligned}
S_{t+h} &\approx S_t + rS_t h + \sqrt{v_t} S_t \sqrt{h}\, \epsilon_S, \\
v_{t+h} &\approx v_t + \kappa(\theta - v_t)h + \xi\sqrt{v_t}\sqrt{h}\, \epsilon_v.
\end{aligned}$$

Generate independent $Z_1, Z_2 \sim N(0,1)$ and take $\epsilon_v = Z_1$, $\epsilon_S = \rho Z_1 + \sqrt{1-\rho^2} Z_2$. The resulting innovations have unit variances and covariance ρ, as required. Repeat the update from S_0, v_0 over the simulation grid. A direct Euler stock step can become negative, and a variance step can be negative even when the exact variance cannot hit zero. Any truncation or alternative positivity treatment is an additional numerical convention; it should be stated and its discretization error assessed.

19.2.4 An alternative using the exact CIR variance transition

Represent the stock Brownian motion as $dW^S = \rho\, dZ + \sqrt{1-\rho^2}\, dB$, with B independent of Z, and apply Itô's lemma:

$$\begin{aligned} d\log S_t &= (r - \tfrac{1}{2} v_t)\, dt + \rho \sqrt{v_t}\, dZ_t + \sqrt{1-\rho^2}\sqrt{v_t}\, dB_t, \\ dv_t &= \kappa(\theta - v_t)\, dt + \xi \sqrt{v_t}\, dZ_t. \end{aligned}$$

For $h > 0$ the CIR transition is available exactly. With

$$q_h = \frac{\xi^2 (1 - e^{-\kappa h})}{4\kappa}, \qquad d = \frac{4\kappa\theta}{\xi^2}, \qquad \ell_h = \frac{e^{-\kappa h} v_t}{q_h},$$

sample $v_{t+h} = q_h X$ where X has a noncentral chi-square distribution with d degrees of freedom and noncentrality ℓ_h. Equivalently, the presentation's C is $1/q_h$, and its noncentrality is $C e^{-\kappa h} v_t$. Specifying which argument is the degree of freedom avoids exchanging two different parameters in a numerical routine.

Let $I = \int_t^{t+h} v_u\, du$. Integrating the variance equation first isolates the correlated stochastic integral:

$$\int_t^{t+h} \sqrt{v_u}\, dZ_u = \frac{v_{t+h} - v_t - \kappa\theta h + \kappa I}{\xi}.$$

Substituting into the integrated log-price equation gives

$$\log S_{t+h} = \log S_t + rh + \frac{\rho}{\xi}(v_{t+h} - v_t - \kappa\theta h) + \left(\frac{\rho\kappa}{\xi} - \frac{1}{2}\right) I + \sqrt{1-\rho^2} \int_t^{t+h} \sqrt{v_u}\, dB_u.$$

Conditional on the *whole variance path*, the last stochastic integral is Gaussian with mean zero and variance I, since B is independent of the variance-driving Brownian motion. Its unconditional distribution is generally a variance mixture, not a single normal distribution. For a simple implementable approximation use

$$\widehat{I} = \frac{h}{2}(v_t + v_{t+h}), \qquad \int_t^{t+h} \sqrt{v_u}\, dB_u \approx \sqrt{\widehat{I}}\, Y, \quad Y \sim N(0,1)$$

with Y independent of the variance simulation. The variance endpoint is sampled exactly, whereas its integrated path and therefore the stock step are still approximated. This distinction explains both the efficiency of the construction and why it is not an exact joint simulation of (S, v).

Algebraic corrections made explicit. The integrated variance equation must retain the coefficient ξ of its Brownian integral and use upper limit $t + h$ throughout. A written upper limit $t + \nu$ cannot replace a time endpoint. The drift in the risk-neutral log-stock equation is r, rather than the physical drift μ. These corrections follow directly by integrating the stated SDEs and are used in the derivation above.

19.3 Merton: diffusion combined with compound-Poisson jumps

Let N_t be a Poisson process with intensity $\lambda \geq 0$. At its jump times T_i, the stock is multiplied by an independent positive random factor $Y_i = S_{T_i}/S_{T_i-}$. Assume the Y_i are identically distributed, independent of one another, of N and of the Brownian motion, and

$$V_i = \log Y_i \sim N(m, \delta^2), \qquad k_J = \mathbb{E}^{\mathbb{Q}}[Y_i - 1] = e^{m+\delta^2/2} - 1.$$

The compound-Poisson process $J_t = \sum_{i=1}^{N_t}(Y_i - 1)$ has expectation $\lambda k_J t$. Under the selected risk-neutral measure the dynamics are

$$dS_t = (r - \lambda k_J)S_t\,dt + \sigma S_t\,dW_t + S_{t-}\,dJ_t.$$

The compensation term has a precise purpose: the expected jump contribution is $\lambda k_J S_t dt$, so the total conditional expected return is $r\,dt$. At a jump the left limit is essential: $\Delta S_{T_i} = S_{T_i-}(Y_i - 1)$ and hence $S_{T_i} = S_{T_i-}Y_i > 0$.

Solving the diffusion between jumps and multiplying the jump factors yields

$$S_t = S_0 \exp\{(r - \lambda k_J - \tfrac{1}{2}\sigma^2)t + \sigma W_t\} \prod_{i=1}^{N_t} Y_i.$$

Consequently, for the number of *new* jumps $N_T - N_t = j$ over $\tau = T - t$,

$$\log(S_T/S_t) \mid (N_T - N_t = j) \sim N\big((r - \lambda k_J - \tfrac{1}{2}\sigma^2)\tau + jm, \sigma^2\tau + j\delta^2\big).$$

Conditioning on $N_T = j$ without accounting for the observed N_t is appropriate only when the valuation starts at zero. At a general valuation date it is the increment that remains random. Mixing these conditional normals over the Poisson count produces richer asymmetry and tail behaviour than one Gaussian log-return distribution.

19.3.1 Why choosing the risk-neutral jump law matters

Adding jump risk generally makes a stock-and-bond market incomplete: different equivalent martingale measures can price the unhedgeable risk differently. The classical choice discussed here treats the jump risk as diversifiable and without its own premium. In that choice the physical dynamics use $(\mu - \lambda k_J)S_t dt$ and the same jump parametrization, while the drift changes to $(r - \lambda k_J)S_t dt$ under the pricing measure; $\mu - r$ compensates diffusive risk. This is a modelling assumption, not a theorem that every jump has zero risk premium. Alternative jump-risk premia lead to different risk-neutral λ, m, δ even when the physical return data are unchanged.

19.3.2 First call-price representation: condition on the jump count

Write $C_{BS}(S, K, \tau; \sigma, r)$ for a no-dividend Black–Scholes call. The tower property gives

$$C_M = \sum_{j=0}^{\infty} w_j \mathbb{E}_t^{\mathbb{Q}}[e^{-r\tau}(S_T - K)^+ \mid N_T - N_t = j], \quad w_j = e^{-\lambda\tau}\frac{(\lambda\tau)^j}{j!}.$$

For $\tau > 0$ define

$$\sigma_j^2 = \sigma^2 + \frac{j\delta^2}{\tau}, \qquad S_t^{(j)} = S_t \exp\{-\lambda k_J \tau + j(m + \tfrac{1}{2}\delta^2)\}.$$

The lognormal variable generated by $S_t^{(j)}$ with rate r and volatility σ_j has precisely the conditional log mean and variance above:

$$\log S_t^{(j)} + (r - \tfrac{1}{2}\sigma_j^2)\tau = \log S_t + (r - \lambda k_J - \tfrac{1}{2}\sigma^2)\tau + jm,$$
$$\sigma_j^2\tau = \sigma^2\tau + j\delta^2.$$

Therefore

$$\boxed{C_M = \sum_{j=0}^{\infty} e^{-\lambda\tau} \frac{(\lambda\tau)^j}{j!} C_{BS}(S_t^{(j)}, K, \tau; \sigma_j, r).}$$

This is a convergent infinite-series representation rather than one elementary Black–Scholes formula. Numerical implementation truncates the count distribution with an explicit tail/error tolerance. When $\lambda = 0$, only $j = 0$ remains, $S_t^{(0)} = S_t$, and the Black–Scholes price is recovered.

19.3.3 Equivalent representation with tilted Poisson weights

Let $a = m + \delta^2/2 = \log(1 + k_J)$ and set

$$\bar{\lambda} = \lambda(1 + k_J), \qquad r_j = r - \lambda k_J + \frac{ja}{\tau}.$$

For $A_j = -\lambda k_J \tau + ja$, homogeneity of the Black–Scholes expression gives

$$C_{BS}(S_t e^{A_j}, K, \tau; \sigma_j, r) = e^{A_j} C_{BS}(S_t, K, \tau; \sigma_j, r_j).$$

Indeed the d arguments coincide and $e^{A_j} e^{-r_j\tau} = e^{-r\tau}$. Multiplying the original weight by this exponential yields

$$e^{-\lambda\tau} \frac{(\lambda\tau)^j}{j!} e^{A_j} = e^{-\bar{\lambda}\tau} \frac{(\bar{\lambda}\tau)^j}{j!}.$$

Hence the second, equivalent formula is

$$\boxed{C_M = \sum_{j=0}^{\infty} e^{-\bar{\lambda}\tau} \frac{(\bar{\lambda}\tau)^j}{j!} C_{BS}(S_t, K, \tau; \sigma_j, r_j).}$$

The r_j are auxiliary rates in the series representation, not a new economy with a different funding rate for each realized jump count. Both formulas price the same claim with the same risk-neutral discounting; the weight transformation is what makes the two presentations agree.

19.4 Calibration in option space and estimation from returns

Suppose N market option quotes give prices C_i^{mkt} and implied volatilities Σ_i at strikes K_i and maturities T_i. Implied volatility is the volatility which reproduces the quote in the Black–Scholes benchmark. It remains a useful quoting coordinate even when the model used to explain the quote is not Black–Scholes.

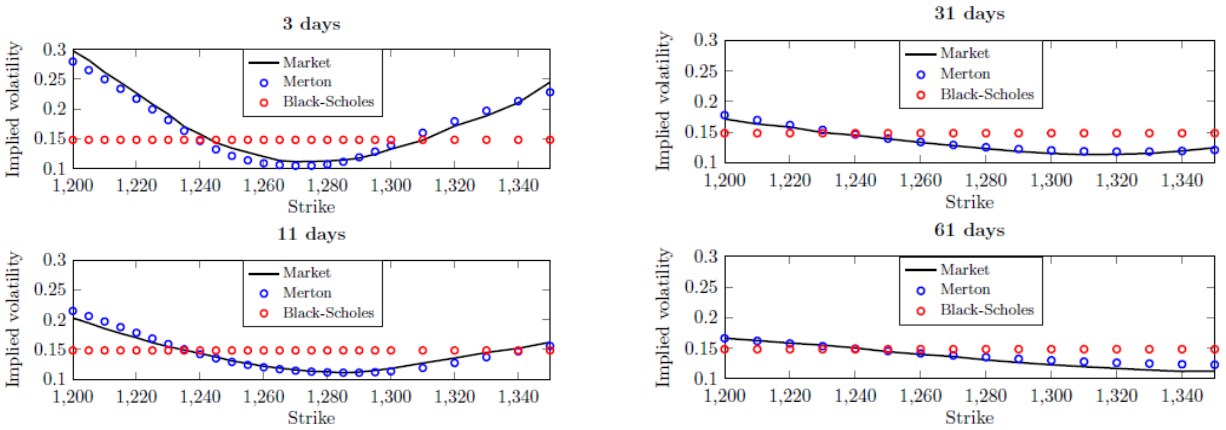
For the constant-volatility benchmark the fitted volatility minimizes

$$\widehat{\sigma} = \arg\min_{\sigma \geq 0} \sum_{i=1}^{N} (\Sigma_i - \sigma)^2.$$

With equally weighted positive implied volatilities the unconstrained solution is their arithmetic mean, as follows by setting $-2\sum_i(\Sigma_i - \sigma) = 0$. For Merton, with risk-neutral parameter vector $\vartheta = (\sigma, \lambda, m, \delta)$, the corresponding objective is

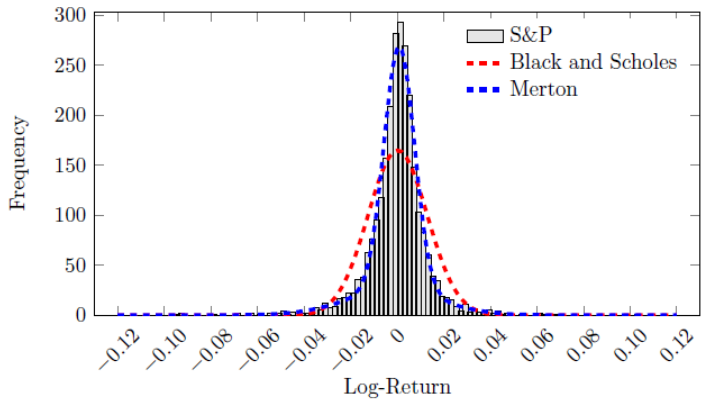
$$\widehat{\vartheta} = \arg\min_{\vartheta} \sum_{i=1}^{N} \left[\Sigma_i - \Sigma_i^M(\vartheta)\right]^2,$$

where each Σ_i^M is obtained by pricing the option in the model and inverting its Black–Scholes quote. Thus one objective evaluation involves model prices and implied-volatility inversion, not direct comparison of σ with every market quote. Constraints such as nonnegative intensity and standard deviations are part of calibration. Heston can likewise be calibrated to a cross-section of vanilla quotes. A fit in volatility units is not numerically identical to an unweighted fit in price units.



The original comparisons plot implied volatility against strike for maturities of 3 and 11 days on the left, and 31 and 61 days on the right. Black lines represent market quotes, blue points Merton and red points Black–Scholes. The short-maturity smile is particularly difficult for one constant volatility; the jump model captures much more of its curvature. At the longer maturities in this example the surface is flatter and the benchmark becomes more acceptable. The illustrations show a substantial improvement rather than an exact fit or a universal claim that all long-maturity smiles are flat.

The parameters under the *physical* measure can instead be estimated from a time series of stock returns, for example by maximum likelihood or generalized method of moments. This answers a different question from calibration of risk-neutral parameters to options. Under the special no-jump-premium assumption one may link the jump parameters across measures; without it their equality is not justified.



The historical S&P 500 return-frequency plot compares the empirical distribution with Black–Scholes and Merton specifications. The extra jump-intensity, jump-mean and jump-dispersion parameters help represent a sharp centre, asymmetry and larger tail mass. The comparison concerns the fitted return distribution, while the previous figures concern option-implied volatility; they should not be treated as the same empirical test.

19.5 Simulating jumps without changing the stated model

Taking logarithms of the exact multiplicative solution gives

$$\boxed{\log S_T = \log S_0 + (r - \lambda k_J - \tfrac{1}{2}\sigma^2)T + \sigma W_T + \sum_{i=1}^{N_T} \log Y_i.}$$

This corrects an inconsistent logarithmic simulation expression: the drift compensator is $\lambda \mathbb{E}[Y-1]$, not $\lambda(m+\delta^2/2)$, and logarithmic jumps are $\log Y_i$, not $Y_i - 1$. The latter are relative *price* jumps.

These distinctions follow from the preceding SDE and product solution and cannot be dropped when coding.

For an exact grid-endpoint simulation with constant coefficients, on each step h independently draw $J \sim \text{Poisson}(\lambda h)$, $Z \sim N(0,1)$, and the jump log-sum conditional on J:

$$L_J = Jm + \delta\sqrt{J}\,Z_J, \qquad Z_J \sim N(0,1),$$

with $L_0 = 0$. Then update

$$\log S_{t+h} = \log S_t + (r - \lambda k_J - \tfrac{1}{2}\sigma^2)h + \sigma\sqrt{h}\,Z + L_J.$$

The Poisson count must allow more than one jump; replacing it by a Bernoulli event is only a small-step approximation. This update is exact for the joint grid-endpoint law under the constant-parameter model, but does not reveal the intrastep jump times needed by some path-dependent payoffs. Generating actual arrival times or refining the path is a separate requirement. The supplied computational implementations belong to Part II; the lecture's invitation to inspect implementation examples does not provide an additional unseen Matlab source file.

20 Credit risk: claims on firm value and default intensity

Credit risk arises when interest or principal is not paid, or is not paid on time. Default is the extreme outcome in which the counterparty cannot or will not fulfil its obligation, but a deterioration in credit quality can reduce the value of a loan before any missed payment. A rating downgrade can therefore change the value of an exposure to a company, another financial institution or a government. Credit risk is considered alongside market risk and operational risk among the major risks of a bank.

The structural approach links a firm's obligations to its total asset value. A reduced-form approach instead models default probabilities or intensities, often inferred from debt prices or credit derivatives. Accounting-based scores provide another way to classify financial health. These are distinct modelling choices, with different information requirements and limitations. The structural Merton model below is a model of corporate capital structure; it should not be confused with the Merton *jump-diffusion stock model* in Section 19.3.

20.1 The option-pricing argument used by structural models

The general Black–Scholes framework is in Section 14. For its credit application the underlying is firm value, the claim is European, and the basic example assumes no payouts, divisible trading, no taxes, transaction costs or margin frictions, continuous rebalancing, constant risk-free rate at all maturities and geometric Brownian asset dynamics. These assumptions describe the model, not all real corporate assets or taxes.

For completeness, the local hedging step needed for the application is as follows. If $dS = \alpha S dt + \sigma S dW$ and $C = C(S,t)$, Itô's formula gives

$$dC = C_S\,dS + (C_t + \tfrac{1}{2}\sigma^2 S^2 C_{SS})dt.$$

A self-financing hedge with $\Delta = C_S$ shares against one written option has local gain $\Delta\,dS - dC$. The random term cancels. Its value $\Pi = \Delta S - C$ must earn the bank rate, so

$$-(C_t + \tfrac{1}{2}\sigma^2 S^2 C_{SS}) = r(SC_S - C), \qquad C_t + \tfrac{1}{2}\sigma^2 S^2 C_{SS} + rSC_S - rC = 0.$$

This is a statement about gains with the current hedge, including the bank account which finances rebalancing; naively differentiating $\Delta S - C$ without its self-financing cash adjustments would add incorrect terms. The physical drift α cancels. For a no-dividend call the terminal condition is $(S_T - K)^+$, the zero-stock boundary is $C(0,t) = 0$, and $C_S \to 1$ as $S \to \infty$. These boundary

conditions accompany the PDE rather than replacing its terminal payoff. The long/short call and put payoff diagrams are precisely the four payoffs already developed earlier, so a second copy of those diagrams is unnecessary here.

In the formula $C = SN(d_1) - Ke^{-r\tau}N(d_2)$, $N(d_2)$ is the risk-neutral probability of exercise. The strike component is $Ke^{-r\tau}\mathbb{Q}(S_T > K)$. The asset component is different:

$$e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{\{S_T>K\}}] = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[S_T \mid S_T > K]N(d_2) = SN(d_1).$$

Since $\mathbb{E}^{\mathbb{Q}}[S_T] = Se^{r\tau}$, division gives

$$N(d_1) = \frac{\mathbb{E}^{\mathbb{Q}}[S_T \mid S_T > K]}{\mathbb{E}^{\mathbb{Q}}[S_T]}N(d_2) > N(d_2)$$

for a nondegenerate lognormal stock and finite positive strike. The extra factor reflects the higher asset value in exercise states; it is not an extra exercise probability under the same measure. Under zero dividends the call delta is $N(d_1)$. This distinction explains why equity sensitivity and default probability involve different normal tails.

20.2 Equity is a call and risky debt contains a short put

Let V_t be total firm asset value, E_t equity value and B_t debt market value, so $V_t = E_t + B_t$. The firm's only debt is a senior zero-coupon promise to pay face value D at time T. Assume V_t follows geometric Brownian motion with constant volatility σ_V. The firm defaults *at maturity* if $V_T < D$. There is no earlier first-passage default in this elementary model.

Limited liability and priority determine both terminal payoffs:

$$E_T = (V_T - D)^+, \qquad B_T = \min(V_T, D) = D - (D - V_T)^+ = V_T - (V_T - D)^+.$$

For $V_T < D$, creditors receive all V_T and shareholders receive zero. For $V_T > D$, creditors receive the fixed D and shareholders the excess. The two claims sum to V_T in every state, with the kink at D. Equity is convex, whereas debt becomes flat above its face value.

Under risk-neutral valuation, let $\tau = T - t$ and

$$d_1 = \frac{\log(V_t/D) + (r + \frac{1}{2}\sigma_V^2)\tau}{\sigma_V\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma_V\sqrt{\tau}.$$

Then

$$\begin{aligned} E_t &= V_tN(d_1) - De^{-r\tau}N(d_2), \\ B_t &= V_t - E_t = De^{-r\tau} - P(V_t, D, r, \tau, \sigma_V) \\ &= V_tN(-d_1) + De^{-r\tau}N(d_2). \end{aligned}$$

Here P is the corresponding put on firm assets. Thus risky debt is risk-free debt *minus a put*, rather than a pure short-put position with no safe component. The put price measures the risk-adjusted discounted shortfall due to default. The debt formula follows equally from payoff subtraction or put–call parity, providing a consistency check.

The payoff subtraction uses $V_T - \max(V_T - D, 0)$, not $V_T - \min(V_T - D, 0)$: the latter printed expression would exceed available assets in default states. The state-by-state payoffs above resolve this sign/function typo directly.

Notation correction. The option price uses r inside both d_1 and d_2. A formula which inserts the physical firm drift α into a risk-neutral equity price mixes forecasting and pricing. The physical drift has a role in a physical default forecast below, but does not replace r in these market values.

20.3 Inferring asset volatility from traded equity

The asset value and its volatility may be unobservable even when equity is traded. Apply Itô's formula to $E(V,t)$. Its Brownian coefficient is $E_V\sigma_V V = N(d_1)\sigma_V V$. If equity's observed percentage volatility is σ_E, the same coefficient is $\sigma_E E$. Hence

$$\boxed{\sigma_E E = N(d_1)\sigma_V V, \qquad \sigma_E = \frac{V}{E}N(d_1)\sigma_V.}$$

The factor V/E connects leverage and equity risk. With observed E_0, σ_E, debt face value and maturity, one can solve simultaneously

$$E_0 = V_0 N(d_1) - De^{-rT}N(d_2), \qquad \sigma_E E_0 = N(d_1)\sigma_V V_0$$

for (V_0, σ_V). These are nonlinear equations because d_1, d_2 depend on both unknowns; the second equation alone is not a direct estimate of σ_V.

20.4 Distance to default and the KMV interpretation

The KMV approach, named for Kealhofer, McQuown and Vasicek, starts from the observation that companies with the same credit rating can have different default risks. Firm-specific, forward-looking information in equity prices and balance sheets can help identify an implied distance to the debt threshold. A diagram of the mechanism has firm value above a debt barrier at the valuation date and a distribution of possible values at T: its mass below D represents default states. The distribution and threshold, rather than one realized sample path alone, determine the probability.

For physical asset drift α, the lognormal model gives distance to default

$$DD^{\mathbb{P}} = \frac{\log(V_t/D) + (\alpha - \frac{1}{2}\sigma_V^2)\tau}{\sigma_V\sqrt{\tau}}, \qquad p^{\mathbb{P}} = N(-DD^{\mathbb{P}}).$$

Replacing α with r gives $DD^{\mathbb{Q}} = d_2$ and

$$\boxed{p^{\mathbb{Q}} = \mathbb{Q}(V_T < D \mid \mathcal{F}_t) = N(-d_2).}$$

The presentation calls this tail an expected default frequency (EDF). Within the displayed normal model it is the model-implied default probability. Do not equate a risk-neutral pricing probability with an observed or empirically calibrated real-world default frequency. The firm's drift and the chosen measure must be specified. The methodology can be extended to several firms and systemic risk; the cited example is Gupta, Lu and Wang (2021), but no further formula or complete bibliographic details were supplied, so those are not inferred.

20.5 Worked example: two observables, two unobservables

The example has equity market value $E_0 = 3$ million, annual equity volatility $\sigma_E = .8$, a single debt payment $D = 10$ million in one year and risk-free rate $r = .05$. Solve

$$\begin{aligned} 3 &= V_0 N(d_1) - 10e^{-.05}N(d_2), \\ 2.4 &= N(d_1)\sigma_V V_0, \\ d_1 &= \frac{\log(V_0/10) + (.05 + \sigma_V^2/2)}{\sigma_V}, \qquad d_2 = d_1 - \sigma_V. \end{aligned}$$

A numerical solution gives approximately $V_0 = 12.40$ million and $\sigma_V = .2123$. Using the unrounded root gives $V_0 \simeq 12.3954$, $\sigma_V \simeq .2123$, and risk-neutral default probability approximately 12.7%. Plugging the root back into both equations is necessary: a plausible equity value with an incorrect volatility equation is not a solved system.

Debt is the residual $B_0 = V_0 - 3 \simeq 9.40$ million. Its promised continuously compounded yield satisfies $B_0 = 10e^{-y}$, so with the rounded debt value

$$y = \log(10/9.40) \simeq .0619, \qquad y - r \simeq .0119.$$

The rounded classroom calculation therefore reports about 1.19 percentage points, or 119 basis points. With the unrounded nonlinear solution, $B_0 = 9.39538719$, $y = .06236625$ and $y - r = .01236625$, or about 123.66 basis points. Showing the rounding path avoids pretending the rounded example is the exact solution. The spread is computed from the promised debt payment and its market value; it is not the firm's physical expected return and is not itself the default probability.

20.6 Capital structure, incentives and the limits of the model

Debt can create a tax shield when interest is deductible before tax whereas equity dividends are paid from after-tax income. This real-world corporate-finance observation extends beyond the frictionless no-tax assumptions of the elementary option-pricing calculation. Its incentive to issue more debt competes with the costs and risk of bankruptcy, which favour less debt.

There is also an agency problem. If managers act for equity holders, increasing asset risk can increase the value of their residual convex call-like claim, potentially at the expense of creditors. The reason is the equity payoff's convexity and limited downside, not a general claim that every risky project creates total firm value. A shareholder benefits from upside after debt is paid while the debt threshold limits the equity holder's losses. This separates the incentive of one claim from the interests of all claimants together.

Structural-model implementation requires estimates of unobserved inputs. A theoretical EDF depends on assumptions about the asset-return distribution, especially its normal log-return form. For a private firm without traded equity, comparable-company and accounting information must substitute for direct equity-market inputs. The simple one-debt model also does not distinguish seniority, collateral, covenants and convertibility of actual long-term bonds. These limitations motivate alternative or richer models rather than silently treating the single-debt calculation as every bond's payoff.

20.7 Reduced-form models and hazard rates

A reduced-form model can infer default information from prices of the firm's debt. Comparing risky and risk-free bond prices reveals a credit spread, which the elementary interpretation attributes entirely to default loss. In practice such an attribution is an assumption: recovery conventions and non-default price components must be considered before calling the whole spread a default probability.

Let τ_d denote default time and $\lambda(t)$ a hazard rate. Conditional on survival to t, for a small h,

$$\Pr(t < \tau_d \le t + h \mid \tau_d > t) \approx \lambda(t)h.$$

This is a first-order local probability; it is not an exact probability for arbitrary intervals. For a deterministic hazard rate, the survival function is

$$Q(t,T) = \exp\left(-\int_t^T \lambda(u)\,du\right), \qquad \Pr(\tau_d \le T \mid \tau_d > t) = 1 - Q(t,T).$$

The exponential follows by solving $\partial_T Q = -\lambda(T)Q$ with $Q(t,t) = 1$. For a constant intensity it reduces to $e^{-\lambda(T-t)}$. A stochastic intensity requires an appropriate conditional expectation of the integrated-intensity exponential, rather than replacing it by the exponential of its mean.

Intensity can be estimated from historical defaults or calibrated from bonds and credit derivatives such as credit default swaps (CDS). A CDS transfers specified default loss against a premium stream, so it gives market information about protection pricing. Calibration to such prices identifies a risk-neutral intensity under the chosen recovery and pricing conventions; it does not

directly identify the historical physical intensity. To illustrate the distinction without imposing an unprovided recovery model, zero-recovery debt with constant deterministic r, $\lambda^{\mathbb{Q}}$ would have price $De^{-(r+\lambda^{\mathbb{Q}})T}$ and spread $\lambda^{\mathbb{Q}}$. Positive recovery generally changes this relation. The lecture introduces hazard modelling and its data sources, rather than a complete CDS-pricing model.

20.8 Altman's accounting and market-variable score

The displayed Z-score combines five ratios:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + .6X_4 + .999X_5,$$

where X_1 is working capital divided by total assets (liquidity), X_2 retained earnings divided by total assets, X_3 earnings before interest and taxes divided by total assets (profitability), X_4 market equity value divided by book liabilities (leverage), and X_5 sales divided by total assets (asset productivity). Ratios must use consistent units; the coefficients do not apply unchanged to percentages multiplied by 100.

Higher scores indicate stronger financial health in this classification. The teaching thresholds are: above 3.0, unlikely default; between 2.7 and 3.0, an alert zone; between 1.8 and 2.7, a substantial chance of distress; below 1.8, very high financial embarrassment. These are model-specific classification cutoffs presented in the course, not a numerical default probability or a universal current banking rule. Boundary conventions at exactly a cutoff must be stated when implementing a classifier.

Two classification errors are distinguished: predicting safety for a firm which subsequently fails (called Type 1 here), and predicting severe default risk for a firm which survives (Type 2 here). The labels depend on which outcome is defined as the positive decision; learn the actual false prediction rather than assuming every statistical text uses the same numbering. The score, structural default tail and calibrated hazard rate each summarize different inputs and assumptions, so agreement between them is an empirical question.

21 Black–Scholes: worked problems

The solutions below assume constant parameters, frictionless trading and the Black–Scholes stock model. Write N for the standard normal distribution function and τ for the remaining life. A proportional dividend yield is denoted by δ. Unless specified otherwise, rates and volatilities are annual decimals, and months are converted to fractions of a year. Monetary answers are per share unless a block size is specified. These are generated worked solutions; ambiguities in a question are identified explicitly.

21.1 Problem 4.1: call, put and parity

Question. The Black–Scholes framework holds. A stock currently sells for 41, its volatility is 30%, and it pays no dividends. An option has strike 40 and expires in three months. The continuously compounded risk-free rate is 8%. (a) Calculate the European call price. (b) Calculate the European put price. (c) Price the put using put–call parity and check the result.

Solution. Here $S = 41$, $K = 40$, $\sigma = .30$, $r = .08$, $\delta = 0$ and $\tau = .25$, so $\sigma\sqrt{\tau} = .15$. The standardized exercise thresholds are

$$d_1 = \frac{\log(41/40) + (.08 + .30^2/2)(.25)}{.15} = .37295075, \qquad d_2 = d_1 - .15 = .22295075.$$

For (a), discount the stock component at the dividend yield and the strike component at the risk-free rate:

$$C = 41N(.37295075) - 40e^{-.08(.25)}N(.22295075) = \boxed{3.399078}.$$

For (b), the put formula uses the lower tails:

$$P = 40e^{-.02}N(-.22295075) - 41N(-.37295075) = \boxed{1.607025}.$$

For (c), a call plus a discounted strike and a put plus a share have the same terminal payoff. No arbitrage therefore gives $C + Ke^{-r\tau} = P + S$. Substitution yields

$$P = C + 40e^{-.02} - 41 = 3.399078 + 39.207947 - 41 = 1.607025.$$

The calculation is independent of the put-tail evaluation and provides a useful sign and discounting check.

21.2 Problem 4.2: the price of a block

Question. Price 100 three-month European calls with strike 25. The stock is 20, volatility is 24%, continuous dividend yield is 3% and the continuously compounded risk-free rate is 5%. The Black–Scholes framework holds. Choices: (A) .04; (B) 1.93; (C) 3.63; (D) 4.22; (E) 5.09.

Solution. Set $\tau = .25$. The denominator is $.24\sqrt{.25} = .12$ and

$$d_1 = \frac{\log(20/25) + (.05 - .03 + .24^2/2)(.25)}{.12} \simeq -1.75786, \qquad d_2 = d_1 - .12.$$

The call is out of the money, but still has positive time value:

$$C = 20e^{-.03(.25)}N(d_1) - 25e^{-.05(.25)}N(d_2) = .036202397.$$

There is no block discount in this model, so the requested block costs $100C = \boxed{3.620240}$. Choice **C** is the nearest offered value; its printed 3.63 differs slightly from a full-precision evaluation. Choice A is approximately the single-option value and misses the requested quantity.

21.3 Problem 4.3: a forward-start call

Question. One year from today a forward-start contract gives its owner a one-year European call whose strike equals the stock price at that time. The stock pays no dividends and has volatility 30%. Its one-year forward price is 100 and the continuously compounded risk-free rate is 8%. Under Black–Scholes, find the contract's value today. Choices: (A) 11.90; (B) 13.10; (C) 14.50; (D) 15.70; (E) 16.80.

Solution. The call is issued at time 1 and expires at time 2. Its strike is the random value S_1, not today's forward price 100. Conditional on time-1 information it is at the money, with one year remaining. Homogeneity of the call formula gives

$$C_1 = S_1 a, \qquad a = N\left(\frac{.08 + .30^2/2}{.30}\right) - e^{-.08}N\left(\frac{.08 - .30^2/2}{.30}\right) = .1571131255.$$

The coefficient is deterministic because the rate, volatility and remaining life are fixed. Under the risk-neutral measure, $e^{-.08}E^{\mathbb{Q}}[S_1] = S_0$, hence the value of the time-1 delivery is

$$V_0 = e^{-.08}E^{\mathbb{Q}}[aS_1] = aS_0.$$

The quoted forward determines $S_0 = 100e^{-.08} = 92.311635$. Therefore

$$V_0 = .1571131255(92.311635) = \boxed{14.503369},$$

which is choice **C**. Discounting aS_0 once more would double-count the discounting already used in valuing delivery at time 1.

21.4 Problem 4.4: prepaying rolling insurance

Question. You own one share of a non-dividend-paying stock currently worth 45. To insure it over a year, you obtain a three-month European put every three months, the first immediately. Each strike is 90% of the then-current stock price. The continuously compounded risk-free rate is 8% and volatility is 30%. A broker charges today for the total cost of all four options. What is the fair prepaid amount? Choices: (A) 1.59; (B) 2.24; (C) 2.86; (D) .48; (E) 3.61.

Solution. The purchases take place at $t_j = 0, .25, .50, .75$; each option lasts $h = .25$. At a purchase date its stock price is s and its strike is $.9s$. The put formula scales linearly with s:

$$P(s, .9s, h) = sb, \quad b = .9e^{-.08h}N(-d_2) - N(-d_1),$$

$$d_1 = \frac{\log(1/.9) + (.08 + .30^2/2)h}{.30\sqrt{h}}, \qquad d_2 = d_1 - .30\sqrt{h}.$$

Numerically $b = .01586801367$. Thus each future premium is the random amount bS_{t_j}. Its present fair funding cost is nevertheless deterministic:

$$e^{-.08t_j}E^{\mathrm{Q}}[bS_{t_j}] = bS_0,$$

because the discounted non-dividend-paying stock is a risk-neutral martingale. Summing four premium claims gives

$$V_0 = 4b(45) = 4(.714060615) = \boxed{2.856242},$$

choice **C**. The calculation values the broker's commitment to fund premiums, not a promise that the stock never falls below one fixed annual floor. Each new strike resets to 90% of that quarter's stock price.

21.5 Problem 4.5: a physical-measure prediction interval

Question. In the Black–Scholes framework, a dividend-paying stock currently costs .25, has volatility .35 and has continuously compounded expected stock-price appreciation of 15%. Calculate the upper endpoint of the 90% lognormal confidence interval for its price in six months. Choices: (A) .393; (B) .425; (C) .451; (D) .486; (E) .529.

Solution. Interpret the stated appreciation as the drift $\mu = .15$ of the ex-dividend stock price under the physical measure. For $dS_t/S_t = \mu\,dt + \sigma\,dW_t$, Itô's formula gives

$$\log(S_\tau/S_0) = (\mu - \sigma^2/2)\tau + \sigma\sqrt{\tau}Z, \qquad Z \sim N(0,1).$$

A central 90% interval leaves 5% in each tail, so its upper endpoint uses $z_{.95} = 1.644853627$, not $z_{.90}$. With $\tau = .5$,

$$U = .25\exp\Big[(.15 - .35^2/2)(.5) + .35\sqrt{.5}(1.644853627)\Big] = \boxed{.392649},$$

choice **A**. This is a distributional interval for a future price, not a confidence interval for an estimated parameter. Its drift is the stated expected appreciation, not the risk-free rate. If the 15% had meant total return including dividends instead, the price drift would require a dividend adjustment; that is not the wording used here.

21.6 Problem 4.6: European values and an American-put conclusion

Question. A non-dividend-paying stock is 40. An option has strike 45 and expires in four months. The continuously compounded risk-free rate is 3% and return volatility is 40%. (a) Find the Black–Scholes call price. (b) Find the European put price. (c) Since the put is worth more alive than exercised now, $45 - 40 = 5 < 6.57586$, can one conclude that the American put has the same value as the European put?

Solution. With $\tau = 1/3$,

$$d_1 = \frac{\log(40/45) + (.03 + .40^2/2)/3}{.40\sqrt{1/3}} = -.35124418, \qquad d_2 = -.58218429.$$

For (a), $C = 40N(d_1) - 45e^{-.01}N(d_2) = \boxed{2.023617}$. For (b), parity gives

$$P = C + 45e^{-.01} - 40 = \boxed{6.575860}.$$

For (c), **no**. Comparing exercise with continuation today tests only the immediate-exercise decision. The American contract also permits exercise at a later date before maturity. Positive interest can make early receipt of the strike valuable in states reached later, even when it is unattractive now.

One can see this without assuming an exercise boundary. At any intermediate time with remaining life $u > 0$, a European put satisfies $P_E(s,u) \le Ke^{-ru}$ because its payoff never exceeds K. If $s < K(1 - e^{-ru})$, immediate exercise gives $K - s > Ke^{-ru} \ge P_E(s,u)$. Such sufficiently low positive stock prices have positive probability under a nondegenerate lognormal model. A strategy that exercises in those states and otherwise keeps the European payoff improves expected discounted value. Thus a current positive European time value does not eliminate the American early-exercise premium.

21.7 Problem 4.7: very long maturity with and without dividends

Question. Let $S = 100$, $K = 120$, $\sigma = .30$, $r = .08$. (a) With $\delta = 0$, calculate the call price at one year and several very long maturities, and find its limit as $T \to \infty$. (b) Repeat with $\delta = .001$ and explain the difference.

Solution. For each maturity recompute d_1 and d_2; maturity is not a parameter which can be changed only in the discount factors.

T	$C, \delta = 0$	$C, \delta = .001$
1	7.896571	7.854234
10	56.237666	55.373317
50	98.095926	93.229577
100	99.963140	90.447088
1000	100.000000	36.787944
10000	100.000000	.004540

In (a), $d_1 = [\log(100/120) + (.08 + .045)T]/(.30\sqrt{T}) \to +\infty$ and d_2 also tends to $+\infty$. Hence the stock term tends to 100 and the discounted-strike term tends to zero: $\boxed{C \to 100}$. In (b), the no-arbitrage bound $0 \le C \le Se^{-.001T}$ immediately gives $\boxed{C \to 0}$. A positive dividend yield reduces the prepaid value of stock delivered only at a very remote date. Initially extra optionality can raise the call value as maturity increases, but ultimately the dividend discount dominates. The table shows the resulting nonmonotonicity; one must not assume that every dividend-paying European call increases in value with maturity.

21.8 Problem 4.8: a positive dividend yield dominates the limit

Question. Let $S = 120$, $K = 100$, $\sigma = .30$, $\delta = .08$. (a) With $r = 0$, compute the call at one year and very long maturities and find the limit. (b) Repeat with $r = .001$. What accounts for the difference?

Solution. Direct evaluation gives

T	$C, r = 0$	$C, r = .001$
1	18.670533	18.728146
10	10.157142	10.287809
50	.293803	.304512
100	.003396	.003602
1000	2.63×10^{-37}	3.94×10^{-37}

For *both* rates, $0 \le C \le 120e^{-.08T} \to 0$, so the limit is zero. A small positive rate makes finite-maturity calls slightly more valuable because the present value of the strike is lower. It cannot overcome the upper bound imposed by the positive dividend yield. The question's request for a "difference" must therefore refer to finite-maturity prices, not to different limiting values. Changing the instrument to a put would change the question and is not done here.

21.9 Problem 4.9: the stock–strike and rate–yield duality

Question. (a) Price a European call with $S = 100$, $K = 90$, $\sigma = .30$, $r = .08$, $\delta = .05$, $T = 1$. (b) Price a European put with $S = 90$, $K = 100$, $\sigma = .30$, $r = .05$, $\delta = .08$, $T = 1$. (c) Explain the connection between the two answers.

Solution. For (a), evaluate

$$C = 100e^{-.05}N\left(\frac{\log(100/90) + (.08 - .05 + .045)}{.30}\right) - 90e^{-.08}N(d_1 - .30) = \boxed{17.698814}.$$

For (b), the put thresholds satisfy $d_1' = -d_2$ and $d_2' = -d_1$ from (a). Indeed,

$$d_1' = \frac{\log(90/100) + (.05 - .08 + .045)}{.30} = -d_2.$$

Consequently

$$P = 100e^{-.05}N(-d_2') - 90e^{-.08}N(-d_1') = 100e^{-.05}N(d_1) - 90e^{-.08}N(d_2) = \boxed{17.698814}.$$

This proves (c): swapping stock and strike and simultaneously swapping the risk-free rate and dividend yield converts the call expression into the put expression. The identity is $C(S, K, r, \delta) = P(K, S, \delta, r)$ at the same volatility and maturity. It is not a statement that a call and put with identical inputs generally have equal prices.

22 Greeks and hedging: worked problems

These generated solutions distinguish a position's marked-to-market change from its financing-adjusted profit. A positive initial investment I means cash must be supplied; its opportunity cost over h is $I(e^{rh} - 1)$. A negative investment is an initial cash receipt. Unless a display unit is explicitly changed, theta is per year and vega and rho are derivatives with respect to a unit decimal change, not a percentage point.

22.1 Problem 5.1: recovering a price from delta

Question. In the Black–Scholes framework, consider a three-month American call with strike 41.5 on a non-dividend-paying stock. The stock is 40, volatility is 30%, and current call delta is .5. Determine the current price. Write $J = \int_{-\infty}^{.15} e^{-x^2/2}\,dx$. The choices are (A) $20 - 20.453J$; (B) $20 - 16.138J$; (C) $20 - 40.453J$; (D) $16.138J - 20.453$; (E) $40.453J - 20.453$.

Solution. For a non-dividend-paying stock and a nonnegative risk-free rate, early exercise of an American call is not optimal: keeping the option preserves its downside protection and postpones paying the strike. Its value and delta equal those of the European call. Using this regime, $\Delta = N(d_1) = .5$ implies $d_1 = 0$. Since $\sigma\sqrt{\tau} = .30(.5) = .15$, $d_2 = -.15$.

The omitted interest rate can be recovered rather than guessed:

$$0 = \log(40/41.5) + (r + .30^2/2)(.25), \quad r = -\frac{\log(40/41.5)}{.25} - .045 = .1022558925.$$

It is positive, consistent with the no-early-exercise regime used above. Thus $Ke^{-r\tau} = 40.45254077$ and

$$C = 40(.5) - 40.45254077N(-.15).$$

The question's integral omits the normalizing factor, so $N(.15) = J/\sqrt{2\pi}$. By symmetry, $N(-.15) = 1 - J/\sqrt{2\pi}$. Consequently

$$C = 20 - 40.45254077 + \frac{40.45254077}{\sqrt{2\pi}}J = \boxed{16.13822886J - 20.45254077},$$

which is choice **D**. Numerically $C = 2.185417$. The positive upper limit in J does not mean that d_2 is positive; the complement transformation is essential.

22.2 Problem 5.2: the size of a break-even daily move

Question. A delta-hedging market-maker sells a three-month at-the-money European call on a non-dividend-paying stock. The Black–Scholes framework holds, $r = .10$, $S = 50$, call delta is .61791, and a year has 365 days. If the market-maker has zero profit or loss after one day, find the stock-price move. Choices: (A) .41; (B) .52; (C) .63; (D) .75; (E) 1.11.

Solution and qualification. A daily move cannot be determined exactly from a first-order delta hedge alone. The intended calculation is a local, financing-adjusted second-order approximation. There is also a second implied-volatility branch in the supplied data, which must be exposed.

Since $N(.3) = .61791142$, the rounded delta implies $d_1 \simeq .3$. At the money with $\tau = .25$,

$$d_1 = \frac{(.10 + \sigma^2/2)(.25)}{.5\sigma} = \frac{.05}{\sigma} + .25\sigma.$$

Setting this equal to .3 gives $.25\sigma^2 - .30\sigma + .05 = 0$, whose roots are $\sigma = .20$ and $\sigma = 1.00$. The multiple-choice answer implicitly uses the 20% branch.

To derive the approximation, sell one call for C, buy Δ shares and put $C - \Delta S$ in the bank. With no rebalancing inside the day, the terminal total is

$$\Pi = -C(S + x, \tau - h) + \Delta(S + x) + (C - \Delta S)e^{rh}, \qquad h = 1/365.$$

Using calendar-time theta, Taylor expansion and cancellation of the delta terms give

$$\Pi \simeq -\Theta h - \tfrac{1}{2}\Gamma x^2 + r(C - \Delta S)h.$$

The Black–Scholes equation is $\Theta + \frac{1}{2}\sigma^2 S^2\Gamma + rS\Delta - rC = 0$. Substitution therefore yields

$$\Pi \simeq \tfrac{1}{2}\Gamma(\sigma^2 S^2 h - x^2).$$

Break-even occurs for $x \simeq \pm\sigma S\sqrt{h}$. On the intended branch, the magnitude is $.20(50)/\sqrt{365} = \boxed{.523424}$, giving choice **B**. On the other volatility branch it is 2.617120, not an offered answer.

These are local estimates, not exact one-day identities. Repricing the call exactly while including financing gives approximately $x = -.51120$ or $x = .53492$ on the 20% branch. Ignoring financing would give yet another approximation, about .69057 in magnitude. Thus choice B requires the usual local funded-hedge interpretation and the lower-volatility branch; neither the direction nor an exact unique move follows from the literal data alone.

22.3 Problem 5.3: portfolio elasticity is value-weighted delta

Question. Under Black–Scholes, the stock, a European call and a European put currently cost 45.00, 4.45 and 1.90. Investor A buys two calls and one put. Investor B buys two calls and writes three puts. A's portfolio elasticity is 5.0 and B's portfolio delta is 3.4. Find the put's elasticity. Choices: (A) $-.55$; (B) -1.15; (C) -8.64; (D) -13.03; (E) -27.24.

Solution. Write d_C, d_P for the individual deltas. A's value is $V_A = 2(4.45) + 1.90 = 10.80$ and its delta is $2d_C + d_P$. By the definition $\Omega = (S/V)\Delta$,

$$5 = \frac{45}{10.80}(2d_C + d_P), \qquad 2d_C + d_P = 1.20.$$

B's short puts enter with a minus sign, so $2d_C - 3d_P = 3.40$. Subtracting A's delta equation from B's gives $-4d_P = 2.20$, hence $d_P = -.55$ and $d_C = .875$. The requested quantity is not the put delta but

$$\Omega_P = \frac{45}{1.90}(-.55) = \boxed{-13.026316},$$

choice **D**. The negative sign reflects that a put loses value when the stock rises; the large magnitude reflects leverage relative to the small put premium.

22.4 Problem 5.4: changing time while holding the stock fixed

Question. A 50–60 bull spread is valued with $r = .05$, no dividends, $S = 50$, $\sigma = .20$ and three months to expiry. How much does its delta change after one month if the stock does not move? Choices: (A) increases .04; (B) increases .02; (C) unchanged to .01; (D) decreases .02; (E) decreases .04.

Solution. A bull call spread buys the 50-strike call and writes the 60-strike call, so

$$\Delta_{\text{spr}}(\tau) = N\left(\frac{\log(50/50) + (.05 + .20^2/2)\tau}{.20\sqrt{\tau}}\right) - N\left(\frac{\log(50/60) + (.05 + .20^2/2)\tau}{.20\sqrt{\tau}}\right).$$

Initially $\tau = 3/12$ gives .51980596. One month later $\tau = 2/12$ gives .53850526. Their difference is $+.01869929$, or an increase of approximately $\boxed{.02}$, choice **B**. The two calls' deltas change differently as time passes. Holding S fixed does not hold delta fixed, and one must not retain the old τ inside d_1.

22.5 Problem 5.5: reading the capped-stock payoff

Question. A one-year European contingent claim is written on a stock currently worth 45, with volatility 25%, continuous dividend yield 3% and continuously compounded risk-free rate 7%. Black–Scholes holds. The payoff graph rises linearly from $(0, 0)$ to $(42, 42)$, then remains at 42: $H(S_1) = \min(S_1, 42)$. Calculate its current elasticity. Choices: (A) .24; (B) .29; (C) .34; (D) .39; (E) .44.

Solution. The graph has the identity $\min(s, 42) = s - (s - 42)^+$. Replicate the terminal stock delivery with a prepaid forward and subtract a 42-strike European call. A stock delivered at time 1 has prepaid value $Se^{-\delta}$, not S, because dividends before delivery are not part of the claim. Thus

$$V = 45e^{-.03} - C(45, 42, 1, .07, .25, .03) = 36.91247834.$$

Differentiate this replication at fixed parameters. The prepaid-forward delta is $e^{-.03}$ and call delta is $e^{-.03}N(d_1)$, so

$$\Delta_V = e^{-.03}[1 - N(d_1)] = .2789142833, \quad d_1 = \frac{\log(45/42) + (.07 - .03 + .25^2/2)}{.25}.$$

Therefore $\Omega_V = (45/36.91247834)(.2789142833) = \boxed{.34002439}$, choice **C**. The cap removes much of the upside sensitivity while leaving positive delta.

22.6 Problem 5.6: a stale hedge and an inferred funding factor

Question. An investor sold 100 one-year European calls on a non-dividend-paying stock several months ago, immediately delta-hedged with shares, and never rebalanced. She now closes every position. The observations are

	Initially	Now
Stock	40.00	50.00
Call	8.88	14.42
Put	1.63	.26
Call delta	.794	—

The put has the same strike and expiry as the call, and the risk-free rate is constant. Find the profit. Choices: (A) 11; (B) 24; (C) 126; (D) 217; (E) 240.

Solution. The initial hedge buys $100(.794) = 79.4$ shares for 3176, funded partly by call receipts of 888. The initial bank balance is $b_0 = 888 - 3176 = -2288$: the strategy borrows 2288. The present share value is 3970 and repurchasing the calls costs 1442. Before interest, the excess over the original investment is $3970 - 1442 - 2288 = 240$, but that omits the borrowing cost.

The elapsed time and rate need not be supplied separately. At each date parity gives the value of the same strike payment:

$$Ke^{-r\tau_0} = 40 - (8.88 - 1.63) = 32.75, \quad Ke^{-r\tau_1} = 50 - (14.42 - .26) = 35.84.$$

Taking the ratio cancels K and yields the funding accumulation $e^{r(\tau_0 - \tau_1)} = 35.84/32.75 = 1.094351145$. The debt is therefore $2288(35.84/32.75)$, and final net profit is

$$79.4(50) - 100(14.42) - 2288\frac{35.84}{32.75} = \boxed{24.124580},$$

choice **B**. The initially delta-neutral hedge is not delta-neutral throughout an unhedged interval; using today's delta instead would change the specified strategy.

22.7 Problem 5.7: the Greeks of a bull spread

Question. Buy a 40-strike call and sell a 45-strike call, with $\sigma = .30$, $r = .08$, $\delta = 0$, $T = .5$. (a) Find delta, gamma, vega, theta and rho at $S = 40$. (b) Repeat at $S = 45$. (c) Explain any differences.

Solution. Both calls have the same expiry and volatility. For $j \in \{40, 45\}$ put

$$d_{1,j} = \frac{\log(S/j) + (.08 + .30^2/2)(.5)}{.30\sqrt{.5}}, \quad d_{2,j} = d_{1,j} - .30\sqrt{.5}.$$

The spread value is $C_{40} - C_{45}$. Linearity of differentiation means each Greek is the long call's Greek minus the short call's Greek, with no additional weighting:

$$\begin{aligned}
\Delta &= N(d_{1,40}) - N(d_{1,45}),\\
\Gamma &= \frac{\phi(d_{1,40}) - \phi(d_{1,45})}{S(.30)\sqrt{.5}},\\
\text{Vega} &= S\sqrt{.5}[\phi(d_{1,40}) - \phi(d_{1,45})],\\
\Theta &= -\frac{S(.30)}{2\sqrt{.5}}[\phi(d_{1,40}) - \phi(d_{1,45})] - .08e^{-.04}[40N(d_{2,40}) - 45N(d_{2,45})],\\
\rho &= .5e^{-.04}[40N(d_{2,40}) - 45N(d_{2,45})].
\end{aligned}$$

Here ϕ is the standard normal density. Numerical substitution answers (a) and (b):

Quantity	$S = 40$	$S = 45$
Spread value	2.024875	3.059556
Delta	.218663	.186438
Gamma	$-.00042721$	$-.01089259$
Vega, per unit	$-.102530$	-3.308625
Theta, per year	$-.506972$	.566174
Rho, per unit	3.360822	2.665084

For per-percentage-point vega or rho divide by 100; for daily theta divide by 365. For (c), the differences arise because the two strikes have different moneyness at each stock price. Although each individual long call has positive gamma and vega, their *difference* need not. In particular, Vega $= S^2\sigma T\Gamma$ for this same-expiry spread, confirming that the two have the same sign. Theta can change sign as the balance of the two options' time decay changes. A bull spread is positive delta here, but is not automatically positive every other Greek.

22.8 Problem 5.8: a delta-hedged short call overnight

Question. Let $S = 40$, $\sigma = .30$, $r = .08$, $\delta = 0$. Sell a 45-strike call with 91 days remaining. (a) Find delta. (b) If the option covers 100 shares, what investment is required to delta-hedge it? (c) Find the overnight profit if tomorrow's stock is 39. (d) Repeat if it is 40.50.

Solution. Use actual days, $\tau = 91/365$, $h = 1/365$. The call thresholds are $d_1 = -.57825035$, $d_2 = -.72804473$. The long call price is .9710267842 and its delta is $N(d_1) = .2815475557$. This is the answer to (a); the *short* call position has the negative of that delta.

For (b), the short contract has delta $-100(.2815475557)$, so buy $q = 28.15475557$ shares. Stock costs $40q = 1126.190223$ and option receipts are $100C = 97.10267842$. The net initial investment is

$$I = qS - 100C = \boxed{1029.087544}.$$

For (c) and (d), the share holding stays fixed during the night. Reprice the call with *90*, not 91, days left. If the initial investment is financed, the net overnight profit is

$$\Pi(S') = q(S' - 40) - 100[C(S', 90/365) - C(40, 91/365)] - I(e^{.08/365} - 1).$$

The last term is .2255781544. Full repricing gives

S'	Marked-position change before funding	Profit after funding
39	-1.596903	$\boxed{-1.822481}$
40.50	.716090	$\boxed{.490512}$

Both conventions are shown because a table which ignores financing reports the middle column. Delta neutrality is local; it does not force zero finite-move profit, and the passage of one day contributes theta as well as funding.

22.9 Problem 5.9: an overnight delta-hedged bull spread

Question. With $S = 40$, $\sigma = .30$, $r = .08$, $\delta = 0$, buy a 40–45 bull spread on 100 shares with 91 days to expiry. (a) What investment is required after delta hedging? (b) What is the overnight profit if the stock becomes 39? (c) Repeat for 40.50.

Solution. Let $V = C_{40} - C_{45}$ for the 91-day spread. Direct evaluation gives $V = 1.809374837$ and $D = \Delta_{40} - \Delta_{45} = .3008566022$. Since this time the options are *long* the spread, delta neutrality requires *shorting* $q = 100D = 30.08566022$ shares. For (a),

$$I = 100V - q(40) = 180.937484 - 1203.426409 = \boxed{-1022.488925}.$$

Thus the hedged position generates an initial receipt of 1022.488925; it does not require that much cash to be paid in. Keep the receipt in the bank for a self-financing zero-initial-wealth strategy.

After one day the spread still has 90 days remaining. Its funded profit is

$$\Pi(S') = 100[V(S', 90/365) - V(40, 91/365)] - q(S' - 40) - I(e^{.08/365} - 1).$$

Because $I < 0$, the funding term adds interest of .2241317231. The answers to (b) and (c) are

S'	Before funding	Including interest on cash receipt
39	.044428	$\boxed{.268560}$
40.50	$-.234297$	$\boxed{-.010165}$

Using a terminal payoff for either call would be incorrect: neither has expired overnight. The portfolio's finite-move sensitivity also cannot be read from the positive delta of the unhedged bull spread, because the stock position deliberately cancels that first-order term.

23 Credit risk: worked problems

The supplied exercise sheet identifies Problems 6.1–6.5 as required for examination preparation. Problems 6.6–6.9 are retained and solved as supplementary material; this preserves their content without changing the stated examination scope. All solutions here are generated. Pricing probabilities are risk-neutral unless another measure is expressly identified.

23.1 The structural calculation used in Problems 6.1–6.4

In the single-debt Merton model, asset value A_T is lognormal under $\mathbb{Q}$:

$$A_T = A_0 \exp[(r - \sigma^2/2)T + \sigma\sqrt{T}Z], \qquad Z \sim N(0,1).$$

There are no dividends or other payouts and one debt payment D at maturity. Debt receives $\min(A_T, D)$, equity receives $(A_T - D)^+$, and default occurs only at T when $A_T < D$. Put

$$d_1 = \frac{\log(A_0/D) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

The threshold transformation gives $p = \mathbb{Q}(A_T < D) = N(-d_2)$. Completing the square in the lower-tail lognormal integral gives

$$E^{\mathbb{Q}}[A_T \mathbf{1}_{\{A_T < D\}}] = A_0 e^{rT} N(-d_1).$$

Thus the conditional recovery amount and the conditional loss fraction are

$$R = E^{\mathbb{Q}}[A_T \mid A_T < D] = \frac{A_0 e^{rT} N(-d_1)}{p}, \qquad \ell = E^{\mathbb{Q}}[\text{LGD} \mid \text{default}] = 1 - \frac{R}{D}.$$

The debt price, continuously compounded promised yield and spread are

$$B_0 = e^{-rT}\{D(1-p) + pR\} = A_0 N(-d_1) + De^{-rT} N(d_2),$$
$$\rho = \frac{1}{T}\log\frac{D}{B_0} = r - \frac{1}{T}\log(1 - p\ell), \qquad s = \rho - r.$$

The physical drift α is not used in these pricing formulas. It would matter for a physical default forecast, which is a different question. The yield is the discount rate matching the *promised* payment to price, not the expected realized return after default losses.

23.2 Problem 6.1: default probability and spread

Question. A firm has assets of 100, volatility 40%, physical drift 15% and no dividends. The risk-free rate is 8%. Its only debt promises 120 in five years. Calculate the risk-neutral bankruptcy probability and credit spread.

Solution. Insert $A_0 = 100, D = 120, T = 5, r = .08, \sigma = .40$. Then

$$d_1 = \frac{\log(100/120) + (.08 + .08)5}{.40\sqrt{5}} = .69058549, \qquad d_2 = -.20384170.$$

The event $A_5 < 120$ is equivalent to $Z < -d_2$, so its probability is $p = N(.20384170) = \boxed{.58076140}$. The debt value is

$$B_0 = 100N(-.69058549) + 120e^{-.4}N(-.20384170) = 58.21418812.$$

Consequently $\rho = \log(120/58.21418812)/5 = .1446725271$, and the requested spread is

$$s = .1446725271 - .08 = \boxed{.06467253},$$

or 6.467253 percentage points, approximately 646.73 basis points. Using the firm's 15% physical drift inside d_2 would answer a forecasting question instead of the requested risk-neutral one.

23.3 Problem 6.2: conditional loss and a yield approximation

Question. A firm has assets of 100, $\sigma = .40$, $\alpha = .15$, no dividends and $r = .08$. Its single zero-coupon bond promises 100 in two years. Under the risk-neutral measure, calculate its yield, default probability and expected loss given default. How many basis points does $\rho_{\text{app}} = r + T^{-1}pE(\text{LGD})$ differ from the exact yield?

Solution. Here $\log(A_0/D) = 0$ and $r = \sigma^2/2 = .08$. Thus $d_2 = 0$ and $d_1 = .40\sqrt{2} = .56568542$. In particular $p = N(0) = \boxed{.5}$; this is a terminal-default model, so it is not a first-passage probability. Conditional recovery is

$$R = \frac{100e^{.16}N(-.56568542)}{.5} = 67.07877853, \quad \ell = 1 - R/100 = \boxed{.32921221}.$$

The expected loss *amount* given default is 32.92122147, while expected unconditional loss as a fraction of face value is $p\ell = .1646061074$. Distinguishing these quantities prevents applying the default probability twice.

$$B_0 = 100e^{-.16}(1 - .1646061074) = 71.18757170, \quad \rho = \frac{1}{2}\log(100/B_0) = \boxed{.16992597}.$$

The approximation replaces $-\log(1-x)$ by x:

$$\rho_{\text{app}} = .08 + \frac{.5(.3292122147)}{2} = .16230305.$$

Hence $(\rho_{\text{app}} - \rho)10^4 = \boxed{-76.22915 \text{ bp}}$: it understates yield by about 76.23 bp. The direction follows from $-\log(1-x) > x$ for $0 < x < 1$, rather than from numerical rounding.

23.4 Problem 6.3: recovery and the exact logarithmic identity

Question. With assets 100, $\sigma = .40$, $\alpha = .15$, no dividends and $r = .08$, a firm issues a single three-year zero-coupon bond promising 110. (a) Compute the risk-neutral default probability, expected recovery, bond price and yield. (b) Verify $\rho = r + T^{-1}\log[1/(1 - pE(\text{LGD}))]$.

Solution. The thresholds are

$$d_1 = \frac{\log(100/110) + .16(3)}{.40\sqrt{3}} = .55525193, \qquad d_2 = -.13756839.$$

For (a), $p = N(.13756839) = \boxed{.55470923}$. The recovery amount conditional on default is

$$R = \frac{100e^{.24}N(-.55525193)}{.55470923} = \boxed{66.31404877}.$$

Its recovery fraction is .60285499, so $\ell = .39714501$. The unconditional recovery paid in default states is pR, whereas the entire expected bond payment is $110(1-p) + pR$. Discounting that latter amount gives

$$B_0 = e^{-.24}[110(1-p) + pR] = \boxed{67.46671139}, \quad \rho = \tfrac{1}{3}\log(110/B_0) = \boxed{.16294868}.$$

For (b), factor the expected payment as

$$110(1-p) + pR = 110[1 - p(1 - R/110)] = 110(1 - p\ell).$$

Taking logarithms of $B_0 = 110e^{-.24}(1 - p\ell)$ yields

$$\frac{1}{3}\log\frac{110}{B_0} = .08 - \frac{1}{3}\log(1 - p\ell) = .16294868.$$

This verifies the identity algebraically and numerically; it is exact within the model, unlike the first-order approximation in Problem 6.2.

23.5 Problem 6.4: maturity, default and recovery

Question. A firm has assets 100, $\sigma = .40$, $\alpha = .15$, no dividends and $r = .08$, and issues one zero-coupon bond. (a) For face value 80, calculate risk-neutral default probabilities at maturities 5, 10 and 20 years. (b) Does default probability rise or fall with maturity? Explain. (c) How does expected recovery change with maturity? Explain.

Solution. Since $r - \sigma^2/2 = 0$,

$$d_2(T) = \frac{\log(100/80)}{.40\sqrt{T}}, \qquad p(T) = N\left(-\frac{\log(1.25)}{.40\sqrt{T}}\right).$$

For (a) and (c), evaluate this threshold and the conditional recovery integral:

T	$p(T)$	$E[A_T \mid A_T < 80]$	Recovery fraction
5	.40149395	46.94047288	.58675591
10	.42998574	38.68787827	.48359848
20	.45036428	30.61350456	.38266881

For (b), $\log(1.25) > 0$ and $d_2(T)$ decreases towards zero as T increases. Therefore $-d_2(T)$ increases and, since N is increasing, default probability increases towards one half. This conclusion uses these parameters. A general structural default probability need not be monotone for every drift and leverage configuration.

For (c), the relevant expectation conditions on default, so it concerns the *lower tail*, not the unconditional mean $100e^{.08T}$. More time spreads the lognormal distribution more widely. Among default states, increasingly severe asset shortfalls reduce the conditional recovery, as the computed values show. Explicitly

$$R(T) = \frac{100e^{.08T}N[-d_2(T) - .40\sqrt{T}]}{N[-d_2(T)]}.$$

The factor $e^{.08T}$ does not make recovery increase: the shrinking lower-tail probability in the numerator must also be included. This is why simply applying expected asset growth to today's asset value is not a recovery calculation.

23.6 Problem 6.5: a jump to zero

Question. The firm has assets 100, $\sigma = .40$, $\alpha = .15$, no dividends and $r = .08$. There is a 3% per-year chance that assets jump to zero. The firm issues five-year zero-coupon debt promising 110. In a Merton jump model, compute the risk-neutral debt price and yield and compare them with the no-jump case.

Solution and probability convention. Interpret "3% per year" as risk-neutral Poisson intensity $\lambda = .03$, independent of the Brownian motion. A quoted physical catastrophe probability would not by itself identify the risk-neutral intensity. Also, an intensity of .03 implies annual jump probability $1 - e^{-.03}$, not exactly .03; if the wording instead means an exact annual probability, use $\lambda = -\log(.97)$ in the formulas below.

Let N_t count catastrophe events, and let zero be absorbing. Before the first jump the risk-neutral dynamics must have drift $r + \lambda$:

$$\frac{dA_t}{A_{t-}} = (r + \lambda)dt + \sigma dW_t^{\mathbb{Q}} - dN_t.$$

The extra λ compensates the expected proportional loss from the jump. Omitting it would make the discounted asset fail the required risk-neutral mean condition. If no jump occurs before T,

$$A_T^{\text{surv}} = A_0 e^{(r+\lambda-\sigma^2/2)T+\sigma W_T};$$

otherwise debt receives zero. Independence therefore gives

$$B_\lambda = e^{-rT} e^{-\lambda T} E^{\mathbb{Q}}[\min(A_T^{\text{surv}}, D)].$$

Set $d_{1,\lambda} = [\log(A_0/D) + (r + \lambda + \sigma^2/2)T]/(\sigma\sqrt{T})$ and $d_{2,\lambda} = d_{1,\lambda} - \sigma\sqrt{T}$. Evaluating the two truncated lognormal pieces yields

$$B_\lambda = A_0 N(-d_{1,\lambda}) + De^{-(r+\lambda)T} N(d_{2,\lambda}).$$

Thus the debt formula is the no-jump formula evaluated with effective rate $r + \lambda$, although the actual bank rate remains r. At $\lambda = .03$,

$$B_\lambda = \boxed{50.24380601}, \qquad \rho_\lambda = \tfrac{1}{5}\log(110/B_\lambda) = \boxed{.15671862}.$$

Without jumps, $B_0 = \boxed{55.27766570}$ and $\rho_0 = \boxed{.13762228}$. Catastrophe risk lowers price by 5.03385970 and raises promised yield by about 190.96 bp. Multiplying the original no-jump debt price only by a survival probability would miss the compensating change in the conditional asset drift.

23.7 Problem 6.6: four-year rating migration (supplementary)

Question. There are three ratings, F (first-rate), FF (future failure?) and FFF (fading, forlorn, and forsaken). The annual transition matrix, with rows denoting the current rating, is

$$M = \begin{pmatrix} .90 & .07 & .03 \\ .15 & .80 & .05 \\ .10 & .30 & .60 \end{pmatrix}.$$

A firm starts in F. (a) Find the probability of F after four years. (b) Find the probability of FF or FFF after four years. (c) Are firms tending towards higher or lower ratings over time? Explain from the matrix.

Solution. This calculation assumes the displayed matrix applies each year and the current rating is the state variable of a time-homogeneous Markov chain. Summing over intermediate states means matrix multiplication, not raising every matrix entry to the fourth power. For example,

$$(M^2)_{11} = .90(.90) + .07(.15) + .03(.10) = .8235.$$

Continuing the multiplication gives

$$M^4 = \begin{pmatrix} .72088975 & .21130150 & .06780875 \\ .40166750 & .50798175 & .09035075 \\ .34531250 & .47053450 & .18415300 \end{pmatrix}.$$

The first row applies to an initial F, so (a) is $\boxed{.72088975}$ and (b) is $.21130150 + .06780875 = \boxed{.27911025}$. These exhaust the alternatives and sum to one.

For (c), the initial state matters. A firm initially in the best state cannot improve; its chance of remaining in F declines from one. A firm in FFF, however, has annual upward transition probability $.10 + .30 = .40$, while an FF firm has .15 upward and .05 downward transition probability. There is therefore no initial-distribution-free claim that every firm simply deteriorates. A long-run comparison can be made precisely by solving $\pi M = \pi$, $\sum_i \pi_i = 1$:

$$\pi = (.58295964, .33183857, .08520179).$$

All entries of M are positive, so the chain is irreducible and aperiodic and this stationary distribution is the limiting distribution. It has more F than FF and FFF, but an initial all-F population still moves down from its initial best possible composition.

23.8 Problem 6.7: other initial ratings (supplementary)

Question. Use the same three ratings and transition matrix

	F	FF	FFF
F	.90	.07	.03
FF	.15	.80	.05
FFF	.10	.30	.60

.

One firm starts in FF and another in FFF. For each, find the probability that after four years it retains its initial rating, and the probability it has moved to one of the other two ratings.

Solution. Four-year transition probabilities are again entries of M^4. For an initial FF, the second row gives probabilities .40166750, .50798175, .09035075. Thus the probability of the same final rating is $\boxed{.50798175}$ and that of a different final rating is $1 - .50798175 = \boxed{.49201825}$. For an initial FFF, the third row gives .34531250, .47053450, .18415300. The corresponding answers are $\boxed{.18415300}$ and $\boxed{.81584700}$.

"Retains" here means the rating observed at year 4 is the initial rating; it permits intermediate changes followed by a return. If instead the question meant never leaving the initial rating during all four annual observations, the probabilities would be $.8^4 = .4096$ and $.6^4 = .1296$. Those are different path events, not diagonal entries of M^4.

23.9 Problem 6.8: independent defaults and a CDO waterfall (supplementary)

Question. Three speculative-grade bonds each promise 100 in one year. Each defaults only at maturity with risk-neutral probability .10, defaults are independent, recovery is 40%, and $r = .06$. Pool the bonds into tranches promising 160 senior, 50 mezzanine and 90 subordinated. Find each tranche's price and yield.

Solution. If J bonds default, each loses $100(1-.40)=60$, so the pool pays $300-60J$. Independence gives $J\sim\text{Binomial}(3,.10)$, with probabilities $\binom{3}{j}(.1)^j(.9)^{3-j}$. Allocate the pool in priority order:

$$H_s(x)=\min(x,160),\quad H_m(x)=\min((x-160)^+,50),\quad H_e(x)=\min((x-210)^+,90).$$

This produces the complete payoff table:

Defaults	Probability	Pool	Senior	Mezzanine	Subordinated
0	.729	300	160	50	90
1	.243	240	160	50	30
2	.027	180	160	20	0
3	.001	120	120	0	0

Each row sums to the pool payoff. Expected tranche cash flows are

$$\begin{aligned}EH_s&=160(.729+.243+.027)+120(.001)=159.96,\\EH_m&=50(.729+.243)+20(.027)=49.14,\\EH_e&=90(.729)+30(.243)=72.90.\end{aligned}$$

Since probabilities are risk-neutral, price is $e^{-.06}$ times expected payoff. A one-year continuously compounded promised yield is $\log(\text{face}/\text{price})$:

Tranche	Price	Yield
Senior	150.64465479	.06025003
Mezzanine	46.27830918	.07734964
Subordinated	68.65463450	.27072103

As a conservation check, expected tranche payoffs sum to 282, equal to $3[.9(100)+.1(40)]$. Tranching reallocates risk but does not create extra pool value.

23.10 Problem 6.9: perfectly correlated defaults (supplementary)

Question. Repeat the preceding one-year pool with three bonds promising 100 each, risk-neutral default probability .10, recovery 40%, $r=.06$ and tranche faces 160, 50 and 90, but now defaults are perfectly correlated. Find tranche prices and yields.

Solution. Perfect correlation of these equal-probability default indicators means all three default together or none does. The pool pays 300 with probability .9 and 120 with probability .1. Apply the same waterfall: senior receives $(160,120)$ in the two states, mezzanine $(50,0)$ and subordinated $(90,0)$. Therefore

$$EH_s=.9(160)+.1(120)=156,\quad EH_m=.9(50)=45,\quad EH_e=.9(90)=81.$$

Discount and compute promised yields:

Tranche	Price	Yield
Senior	146.91526724	.08531781
Mezzanine	42.37940401	.16536052
Subordinated	76.28292722	.16536052

The equity-like subordinated tranche is worth *more* than under independent defaults: it receives its full 90 more often (.9 rather than .729), although it receives zero in the common-default state. Senior and mezzanine are worth less because simultaneous losses penetrate the waterfall. The two lower tranches have equal promised yields here because each receives its full promised amount with probability .9 and zero otherwise. Total expected cash remains $156 + 45 + 81 = 282$, so total pool price is unchanged; the correlation change redistributes value among tranches.

Part II
Code

24 Reading the MATLAB payoff explorer

The supplied `code1.txt` defines `TradingStrategyGUI`, a MATLAB graphical interface for building portfolios of at most eight positions from stock, a zero-coupon bond, a forward, a European call and a European put. The code is preserved verbatim in the project as `code/TradingStrategyGUI.m`; this section explains its logic rather than replacing it with a new program. It was inspected as source but *not executed in MATLAB* on this PC, so numerical and GUI compatibility are not claimed as tested.

24.1 Inputs, positions and the terminal-payoff curve

The window asks for a stock-price plotting range, today's stock price S_0, continuously compounded interest rate R, volatility σ, and time to expiry T. Each row selects a security type, a strike or bond face value K, and signed units. Positive units buy a position; negative units write or short it. The arrays `pT` and `p0` are the central translation table:

Position	`pT`: payoff at T	`p0`: negative model cost at 0
Stock	S_T	$-S_0$
Bond	K	$-Ke^{-RT}$
Forward	$S_T - K$	$Ke^{-RT} - S_0$
Call	$(S_T - K)^+$	$-c_{BS}(S_0, K, R, T, \sigma)$
Put	$(K - S_T)^+$	$-p_{BS}(S_0, K, R, T, \sigma)$

For each filled row, `evalSchedule` multiplies these expressions by signed units and adds them to two strings. It then replaces symbols with the GUI inputs. The payoff buttons evaluate the combined expression over 100 stock prices between the selected minimum and maximum. The vertical value therefore depends on a hypothetical S_T; the horizontal axis is not time.

24.2 Why the second curve is payoff minus financed cost

In the source code `p0` is negative of each position's model cost. The routine multiplies it by e^{RT}, so its `e` expression is $-e^{RT}C_0$ for total initial cost C_0. The `Cost` and `Payoff & Cost` buttons plot $p + e$:

$$\text{terminal profit in the code's convention} = \text{terminal payoff} - e^{RT} \times \text{initial model cost.}$$

This puts the payoff and premium on the same future date, as in the option examples of Section 3. It is not merely the option's intrinsic payoff. For a multi-leg strategy, signed costs and payoffs of all rows are summed before plotting.

Worked example

To reproduce the long call butterfly in Problem 9, choose a common T and enter three rows: $+1$ call with $K = 90$, -2 calls with $K = 100$, and $+1$ call with $K = 110$. The `Payoff` curve is $(S_T - 90)^+ - 2(S_T - 100)^+ + (S_T - 110)^+$. The adjusted curve subtracts the future value of the net Black–Scholes cost $c_{90} - 2c_{100} + c_{110}$. A payoff diagram can be identified without σ, but the priced-profit curve cannot be drawn without the valuation inputs.

24.3 Pricing and implementation limits

The helper functions `blscall` and `blsput` implement European Black–Scholes formulas using `normcdf`. In particular, they model a non-dividend-paying stock with fixed R, σ and T. The GUI's positive-input callback merely beeps on some invalid values; it does not enforce $S_0, K, T, \sigma > 0$, which the formulas need. Expression strings are evaluated with MATLAB `eval`, so inputs should be trusted and numeric. The code does not model bid–ask spreads, commissions, discrete dividends, early exercise, margin calls or cross-hedge basis risk.

The program's actual on-screen behavior, availability of `normcdf` and required MATLAB toolbox should be checked in a MATLAB installation before treating it as a validated teaching tool. The unmodified 225-line source is included as `code/TradingStrategyGUI.m` in the accompanying Overleaf project and is also reproduced in the source-faithful `Derivatives_Notes`; the study PDF keeps the explanation rather than four pages of tiny code text.

25 Efficient binomial implementation: European and American puts

The PDF listings use $S = 8, K = 10, T = 1, r = 0.03, \sigma = 0.3, M = 256$; the separately supplied script starts from $S = 5, r = 0.06$. These are different numerical examples, not conflicting descriptions of one run. Source code is preserved; the discussion below does not modify its behaviour.

25.1 From backward induction to a terminal weighted sum

Let $h = T/M$, $u = e^{rh+\sigma\sqrt{h}}$, $d = e^{rh-\sigma\sqrt{h}}$ and $p = (e^{rh} - d)/(u - d)$ for the non-dividend example. Number the terminal nodes by $j = 0, \ldots, M$ upward moves:

$$S_{M,j} = S_0 d^{M-j} u^j, \qquad W_j = (K - S_{M,j})^+, \qquad V_0 = e^{-rT} \sum_{j=0}^{M} \binom{M}{j} p^j (1-p)^{M-j} W_j.$$

Code 1 builds the terminal prices by nested updates and rolls option values back with nested loops. Code 2 computes terminal prices directly. Code 3 vectorizes terminal payoffs. Code 4 moves the common discount factor outside the backward recursion for a European payoff. Code 5 vectorizes each rollback. This algebraic movement of discounting uses a constant deterministic rate; do not carry it over indiscriminately to early-exercise decisions. Code 6 evaluates binomial coefficients with `nchoosek`; the source disables warnings and identifies inefficiency. The following implementations avoid forming large factorials.

25.2 Recursive and logarithmic probabilities

Define $a_j = \binom{M}{j} p^j (1-p)^{M-j}$. Then

$$a_0 = (1-p)^M, \qquad a_j = a_{j-1} \frac{M-j+1}{j} \frac{p}{1-p}, \quad j = 1, \ldots, M.$$

Code 7 uses `cumprod`. Code 8 computes

$$\log a_j = \log(M!) - \log(j!) - \log((M-j)!) + j\log p + (M-j)\log(1-p)$$

with cumulative sums of logarithms, reducing overflow/underflow problems associated with factorials. The addendum uses $k = j+1$; its initial binomial coefficient should be read as $\binom{M}{0} = 1$, rather than the printed $\binom{M}{1} = 1$. For Code 9, nonzero put payoffs satisfy $S_0 d^{M-j} u^j < K$. At $u > d > 0$ this is

$$j < \frac{\log(K/(S_0 d^M))}{\log(u/d)}.$$

Only these nodes need to be summed. At exact equality the payoff is zero. The source cutoff expression and clamping are preserved in its listing; no claim is made that every numerical edge case of that implementation is validated.

25.3 Black–Scholes comparison and early exercise

Code 10 provides the European put formula:

$$P_0 = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1), \quad d_1 = \frac{\log(S_0/K) + (r+\sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

The binomial value converges towards the continuous-time value as the discretization is refined under the stated assumptions; finite-M equality is not required. Code 11 values an American put by taking the maximum of immediate exercise and discounted continuation at each rollback. It precomputes powers of u, d and discounted probability weights, but retains the early-exercise comparison.

25.4 Reported computational times

These are the source's historical times, not benchmarks measured on this PC:

Code	1	2	3	4	5	6	7	8	9	10	11
Seconds	.0102	.0073	.0062	.0042	.0068	.1232	.0039	.0036	.0033	.0006	.0092

Codes 1–10 refer to the European put and Code 11 to the American put. Vectorization changes implementation cost; it does not change the payoff or the no-arbitrage pricing rule. The original PDF and separate MATLAB script are the provenance for these examples.

25.5 Supplied comparison script

```
clc

% Parameters
S = 5;
K = 10;
T = 1;
r = 0.06;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%

% CODE 1 ============================================================
tic

W = zeros(M+1,1);
```

```
W(1) = S;

% Asset prices at time T
for i = 1:M
for n = i:-1:1
W(n+1) = u*W(n);
end
W(1) = d*W(1);
end

% Option values at time T
for n = 1:M+1
W(n) = max(K-W(n),0);
end

% Re-trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end

t1 = toc
disp('Option value is'), disp(W(1))
%==========================================================================
clear W

% CODE 2 ==================================================================
tic

W = zeros(M+1,1);
% Asset prices at time T

for n = 1:M+1
W(n) = S*(d^(M+1-n))*(u^(n-1));
end

% Option values at time T
for n = 1:M+1
W(n) = max(K-W(n),0);
end

% Re-trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end

t2 = toc
disp('Option value is'), disp(W(1))
%==========================================================================
clear W

% CODE 3 ==================================================================
tic

% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% Re-trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end

t3 = toc
disp('Option value is'), disp(W(1))
%==========================================================================
clear W

% CODE 4 ==================================================================
tic
```

```
% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% Re-trace to get option value at time zero
q = 1-p;
for i = M:-1:1
for n = 1:i
W(n) = p*W(n+1) + q*W(n);
end
end
W(1) = exp(-r*T)*W(1);

t4 = toc
disp('Option value is'), disp(W(1))
%=========================================================================
clear W

% CODE 5 ==================================================================
tic

% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% Re-trace to get option value at time zero
q = 1-p;
for i = M:-1:1
W = p*W(2:i+1) + q*W(1:i);
end
W(1) = exp(-r*T)*W(1);

t5 = toc
disp('Option value is'), disp(W(1))
%=========================================================================
clear W

% CODE 6 ==================================================================
tic

% nchoosek produces numerical warning
warning off

% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% Binomial coefficients ( nchoosek =  (n!)/(k!(n-k)! )
v = zeros(M+1,1);
for k = 1:M+1
v(k) = nchoosek(M,k-1);
end

value = exp(-r*T)*sum(v.*(p.^([0:M])).*((1-p).^([M:-1:0])).*W)';

t6 = toc
disp('Option value is'), disp(value)
%=========================================================================
clear W v value

% CODE 7 ==================================================================
tic

% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% Recursion for coefficients
a = cumprod([(1-p)^M,(p/(1-p))*[M:-1:1]./[1:M]]);

value = exp(-r*T)*sum(a'.*W);

t7 = toc
disp('Option value is'), disp(value)
%=========================================================================
clear W a value

% CODE 8 ==================================================================
tic
```

```
% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% log/cumsum version
csl = cumsum(log([1;[1:M]']));
tmp = csl(M+1) - csl - csl(M+1:-1:1) + log(p)*([0:M]') + log(1-p)*([M:-1:0]');

value = exp(-r*T)*sum(exp(tmp).*W);

t8 = toc
disp('Option value is'), disp(value)
%========================================================================
clear W csl tmp value

% CODE 9 ================================================================
tic

% cut-off index (for nodes > z, option payoff is zero)
z = max(1,min(M+1,floor( log((K*u)/(S*d^(M+1)))/(log(u/d)) )));

% Option values at time T
W = K - S*d.^([M:-1:M-z+1]').*u.^([0:z-1]');

% log/cumsum version using cut-off index z
tmp1 = cumsum(log([1;[M:-1:M-z+2]'])) - cumsum(log([1;[1:z-1]']));
tmp2 = tmp1 + log(p)*([0:z-1]') + log(1-p)*([M:-1:M-z+1]');

value = exp(-r*T)*sum(exp(tmp2).*W);

t9 = toc
disp('Option value is'), disp(value)
%========================================================================
clear W tmp1 tmp2 value

% CODE 10 ===============================================================
tic

d1 = (log(S/K) + (r + 0.5*sigma^2)*T)/(sigma*sqrt(T));
d2 = d1 - sigma*sqrt(T);
N1 = 0.5*(1+erf(-d1/sqrt(2)));
N2 = 0.5*(1+erf(-d2/sqrt(2)));

value = K*exp(-r*T)*N2 - S*N1;

t10 = toc
disp('Option value is'), disp(value)
%========================================================================
clear d1 d2 N1 N2 value

% CODE 11 ===============================================================
tic

% Re-usable computations
dpowers = d.^([M:-1:0]');
upowers = u.^([0:M]');
scale1 = p*exp(-r*dt);
scale2 = (1-p)*exp(-r*dt);

% Option values at time T
W = max(K - S*dpowers.*upowers,0);

% Re-trace to get option value at time zero
for i = M:-1:1
Si = S*dpowers(M-i+2:M+1).*upowers(1:i);
% Check for early exercise
W = max(max(K - Si, 0), scale1*W(2:i+1) + scale2*W(1:i));
end

t11 = toc
disp('Option value is'), disp(W)
%========================================================================
clear dpowers upowers scale1 scale2 Si W
```

```
%=========================================================================
tt = [t1 t2 t3 t4 t5 t6 t7 t8 t9 t10];
fprintf('\n\nComputational time in seconds for a European put option:\n\n');
for i=1:10,
    fprintf('code %2.0d:\t %2.4f\n',[i tt(i)]);
end
fprintf('\n\nComputational time in seconds for an American put option:\n\n');
fprintf('code 11:\t %2.4f\n',t11);
```

25.6 All eleven source addendum listings

The PDF listings below remove displayed line numbers, replace typographic apostrophes by ASCII apostrophes and visible space glyphs by spaces. The original PDF is the authority for any extraction doubt; the separate `BinomialComparison.m` remains byte-identical to `code2.txt`.

Addendum Code 1

```
% Code 1: Binomial method for a European put.
%
% Unvectorized version.
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
W = zeros(M+1,1);
W(1) = S;
% Asset prices at time T
for i = 1:M
for n = i:-1:1
W(n+1) = u*W(n);
end
W(1) = d*W(1);
end
% Option values at time T
for n = 1:M+1
W(n) = max(K-W(n) ,0);
end
% Re -trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end
disp('Option value is'), disp(W(1))
```

Addendum Code 2

```
% Code 2: Binomial method for a European put.
%
% Initial tree computation improved.
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
```

```
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
W = zeros(M+1,1);
% Asset prices at time T
for n = 1:M+1
W(n) = S*(d^(M+1-n))*(u^(n-1));
end
% Option values at time T
for n = 1:M+1
W(n) = max(K-W(n) ,0);
end
% Re -trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end
disp('Option value is'), disp(W(1))
```

Addendum Code 3

```
% Code 3: Binomial method for a European put.
%
% Partially vectorized
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% Re -trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end
disp('Option value is'), disp(W(1))
```

Addendum Code 4

```
% Code 4: Binomial method for a European put.
%
% Partially vectorized , improved
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Option values at time T
```

```
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% Re -trace to get option value at time zero
q = 1-p;
for i = M:-1:1
for n = 1:i
W(n) = p*W(n+1) + q*W(n);
end
end
W(1) = exp(-r*T)*W(1);
disp('Option value is'), disp(W(1))
```

Addendum Code 5

```
% Code 5: Binomial method for a European put.
%
% Partially vectorized , improved
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% Re -trace to get option value at time zero
q = 1-p;
for i = M:-1:1
W = p*W(2:i+1) + q*W(1:i);
end
W(1) = exp(-r*T)*W(1);
disp('Option value is'), disp(W(1))
```

Addendum Code 6

```
% Code 6: Binomial method for a European put.
%
% Partially vectorized
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% nchoosek produces numerical warning
warning off
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% Binomial coefficients ( nchoosek =  (n!)/(k!(n-k)! )
v = zeros(M+1,1);
for k = 1:M+1
v(k) = nchoosek(M,k-1);
end
value = exp(-r*T)*sum(v.*(p.^([0:M]')).*((1 -p).^([M: -1:0] ')).*W) ';
disp('Option value is'), disp(value)
```

Addendum Code 7

```
% Code 7: Binomial method for a European put.
%
% Vectorized (recursive coefficients )
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% Recursion for coefficients
a = cumprod ([(1-p)^M,(p/(1-p))*[M: -1:1]./[1:M]]);
value = exp(-r*T)*sum(a'.*W);
disp('Option value is'), disp(value)
```

Addendum Code 8

```
% Code 8: Binomial method for a European put.
%
% Vectorized (log recursive coefficients )
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% log/cumsum version
csl = cumsum(log ([1;[1:M]']));
tmp = csl(M+1) - csl - csl(M+1: -1:1) + log(p)*([0:M]') + log(1-p)*([M: -1:0] ');
value = exp(-r*T)*sum(exp(tmp).*W);
disp('Option value is'), disp(value)
```

Addendum Code 9

```
% Code 9: Binomial method for a European put.
%
% Vectorized (log recursive coefficients and null terms)
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
```

```
% cut -off index (for nodes > z, option payoff is zero)
z = max(1,min(M+1, floor( log((K*u)/(S*d^(M+1)))/(log(u/d)) )));
% Option values at time T
W = K - S*d.^([M:-1:M-z+1]').*u.^([0:z-1] ');
% log/cumsum version using cut -off index z
tmp1 = cumsum(log ([1;[M:-1:M-z+2] ])) - cumsum(log ([1;[1:z-1] ']));
tmp2 = tmp1 + log(p)*([0:z-1]') + log(1-p)*([M:-1:M-z+1] ');
value = exp(-r*T)*sum(exp(tmp2).*W);
disp('Option value is'), disp(value)
```

Addendum Code 10

```
% Code 10: Black -Scholes formula for a European put.
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
%
d1 = (log(S/K) + (r + 0.5* sigma ^2)*T)/( sigma*sqrt(T));
d2 = d1 - sigma*sqrt(T);
N1 = 0.5*(1+ erf(-d1/sqrt (2)));
N2 = 0.5*(1+ erf(-d2/sqrt (2)));
value = K*exp(-r*T)*N2 - S*N1;
disp('Option value is'), disp(value)
```

Addendum Code 11

```
% Code 11: Binomial method for an American put.
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Re -usable computations
dpowers = d.^([M: -1:0] ');
upowers = u.^([0:M]');
scale1 = p*exp(-r*dt);
scale2 = (1-p)*exp(-r*dt);
% Option values at time T
W = max(K - S*dpowers .* upowers ,0);
% Re -trace to get option value at time zero
for i = M:-1:1
Si = S*dpowers(M-i+2:M+1).* upowers (1:i);
% Check for early exercise
W = max(max(K - Si , 0), scale1*W(2:i+1) + scale2*W(1:i));
end
disp('Option value is'), disp(W)
```

26 Hedging and advanced-model implementations

The supplied Matlab scripts are reproduced without changing executable statements or source comments. The theoretical arguments are in Sections 17, 19.2 and 19.3; this Part explains the experiments, input choices and implementation limits. These scripts were not executed in Matlab

for this update, and no new numerical Matlab output is claimed. Source functions, toolbox availability and the issues identified below must be checked before interpreting a run.

26.1 Discrete delta and delta–gamma hedging

The first script sells a 110-strike half-year call on a stock at 100, with volatility 20%, physical drift 5% and bank rate 3%. It uses a 115-strike one-year call to hedge gamma, 20,000 paths and 100 steps. It compares the two final-balance histograms and their sample mean, standard deviation and fifth percentile. A second experiment varies 100:50:1000 steps and compares zero transaction costs with 0.1% proportional costs.

`GBMPaths` generates log-price increments, cumulatively sums them and exponentiates; its returned path matrix begins after the first step rather than including the initial stock. `BSCall` is vectorized across paths and returns price, delta and gamma. `hedgingSimulator` prices the second option with its own remaining maturity, clips the gamma ratio to $[-100, 100]$, pays trade costs on stock and second-option adjustments, and updates the bank balance in the order derived in Section 17.

At maturity the code sells both remaining hedge positions and pays the written call's payoff. It does not deduct a final liquidation fee. When the second maturity exceeds the first, its remaining value is a Black–Scholes price, not its payoff. The input comment requests $T_2 > T$; a same-expiry contract would need a separate expiry branch in the pricing function. Also, the literal character-vector test `gammaHedge == 'yes'` is not a robust general string comparison: `'not'` and `'yes'` have equal length in the supplied calls, but arbitrary input lengths or vector logical conditions can behave differently. A maintained implementation would validate the flag, for example using `strcmp`; the original is preserved here. Very small gammas can cause extreme or undefined ratios even before clipping; a cap does not repair a zero-over-zero result. These operational limitations are distinct from the mathematical hedge equations.

```
clc;

% Sold option to hedge, stock price and bank account parameters:
K = 110;
T = 0.5;
S0 = 100;
sigma = 0.2;
mu = 0.05;
r = 0.03;
% Additional option to gamma-hedge:
K2 = 115;
T2 = 1;
% Simulation parameters:
nPaths = 20000;
nSteps = 100;

% Price charged to customer:
[charge,~] = BSCall(K,T,S0,r,sigma);

% Simulate delta-hedging strategy:
pl = hedgingSimulator(  ...
    charge, K, T, S0, r, mu, sigma, nPaths, nSteps, 0, 'not', K2, T2  );
fprintf('\n\nDelta-Hedging - Profit & loss:\n');
fprintf('mean\t std\t 5%% VaR\n%2.4f\t %2.4f\t %2.4f\t',...
    [mean(pl) std(pl), prctile(pl,5)]);
figure(1),subplot(1,2,1),hist(pl,50),xlabel('Final balance');
title('Delta-hedging');

% Simulate delta-gamma-hedging strategy:
pl = hedgingSimulator(  ...
    charge, K, T, S0, r, mu, sigma, nPaths, nSteps, 0, 'yes', K2, T2  );
fprintf('\n\nDelta-Gamma-Hedging - Profit & loss:\n');
fprintf('mean\t std\t 5%% VaR\n%2.4f\t %2.4f\t %2.4f\t',...
    [mean(pl) std(pl), prctile(pl,5)]);
subplot(1,2,2),hist(pl,50),xlabel('Final balance');
title('Delta-gamma-hedging');
```

```
% delta-hedging: impact of transaction costs and rebalancing frequency
std1 = []; std2 = []; var1 =[]; var2 = [];
for ii = 100:50:1000;
    pl = hedgingSimulator(  ...
    charge, K, T, S0, r, mu, sigma, nPaths, ii, 0, 'not', K2, T2  );
    std1 = [std1; std(pl)];
    var1 = [var1; prctile(pl,5)];
    pl = hedgingSimulator(  ...
    charge, K, T, S0, r, mu, sigma, nPaths, ii, 0.001, 'not', K2, T2  );
    std2 = [std2; std(pl)];
    var2 = [var2; prctile(pl,5)];
end
figure(2),
subplot(1,2,1),plot(100:50:1000,std1,'b',100:50:1000,std2,'r');
xlabel('# steps'),title('Profit & loss: Std Dev')
legend('no trans. costs','trans. costs')
subplot(1,2,2),plot(100:50:1000,var1,'b',100:50:1000,var2,'r');
xlabel('# steps'),title('Profit & loss: 5% VaR')
legend('no trans. costs','trans. costs')
%=========================================================================



%=========================================================================
function [ S, times ] = GBMPaths( ...
    T, S0, mu, sigma,nPaths, nSteps  )

dt = T/nSteps;
logS0 = log( S0);
eps = randn( nPaths, nSteps );
dlogS = (mu-0.5*sigma^2)*dt + sigma*sqrt(dt)*eps;
logS = logS0 + cumsum( dlogS, 2);
S = exp(logS);
times = dt:dt:T;

end
%=========================================================================

%=========================================================================
function [ price, delta, gamma ] = BSCall( K, T, S0, r, sigma )
% Computes the price of an option using
% the Black Scholes formula.

numerator = log(S0./K) + (r+0.5*sigma.^2).*T;
denominator = sigma.*sqrt(T);
d1 = numerator./denominator;
d2 = d1 - denominator;
price = S0 .* normcdf(d1) - exp(-r.*T).*K.*normcdf(d2);

delta = normcdf(d1);
gamma = normpdf(d1) ./ (S0.*denominator);

end
%=========================================================================

%=========================================================================
function [ finalBalance] = hedgingSimulator(chargeToCustomer, ...
    K, T, S0, r, mu, sigma, nPaths, nSteps, TC, gammaHedge, K2, T2  )
%
% INPUTS:
%
% Selling price of the call option to hedge: chargeToCustomer
% Stock: S0 mu sigma
% Bank account: r
% Call option to be hedged: K T
% Gamma-hedging: gammaHedge (yes or not)
% Call option to gamma-hedge: K2 T2 (T2 > T)
% Proportional transaction cost: TC (>=0)
% Simulation: nPaths, nSteps
%
% OUTPUT:
%
% Profit & Loss: finalBalance
```

```
% Simulate the stock price ================================================
dt = T/nSteps;
times = dt:dt:T;
paths = GBMPaths( T, S0, mu, sigma, nPaths, nSteps );
%==========================================================================

% Initial Portfolio ========================================================
[~, delta1, gamma1] = BSCall(K,T,S0,r,sigma);
if gammaHedge == 'yes', GH = 1; else GH = 0; end
[p2, delta2, gamma2] = BSCall(K2,T2,S0,r,sigma);
option2Quantity = max(min(gamma1 ./ gamma2, 100), -100);
stockQuantity = delta1 - GH * option2Quantity .* delta2;
stockCost = stockQuantity .* S0;
optionCost = GH * option2Quantity .* p2;
transactionCost = TC*(abs(stockQuantity) .* S0 + ...
                  GH * abs(option2Quantity) .* p2);
balance = chargeToCustomer - stockCost - optionCost - transactionCost;
%==========================================================================

% Dynamic Portfolio Adjustment =============================================
for t=1:(nSteps-1)
    S = paths(:,t);
    timeToMaturity = T-times(t);
    [~,delta1,gamma1] = BSCall(K,timeToMaturity,S,r,sigma);
    timeToMaturity2 = T2-times(t);
    [p2,delta2,gamma2] = BSCall(K2,timeToMaturity2,S,r,sigma);
    newOption2Quantity = max(min(gamma1./gamma2,100),-100);
    newStockQuantity = delta1 - GH * newOption2Quantity .* delta2;
    stockCost = (newStockQuantity - stockQuantity).* S;
    optionCost = GH * (newOption2Quantity - option2Quantity).* p2;
    transactionCost = TC*(abs(newStockQuantity - stockQuantity) .* S + ...
                      GH * abs(newOption2Quantity - option2Quantity) .* p2);
    balance = exp(r*dt)*balance - stockCost - optionCost - transactionCost;
    stockQuantity = newStockQuantity;
    option2Quantity = newOption2Quantity;
end
%==========================================================================

S = paths(:,nSteps);
timeToMaturity2 = T2-T;
[p2,delta2,gamma2] = BSCall(K2,timeToMaturity2,S,r,sigma);
stockPosition = stockQuantity .* S;
option2Position = GH * option2Quantity .* p2;
liability = max(S-K, 0);
balance = exp(r*dt)*balance + stockPosition + option2Position - liability;

finalBalance = balance;
end
%==========================================================================
```

26.2 Heston call pricing, Monte Carlo comparison and calibration

The next script uses $S_0 = K = 100$, $v_0 = .04$, $T = 1$, $r = .02$, $(\kappa, \theta, \xi, \rho) = (3, .04, .30, -.50)$. It compares the Fourier-integral call value with 10,000 Euler-based discounted-payoff samples. It then constructs an 11-by-11 implied-volatility surface with strikes from $0.8S_0$ to $1.2S_0$ and maturities from one month to one year. Synthetic quotes multiply each model implied volatility by $\exp(.10Z)$ with independent normal perturbations; these are simulated quotes, not retrieved market observations.

Calibration starts at $(1.5, .03, .30, -.75)$ and bounds the four parameters by $(.5, .01, .10, -1)$ and $(4, .10, .80, 1)$. The objective is $\log(1 + \sum(\Sigma^{model} - \Sigma^{quote})^2)$; since the logarithm is increasing, it has the same minimizers as the sum of squares. It is not the same as changing the data into log-volatility residuals. No Feller-condition constraint is imposed by these box bounds.

The Fourier routine integrates from 0 to 500 and switches coefficients according to the probability index. This finite truncation needs accuracy checks. At zero frequency the written division by iu has a removable-limit issue, and suppressing a divide-by-zero warning does not evaluate that limit. Complex-log branches can also affect the result. The Euler variance is floored at 10^{-20},

an explicit numerical modification. A further timing issue is that $N = \lfloor T/dt \rfloor$ allocated entries include the initial state, while the loop runs from 2 to N: it generates only $N - 1$ increments of length T/N. Consequently the final simulated state is one step short of T, although the outer call discounts at T. Preserving the script does not endorse that mismatch; a corrected application would use $N + 1$ states or match its terminal time.

```
% MC pricing of the European Call
clc

S0 = 100;
V0 = 0.04;

K    = S0;
T    = 1;
rate = 0.02;

kappa = 3.00;
theta = 0.04;
sigma = 0.30;
rho   = -0.50;

params = [kappa, theta, sigma, rho];

% closed form price
tic
call1 = Heston_Call(params, S0, K, T, rate, V0)
toc

% monte carlo price
Nsim = 10000;
payoff = zeros(Nsim, 1);
tic
for i = 1:Nsim
    S = Heston_Sim_Euler([rate,params], S0, V0, T, 1/250);
    payoff(i) = max(S(end)-K, 0);
end
call2 = exp(-rate*T)*mean(payoff)
toc

% implied volatility surface (nxn)
n = 11;
Strike = S0*linspace(.8,1.2,n);
Maturity = linspace(1/12,1,n);
ivs = HestonSurface(Strike,Maturity, S0, V0, rate,params);
[X,Y] = meshgrid(Strike/S0,Maturity);
figure(1)
surf(X,Y,ivs')
xlabel('K/S'),ylabel('Maturity')
title('Implied Volatility Surface with the Heston Model')


% Assume market implied volatilities are equal to Heston implied volatilies
% up to a perturbation:
Quotes = ivs.*exp(.10*randn(n,n));
figure(2)
surf(X,Y,Quotes')
xlabel('K/S'),ylabel('Maturity')
title('Implied Volatility Surface (Quotes)')
% Calibrate the Heston model by minimizing the distance from market implied
% volatilities.
% Guess parameters:
p0 = [1.5 0.03 0.30 -0.75]; % kappa theta sigma rho
% Lower and upper bounds
lb = [0.5 0.01 0.10 -1.0];
ub = [4.0 0.10 0.80  1.0];
% Minimze squared errors (it takes a little while)
popt = fmincon(@(p)HestonError(Strike,Maturity, S0, V0, rate,p,Quotes),...
            p0,[],[],[],[],lb,ub,[],optimset('Display','iter'));
% Optimal parameters:
popt
% Original parameters
params
% Calibrated Surface
```

```
ivs = HestonSurface(Strike,Maturity, S0, V0, rate,popt);
[X,Y] = meshgrid(Strike/S0,Maturity);
figure(3)
surf(X,Y,ivs')
xlabel('K/S'),ylabel('Maturity')
title('Implied Volatility Surface (Calibrated)')



%-------------------------------------------------------------------------
function res = HestonSurface(strike,maturity, S0, V0, rate,params)
res = zeros(length(strike),length(maturity));
for i=1:length(strike)
    for j=1:length(maturity)
        call = Heston_Call(params, S0, strike(i), maturity(j), rate, V0);
        res(i,j) = blsimpv(S0,strike(i),rate,maturity(j),call);
    end
end
end
%-------------------------------------------------------------------------
function res = HestonError(strike,maturity, S0, V0, rate,params,Quotes)
res = HestonSurface(strike,maturity, S0, V0, rate,params);
res = sum(sum((res - Quotes).^2));
res = log(1+res);
end
%-------------------------------------------------------------------------
function [S, V] = Heston_Sim_Euler(params, S0, V0, T, dt)

mu    = params(1);
kappa = params(2);
theta = params(3);
sigma = params(4);
rho   = params(5);

N  = floor(T/dt);
dt = T/N;

V   = zeros(N, 1);
lnS = zeros(N, 1);

V(1)   = V0;
lnS(1) = log(S0);

w = randn(N, 1);
z = rho*w + sqrt(1 - rho^2)*randn(N, 1);

for t = 2:N

    lnS(t) = lnS(t-1) + (mu - 0.5*V(t-1))*dt + sqrt(dt*V(t-1))*w(t);

    V(t)   = max(1e-20, kappa*theta*dt + (1 - kappa*dt)*V(t-1) + sigma*sqrt(V(t-1)*dt)*z(t));

end

S = exp(lnS);

end
%-------------------------------------------------------------------------
function call = Heston_Call(params, S, K, tau, r, v0)
% compute call option price with the closed-form formula (Heston, 1993)
%
% Heston' model: risk-neutral
% dS/S = r dt + sqrt(vt) dWt
% dvt = kappa(theta - vt) dt + sigma sqrt(vt) dZt
% cov(dWt, dZt) = rho dt
%
% params = [kappa, theta, sigma, rho]
%
warning off MATLAB:divideByZero;

call = S*Heston_P(params, S, K, tau, r, v0, 1) - K*exp(-r*tau)*Heston_P(params, S, K, tau, r, v0, 2);
end
%-------------------------------------------------------------------------
function P = Heston_P(params, S, K, tau, r, v0, type)
```

```
P = 0.5 + 1/pi * quadl(@Heston_P_integrand, 0, 500, [], [], params, S, K, tau, r, v0, type);
end
%-------------------------------------------------------------------------
function integrand = Heston_P_integrand(u, params, S, K, tau, r, v0, type)

integrand = real(exp(-i*u*log(K)).*Heston_f(u, params, S, tau, r, v0, type)./(i*u));
end
%-------------------------------------------------------------------------
function f = Heston_f(u, params, S, tau, r, v0, type)

kappa = params(1);
theta = params(2);
sigma = params(3);
rho   = params(4);

if type == 1
    uu = 0.5;
    bb = kappa - rho*sigma;
else
    uu = -0.5;
    bb = kappa;
end

aa = kappa*theta;
dd = sqrt((i*u*rho*sigma - bb).^2 - sigma^2*(i*u*2*uu - u.^2));
gg = (bb - rho*sigma*u*i + dd)./(bb - rho*sigma*u*i - dd);

CC = r*u.*i*tau + aa/sigma^2.*((bb - rho*sigma*u*i + dd)*tau - 2*log((1-gg.*exp(dd*tau))./(1-gg)));
DD = (bb - rho*sigma*u*i + dd)./sigma^2.*((1-exp(dd*tau))./(1-gg.*exp(dd*tau)));

f = exp(CC + DD*v0 + i*u*log(S));
end
%-------------------------------------------------------------------------
```

26.3 Comparing Euler and exact-variance-step Heston paths

The path-comparison script uses $S_0 = 100, v_0 = .04, T = 1, dt = 1/250, r = .02$ and $(\kappa, \theta, \xi, \rho) = (1.5, .05, .20, -.50)$ for three paths per method. Its four panels show stock and variance from each construction. The efficient scheme samples the CIR endpoint using `ncx2rnd`, approximates integrated variance by the endpoint trapezoid, and adds a conditionally normal independent-price component. The exact endpoint distribution does not make the full stock/variance path simulation exact; see Section 19.2.

The same state-count versus step-count mismatch appears here. The call uses `Heston_Sim_Efficient` while the local function is written `Heston_Sim_efficient`; match spelling and case when packaging the function in an environment where resolution distinguishes them. Independently generated paths are not a pathwise accuracy comparison unless random innovations are coupled; a method's visibly smoother path alone does not prove a smaller pricing error.

```
clc

S0 = 100;
V0 = 0.04;

T  = 1;
dt = 1/250;

rate = 0.02;
kappa = 1.50;
theta = 0.05;
sigma = 0.20;
rho   = -0.50;

params = [kappa, theta, sigma, rho];

Nsim = 3;
St1 = zeros(floor(T/dt),Nsim);
St2 = St1;
Vt1 = St1;
```

```
Vt2 = St1;
for i = 1:Nsim
    [St1(:,i), Vt1(:,i)] = Heston_Sim_Euler([rate,params], S0, V0, T, dt);
    [St2(:,i), Vt2(:,i)] = Heston_Sim_Efficient([rate,params], S0, V0, T, dt);
end

subplot(2,2,1),plot(St1),title('Euler scheme: Stock price');
subplot(2,2,2),plot(Vt1),title('Euler scheme: Variance');
subplot(2,2,3),plot(St2),title('Efficient scheme: Stock price');
subplot(2,2,4),plot(Vt2),title('Efficient scheme: Variance');


%-------------------------------------------------------------------------
function [S, V] = Heston_Sim_Euler(params, S0, V0, T, dt)

mu    = params(1);
kappa = params(2);
theta = params(3);
sigma = params(4);
rho   = params(5);

N  = floor(T/dt);
dt = T/N;

V   = zeros(N, 1);
lnS = zeros(N, 1);

V(1)   = V0;
lnS(1) = log(S0);

w = randn(N, 1);
z = rho*w + sqrt(1 - rho^2)*randn(N, 1);

for t = 2:N

    lnS(t) = lnS(t-1) + (mu - 0.5*V(t-1))*dt + sqrt(dt*V(t-1))*w(t);

    V(t)   = max(1e-20, kappa*theta*dt + (1 - kappa*dt)*V(t-1) + sigma*sqrt(V(t-1)*dt)*z(t));

end

S = exp(lnS);
end
%-------------------------------------------------------------------------
function [St, vt] = Heston_Sim_efficient(params, S0, V0, T, dt)

% simulated Heston stochastic volatility model with an efficien scheme

% Heston Stochastic Volatility Model
% dlnS  = (mu - vt/2) dt + rho*sqrt(vt) dZ + sqrt(1-rho^2)*sqrt(vt) dW
% dVt = kappa*(theta - vt) dt + sigma*sqrt(vt) dZ


mu    = params(1);
kappa = params(2);
theta = params(3);
sigma = params(4);
rho   = params(5);

N  = floor(T/dt);
dt = T/N;

% First simulate volatility process from the scaled non-central chi-square
% distribution
d = 4*kappa*theta/sigma^2; % degree of freedom
C = 4*kappa/(sigma^2*(1 - exp(-kappa*dt))); % inversion of scale parameter
x = C*exp(-kappa*dt); % coefficient of non-centrality parameter
vt = zeros(N, 1);
vt(1) = V0;
% Then simulate stock price process
st = zeros(N, 1); % s = log(S)
st(1) = log(S0);
y = randn(N, 1);
for i = 1:N-1
    D = x*vt(i);
```

```
    vt(i+1) = ncx2rnd(d, D)/C;
    vt_int   = dt*(vt(i+1) + vt(i))/2;
    vt_int_W = sqrt(vt_int)*y(i);
    st(i+1) = st(i) + mu*dt + (vt(i+1) - vt(i) - kappa*theta*dt)*rho/sigma + ...
           (kappa*rho/sigma - 0.5)*vt_int + sqrt(1-rho^2)*vt_int_W;
end
St = exp(st); % back to stock price level
end
%-------------------------------------------------------------------------
```

26.4 The Merton call/put and implied-volatility surface

The option-surface script uses $S = 100$, strikes 80 to 120, maturities .1, .2, . . . , 1, $r = .0075$, dividend yield $q = 0$, diffusion volatility .2, intensity .02, jump log-mean $-.15$, jump log-standard-deviation .3, and 50 count terms. `MertonOptionPrice` uses the tilted-intensity series from Section 19.3. Its third array dimension carries jump counts$0, \ldots, n-1$; summing over that dimension produces the maturity-by-strike price grid. Input $cp = -1$ converts the call to a put with dividend-adjusted put–call parity. The code uses an error-function normal CDF and converts the price surface to Black–Scholes implied volatilities with `blsimpv`.

The comment limits the count at 170 because of direct factorial arithmetic; this is a numerical implementation limit, not a mathematical bound on possible jump events. A chosen finite count needs a truncation check, especially when intensity or maturity increases. The general pricing function accepts a dividend yield, but the illustrated implied-volatility inversion omits that argument consistently with its chosen $q = 0$; a nonzero-yield experiment would need matching inversion conventions.

```
clc
% option inputs
S = 100;
K = (80:1:120)';
T = (0.1:0.1:1)';
r = 0.0075;
q = 0;
cp = 1;
% dynamics parameters
sigma = 0.2;   lambda = 0.02; m = -0.15; delta = 0.3;
% sum truncation
n = 50;

optionPrice = MertonOptionPrice(cp,S,K,T,sigma,r,q,lambda,m,delta,n);

[mK,mT] = meshgrid(K,T);
[impliedVol] = blsimpv(S,mK,r,mT,optionPrice);

subplot(1,2,1); mesh(mK/S,mT,optionPrice);
ylabel('Maturity');xlabel('K/S');title('Merton option price')
subplot(1,2,2); mesh(mK/S,mT,impliedVol);
ylabel('Maturity');xlabel('K/S');title('Merton implied volatility')







function optionPrice = MertonOptionPrice(cp,S,K,T,sigma,r,q,lambda,m,delta,n)
%
% Inputs:
%   cp          [1,-1] Call,Put
%   S           Stock Price
%   K           Strike (Vector)
%   T           Maturity (Vector)
%   sigma       Volatility of Diffusion
%   r           Risk-free-Rate
%   q           Div Yield
%   lambda      Poisson intensity
%   m           Jump Mean
%   delta       Jump Std Deviation
```

```
%   n           Event Count (Limited to 170)
%

[K,T] = meshgrid(K,T);
[u,v] = size(K);
K = K(:,:,ones(1,1,n)); T = T(:,:,ones(1,1,n));
n = ones(1,1,n); n(:)=0:size(n,3)-1; factn = factorial(n);
n = n(ones(u,1),ones(1,v),:); factn = factn(ones(u,1),ones(1,v),:);

M = m+0.5*delta.^2;
lambda_bar = lambda.*exp(M);
r_n = r - lambda*(exp(M)-1) + n.*(M)./T;
sigma_n = sqrt(sigma.^2 + (n*delta^2)./T);

dfcn = @(z,sigma,r)((1./(sigma.*(T.^(0.5)))).*(log(S./K) + ...
                    (r - q + z.*0.5*(sigma.^2)).*T));
callfcn = @(sigma,r)(((1./factn).*exp(-lambda_bar.*T).*(lambda_bar.*T).^n).* ...
                     (exp(-q.*T).*S.*Nfcn(dfcn(1,sigma,r)) - K.*exp(-r.*T).* ...
                      Nfcn(dfcn(-1,sigma,r))));

optionPrice = sum(callfcn(sigma_n, r_n),3); % Eqn 19 of Merton 1976

if cp==-1
    optionPrice = optionPrice - S.*exp(-q.*T(:,:,1)) + ...
      K(:,:,1).*exp(-r.*T(:,:,1)); % Convert Call to Put by Parity Relation
end

end

%% NORM INVERSE
function p = Nfcn(x)
p=0.5*(1.+erf(x./sqrt(2)));
end
```

26.5 Physical-measure jump-diffusion return histograms

The final script generates 10,000 paths over one year with 100 steps, $S_0 = 1$, physical expected-return parameter .05, diffusion volatility .2, intensity .2, jump log-mean $-.2$ and dispersion .2. It separately generates diffusion log-increments and Poisson counts. Conditional on a count J, the jump sum is $J\mu_J + \sigma_J\sqrt{J}Z$, and the compensator is $\lambda(e^{\mu_J+\sigma_J^2/2} - 1)$. This corresponds to the corrected logarithmic dynamics in Section 19.3, using a physical expected return instead of r.

The price matrix includes its initial state. Columns 21, 51 and 101 therefore correspond to times .2, .5, 1. The plots are densities of *log returns* X, not prices S, even though both are simulated. The script reports sample skewness and kurtosis and uses fixed histogram edges from -1 to 1, so it does not display tails outside that range. Those finite-window plots must not be treated as proof that no observations can lie beyond the displayed interval. Its skewness and kurtosis outputs require the appropriate Matlab statistics functions.

```
%% Simulation of the Merton Jump Diffusion Process under P
%
%  The MJD process is:
%  X(t) = (mu_S - 0.5*sigma_S^2)*t + sigma*W(t)
%           + sum_{i=1}^{N(t)} Z_i - jump_compensator*t
%
clear all
close all
%
% Parameters
npaths = 10000 ; % Number of paths to be simulated
T = 1 ; % Time horzion
nsteps = 100 ; % Number of timesteps
dt = T/nsteps ; % Size of timesteps
t = 0:dt:T ; % Discretization of our grid
muS = 0.05 ; % Drift for ABM
sigmaS = 0.2 ; % Diffusion for ABM
lambda = 0.2 ; % Rate of arrival for Poisson Process
muJ = -0.2 ;  % Drift for Jumps
sigmaJ = 0.2 ; % Diffusion for Jumps
```

```
S0 = 1 ; % Initial stock price

%% Monte Carlo Simulation - npaths x nsteps
%
% Continuous component
dW = (muS - 0.5*sigmaS^2)*dt + sigmaS*sqrt(dt)*randn([npaths,nsteps]) ;

% Discontinuous component
compensator = lambda*(exp(muJ + 0.5*sigmaJ^2)-1) ;
dN = poissrnd(lambda*dt,[npaths,nsteps]) ;
dJ = -compensator*dt + muJ*dN + sigmaJ*sqrt(dN).*randn([npaths,nsteps]) ;

% Full MJD process
dX = dW + dJ ;
X = [ zeros([npaths,1]) , cumsum(dX,2)] ;

% Stock price
S = S0*exp(X) ;

close all
figure(1)


%% Probability Density Function at different times

% Parameters for x-axis
dx = 0.02 ;
x = -1:dx:1 ;
xx = x + dx/2 ; % Shift required for bar chart

h1 = histogram(X(:,21),'BinEdges',x,'Normalization','pdf') ;
H1 = [h1.Values,0] ;

h2 = histogram(X(:,51),'BinEdges',x,'Normalization','pdf') ;
H2 = [h2.Values,0] ;

h3 = histogram(X(:,101),'BinEdges',x,'Normalization','pdf') ;
H3 = [h3.Values,0] ;

subplot(3,1,1)
bar(xx,H1)
ylabel('f_X(x, t = 0.2)')
xlim([-1,1])
ylim([0,5])
title('Probability density function of a Merton jump-diffusion process at different times');
lgd = ['skewness: ' num2str(skewness(X(:,21)),'%2.2f') ' kurtosis: ' num2str(kurtosis(X(:,21)),'%2.2f')];
legend(lgd)


subplot(3,1,2)
bar(xx,H2)
xlim([-1,1])
ylim([0,5])
ylabel('f_X(x, t = 0.5)')
lgd = ['skewness: ' num2str(skewness(X(:,51)),'%2.2f') ' kurtosis: ' num2str(kurtosis(X(:,51)),'%2.2f')];
legend(lgd)

subplot(3,1,3)
bar(xx,H3)
xlim([-1,1])
ylim([0,5])
xlabel('x')
ylabel('f_X(x, t = 1)')
lgd = ['skewness: ' num2str(skewness(X(:,101)),'%2.2f') ' kurtosis: ' num2str(kurtosis(X(:,101)),'%2.2f')];
legend(lgd)
```

A Course references

Class materials:
Slides
Textbooks:
1. Robert McDonald, Derivatives Markets, Pearson Addison Wesley
2. John Hull, Options, Futures, and Other Derivatives, Prentice Hall

My contact information:
roberto.marfe@unito.it

Historical market statistics

The BIS series provide historical context on exchange-traded and OTC markets. Contract count, notional amount and market value measure different objects. For optional historical data, see `http://www.bis.org/statistics/extderiv.htm` and `http://www.bis.org/statistics/derstats.htm`.